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First-line cadonilimab plus chemotherapy in HER2-negative advanced gastric or gastroesophageal junction adenocarcinoma: a randomized, double-blind, phase 3 trial

In the format provided by the authors and unedited

Supplementary Information

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Supplementary Table 1. Subsequent therapies

Therapy	Cadonilimab Group (N = 305)	Placebo Group (N = 305)
	No. of patients (%)	
Radiotherapy	9 (3.0)	5 (1.6)
Surgery	3 (1.0)	2 (0.7)
Chemotherapy	105 (34.4)	146 (47.9)
Immunotherapy	35 (11.5)	68 (22.3)
Targeted therapy	34 (11.1)	62 (20.3)

Supplementary Table 2. Duration of treatment

	Cadonilimab Group			Placebo Group		
	Cadonilimab (N=305)	**Oxaliplatin (N=304)**	**Capecitabine (N=302)**	**Placebo (N=304)**	**Oxaliplatin (N=304)**	**Capecitabine (N=301)**
Patients dosed with study drugs－no. (%)	305 (100.0)	304 (100.0)	302 (100.0)	304 (100.0)	304 (100.0)	301 (100.0)
Median duration of treatment－months	5.62	4.14	4.17	4.73	4.14	4.14
(range)	(0.5－22.2)	(0.5－7.8)	(0.2－7.7)	(0.6－21.4)	(0.6－7.1)	(0.2－7.1)
Median number of cycles of treatment	8.14	6.00	6.05	6.86	6.00	6.00
(range)	(0.8－32.1)	(0.8－11.2)	(0.3－11.2)	(0.9－31.0)	(0.9－10.3)	(0.3－10.3)
Total drug exposure－mg						
Mean (SD)	5494.54	1001.91	207772.19	4470.13	1004.89	206294.02
	(4818.398)	(391.580)	(86190.045)	(3584.635)	(399.427)	(89450.382)
Median	4000.00	1133.52	221500.00	3627.50	1069.95	225500.00

CLINICAL STUDY PROTOCOL

A Randomized, Double-blind, Multicenter, Phase III Clinical Study of Comparing the Efficacy and Safety of AK104 Plus Oxaliplatin and Capecitabine (XELOX) Versus Placebo Plus XELOX as First-line Treatment for Locally Advanced Unresectable or Metastatic Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma

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PROTOCOL SYNOPSIS

TITLE

A Randomized, Double-blind, Multicenter, Phase III Clinical Study of Comparing the Efficacy and Safety of AK104 Plus Oxaliplatin and Capecitabine (XELOX) Versus Placebo Plus XELOX as First-line Treatment for Locally Advanced Unresectable or Metastatic Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma

Objectives

Primary Objective

- To compare the overall survival (OS) of AK104 plus XELOX and placebo plus XELOX in the intent-to-treat (ITT) population.

Secondary Objectives

- To compare OS of AK104 plus XELOX and placebo plus XELOX in the integrated positive score (CPS) ≥ 5 population.
- To compare the progression-free survival (PFS), objective response rate (ORR), disease control rate (DCR), duration of response (DoR), and time to disease response (TTR) of AK104 plus XELOX and placebo plus XELOX in the PD-L1 CPS ≥ 5 population and the ITT population, as assessed by the Investigator per RECIST v1.1.
- To compare the safety and tolerability of AK104 plus XELOX and placebo plus XELOX.
- To evaluate the pharmacokinetic (PK) profile of AK104 plus XELOX.
- To evaluate the immunogenicity of AK104 plus XELOX.
- To compare the health-related quality of life (HRQoL) of AK104 plus XELOX and placebo plus XELOX.

ENDPOINTS

Primary Endpoints

- OS, based on the ITT population.

Secondary Endpoints

- OS, based on the CPS ≥ 5 population.
- PFS, ORR, DCR, DoR, TTR assessed by Investigator per RECIST v1.1 in the PD-L1 CPS ≥ 5 population and the ITT population.
- Safety assessment: Incidence and severity of adverse events (AEs) and serious adverse events (SAEs), and clinically significant abnormal laboratory results.
- PK characteristics: Serum drug concentrations at different time points after AK104 administration.
- Immunogenicity assessment: Number and percentage of subjects with detectable anti-drug antibodies (ADA).
- HRQoL assessment: Measurements will be performed using the EORTC QLQ-30 and EORTC QLQ-ST022 Quality of Life Questionnaire Module.

STUDY DESIGN

This is a Phase III, randomized, double-blind, placebo-controlled, multicenter clinical study to compare AK104 plus oxaliplatin + capecitabine (XELOX) and placebo plus XELOX in subjects with unresectable locally advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma. The study plans to include 588 eligible subjects and randomize in a ratio of 1:1 according to the stratified block randomization scheme to:

Study group: AK104 10 mg/kg + XELOX, once every 3 weeks (Q3W) for up to 6 cycles; followed by AK104 monotherapy, Q3W.

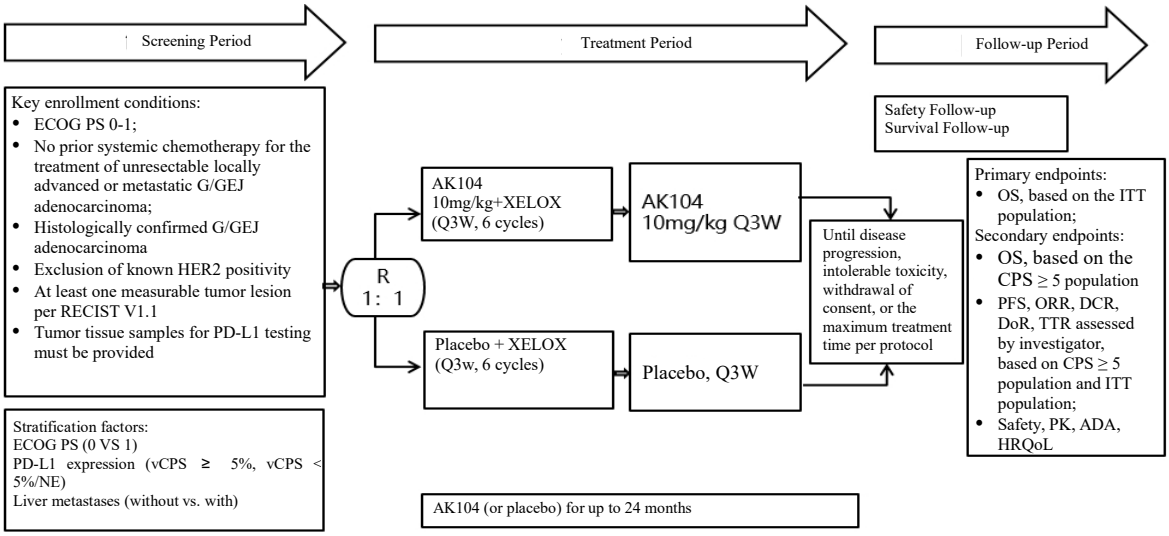
Control group: Placebo + XELOX, Q3W, up to 6 cycles; followed by placebo monotherapy, Q3W.

Randomization will be stratified according to the following specified stratification factors:

- ECOG PS (0 vs 1);
- PD-L1 expression (vCPS ≥ 5% vs. vCPS < 5%);
- Liver metastases (without vs. with).

The study design is detailed in the following figure:

Study Design Diagram



The study consists of a screening period (no more than 28 days after the subject signs the informed consent form up till the first dose), a treatment period (until the Investigator assesses no more clinical benefit, intolerable toxicity, or withdrawal of consent, whichever occurs first), and a follow-up period (including safety follow-up and survival follow-up). Subjects will be screened and evaluated 28 days before the first dose to determine if they meet the study eligibility criteria. Tumor samples from subjects should be collected during screening. Eligible subjects will receive the AK104 or placebo plus XELOX regimen based on randomization results. XELOX will be administered for up to 6 cycles. XELOX should be discontinued for disease progression prior to the end of XELOX in Cycle 6. After 6 cycles of XELOX treatment, subjects with stable disease or responders will continue to receive AK104 or placebo. AK104 (or placebo) will be administered once every 3 weeks until any of the following occurs: the subject is unable to continue to benefit as determined by the Investigator, intolerable toxicity, concomitant diseases that affect further treatment, Investigator decision, subject's withdrawal of consent, or other reasons as specified in the protocol (whichever occurs first). Subjects will receive AK104 (or placebo) for up to 24 months in the absence of withdrawal of consent, confirmed

imaging progression or clinical non-benefit, and intolerable toxicity.

Regular tumor response assessments will be performed by the Investigator for all subjects per RECIST 1.1. Subjects will undergo tumor assessments every 6 weeks (± 7 days) within 54 weeks after enrollment, and then every 9 weeks (± 7 days). For subjects receiving the treatment, the clinical decision will be based on tumor response assessed by the Investigator per RECIST v1.1. After radiographic disease progression as assessed by the Investigator per RECIST v1.1, the subject will be allowed to continue study drug AK104/placebo, if in the opinion of the Investigator, the subject can still clinically benefit from continued study treatment and tolerate the treatment of study drug. Once the study treatment is completed, the survival assessment will be performed every 1 month (± 7 days) after the end of treatment to obtain survival information, and, whenever possible, anti-tumor therapy information after the end of treatment.

Study Population

Inclusion Criteria

Subjects must meet all of the following inclusion criteria to be enrolled in the study:

1. Ability to understand and voluntarily sign a written informed consent form (ICF), which must be signed before the specified study procedures required for the study are performed.
2. Males or females aged ≥ 18 years and ≤ 75 years at the time of signing the ICF.
3. Histopathologically confirmed gastric adenocarcinoma or gastroesophageal junction (GEJ) adenocarcinoma.
4. Unresectable locally advanced or metastatic gastric adenocarcinoma or GEJ adenocarcinoma (a tumor centered within 5 cm proximal and distal to the anatomical location of the cardia as described in the Siewert Classification System).
5. Subjects have not received prior systemic therapy for locally advanced or metastatic gastric adenocarcinoma or GEJ adenocarcinoma; for subjects who have received prior neoadjuvant/adjuvant chemotherapy or chemoradiotherapy for curative intent, the time between disease progression and last treatment should be at least 6 months.
6. Subjects have at least one measurable tumor lesion per RECIST v1.1; lesions that received radiotherapy are not selected as target lesions, unless the lesion is the only measurable lesion and has unequivocal progression as judged by imaging, it can be considered as a target lesion.
7. All subjects must provide archival or freshly obtained tumor tissue samples within 2 years prior to signing the ICF.
8. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.
9. Expected survival ≥ 3 months.
10. Patients who have adequate organ function:

Test items	Inclusion requirements (based on the normal range of test values in corresponding study site)
Hematology (subjects are not allowed to receive blood transfusion or growth factor support within 7 days prior to the first dose)	
Absolute neutrophil count	$\geq 1.5\times10^9$/L
Platelet count	$\geq 100\times10^9$/L

Hemoglobin	≥ 90 g/L or ≥ 5.6 mmol/L
Liver and kidney function	
Creatinine (Cr) or calculated serum creatinine clearance (CrCl)	Cr ≤ 1.5× ULN or CrCl ≥ 50 mL/min (Calculated according to Cockcroft-Gault formula, see Appendix 3)
Albumin	≥ 28 g/L (albumin infusion is not allowed within 14 days prior to the first dose)
Total bilirubin (TBIL)	≤ 1.5× ULN; TBIL ≤ 3× ULN for subjects with documented/suspected Gilbert's syndrome
Alanine transaminase (ALT) and aspartate aminotransferase (AST)	≤2.5×ULN; ALT and AST ≤ 5 × ULN for subjects with liver metastases
Coagulation function	
International normalized ratio (INR), Activated partial thromboplastin time (APTT), Prothrombin time (PT)	≤ 1.5×ULN

11. Women of childbearing potential must have a negative serum pregnancy test within 7 days prior to the first dose and agree to take effective contraception measures during the study drug administration and within 120 days after the last dose. In this protocol, women of childbearing potential are defined as sexually mature women:

    1) No hysterectomy or bilateral oophorectomy,

    2) Spontaneous amenorrhea does not last for at least 24 consecutive months (amenorrhea caused by cancer treatment does not preclude fertility) (i.e., have menstruation at any time during the preceding 24 consecutive months).

    Male patients with female partners of childbearing potential must agree to take effective contraception measures during the study drug administration and within 120 days after the last dose.

Exclusion Criteria

Subjects who meet any of the following criteria are not eligible to participate in this study:

1. Subjects with known HER2-positive gastric or GEJ adenocarcinoma.
2. Histopathologically confirmed other pathological types, such as squamous cell carcinoma, sarcoma or undifferentiated carcinoma.
3. Subjects who received palliative local therapy for non-target lesions within 2 weeks prior to the first dose; systemic nonspecific immunomodulatory therapy (e.g., interleukin, interferon, thymosin, etc.) within 2 weeks prior to the first dose; and Chinese herbal medicine or traditional Chinese medicinal products with anti-tumor indications within 2 weeks prior to the first dose.
4. Subjects who received any prior treatments targeting the mechanism of tumor immunity, such as

immune checkpoint blockades (e.g., anti-PD-1 antibody, anti-PD-L1 antibody, anti-CTLA-4 antibody, etc.), immune checkpoint agonists (e.g., antibodies against ICOS, CD40, CD137, GITR, OX40 targets, etc.), immune cell therapy, etc.

5. Medical history of gastrointestinal perforation or gastrointestinal fistula within 6 months prior to the first dose. If the perforation or fistula has been treated with resection or repair and the disease has recovered or resolved as judged by the Investigator, enrollment may be allowed.
6. Active or previously documented inflammatory bowel disease (e.g., Crohn's disease or ulcerative colitis). Subjects who can't swallow or have malabsorption syndrome, uncontrolled nausea, vomiting, diarrhea, or other gastrointestinal disorders that severely affect drug intake and absorption.
7. Active malignancies within the past 3 years, with the exception of tumors in this study and cured local tumors, such as basal cell carcinoma of skin, squamous cell carcinoma of skin, superficial bladder cancer, carcinoma in situ of cervix, carcinoma in situ of breast, localized prostate cancer, etc.
8. Known active or untreated brain metastases, meningeal metastases, spinal cord compression, or leptomeningeal disease. However, subjects who meet the following requirements and have measurable lesions outside the central nervous system are allowed to be enrolled: asymptomatic after treatment and radiographically stable for at least 4 weeks prior to the start of study treatment (e.g., no new or enlarged brain metastases), and the systemic glucocorticoids and anticonvulsant medications have been discontinued for at least 2 weeks.
9. Presence of clinically symptomatic pleural effusion, pericardial effusion, or ascites requiring frequent drainage (≥ 1/month).
10. Active autoimmune disease requiring systemic treatment within 2 years prior to the start of study treatment, or autoimmune diseases that may relapse or require scheduled treatment as judged by the Investigator. With the following exceptions:
    a) Skin diseases not requiring systemic treatment (e.g., vitiligo, alopecia, psoriasis, or eczema);
    b) Hypothyroidism due to autoimmune thyroiditis, which requires only stable doses of hormone replacement therapy;
    c) Type I diabetes mellitus requiring only stable doses of insulin replacement therapy;
    d) Asthma which has resolved in childhood and does not require any intervention after reaching adulthood;
    e) The Investigator judges that the disease will not recur without external triggering factors.
11. Presence of any of the following cardiovascular and cerebrovascular diseases or cardiovascular and cerebrovascular risk factors:
    a) Myocardial infarction, unstable angina, cerebrovascular accident, transient ischemic attack, acute or persistent myocardial ischemia, symptomatic heart failure (Grade 2 or higher according to New York Heart Association functional classification), symptomatic or poorly controlled arrhythmia, or any arterial thromboembolism within 6 months prior to the first dose.
    b) History of deep vein thrombosis, pulmonary embolism, or other serious thromboembolism within 3 months prior to the first dose.
    c) Presence of major vascular diseases such as aortic aneurysm, dissecting aortic aneurysm,

internal carotid artery stenosis that may be life-threatening or require surgery within 6 months.

d) Previous history of myocarditis and cardiomyopathy.

e) Left ventricular ejection fraction (LVEF) < 50%.

12. Unresolved toxicities during prior anti-tumor therapy are defined as the toxicities that do not resolved to National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) (NCI CTCAE v5.0) Grade 0 or 1, or to the levels specified in the inclusion/exclusion criteria, with the exception of alopecia/pigmentation. Subjects with irreversible toxicity that are not expected to worsen after study drug administration (e.g., hearing loss) may be included in the study after consultation with the medical monitor. Subjects with long-term toxicity due to radiotherapy that cannot be recovered at the discretion of the Investigator may be included in the study after consultation with the medical monitor.
13. Presence of ≥ Grade 2 peripheral neuropathy as defined by NCI CTCAE v5.0.
14. Known presence of interstitial lung disease or noninfectious pneumonitis that is currently symptomatic or requires prior systemic glucocorticoid therapy that, in the judgment of the Investigator, may affect the assessment or management of toxicity related to study treatment.
15. Known active pulmonary tuberculosis. Subjects with suspected active pulmonary tuberculosis should be examined through chest imaging, sputum, and clinical symptoms and signs.
16. Systemic anti-infective therapy (excluding antiviral therapy for hepatitis B or C) within 2 weeks prior to the first dose.
17. Known history of allogeneic organ transplantation and allogeneic hematopoietic stem cell transplantation.
18. Presence of clinically active hemoptysis, active diverticulitis, abdominal abscess, and gastrointestinal obstruction.
19. History of clinically significant hemorrhage symptom or clear hemorrhagic diathesis within 1 month prior to the first dose, such as digestive tract hemorrhage, gastirc ulcer hemorrhage or vasculitis.
20. Endoscopic signs of active bleeding of lesions are known.
21. Major surgery other than the diagnosis of gastric cancer within 28 days prior to the first dose or major surgery is expected during the study.
22. Subjects with active hepatitis B or active hepatitis C.
    a) Subjects with hepatitis B surface antigen positive (HBsAg +) and/or hepatitis B core antibody positive (HBcAb +) requiring hepatitis B virus deoxyribonucleic acid (HBV DNA) testing;
    b) Those who are stable with HBV DNA copy number < 1000 cps/mL (200 IU/ml) or below the lower limit of detectable value at the study site after antiviral therapy may be enrolled, and subjects are required to continue anti-hepatitis B virus therapy during the study treatment;
    c) Patients who are hepatitis C antibody positive (HCV Ab+) require HCV RNA testing and are eligible to participate in the study only if they are HCV RNA negative (defined as below the lower limit of detectable value at the study site);
23. Known medical history of immunodeficiency or positive HIV test.
24. Known active syphilis infection.

25. Concurrent participation in another clinical study, unless it is an observational, non-interventional clinical study or the follow-up period of an interventional study.
26. Subjects who require systemic treatment with glucocorticoids (> 10 mg/day prednisone or equivalent) or other immunosuppressive drugs within 14 days prior to the first dose. With the following exceptions:
    a) Inhaled, ophthalmic, or topical glucocorticoids or other glucocorticoids at doses ≤ 10 mg/day prednisone or equivalent are permitted if there is no active autoimmune disease.
    b) Physiologic doses of systemic glucocorticoids ≤ 10 mg/day prednisone, or equivalent doses of other glucocorticoids.
    c) Glucocorticoids are used as pretreatment for infusion-related reactions or allergic reactions (e.g., before CT examination, etc.).
27. Subjects with serious neurological or psychiatric disorders, including dementia and epileptic seizure.
28. Live vaccines inoculated within 30 days prior to first dose or planned to be inoculated during the study.
29. Subjects with known medical history of severe hypersensitivity reactions to other monoclonal antibodies.
30. Subjects with known contraindications to oxaliplatin and capecitabine chemotherapy (see instructions for oxaliplatin and capecitabine).
31. Subjects with a known history of allergy or hypersensitivity to AK104, oxaliplatin, capecitabine, or any of its ingredients.
32. Subjects with known dihydropyrimidine dehydrogenase (DPD) deficiency.
33. Pregnant or lactating women.
34. Any condition that, in the opinion of the Investigator, may result in a risk when receiving the study drug, or would interfere with the evaluation of the study drug or the safety of subjects, or the interpretation of the study results.

Study Duration

The number of OS events from the date of randomization of the first subject until the number of OS events required for the final analysis is expected to be 443 in this study. The specific study duration will also depend on the time reaching the required number of OS events for the final analysis. The current estimated time is 39 months.

STATISTICAL METHODS

Sample Size Estimation

OS in the ITT population will be used as primary endpoint in this study. A total of 588 subjects will be enrolled and randomized in a 1:1 ratio. It is assumed that a total of 588 subjects will be enrolled over an 18-month period. The number of events required for OS in the ITT population is estimated based on the following assumptions (assuming an exponential distribution of event time):

- The one-sided significance level is 0.025. A test will be performed in the ITT population;
- The power to test an OS HR of 0.75 (equivalent to a median OS increase from 11 to 14.7 months) in the ITT population is at least 85%;

- An interim analysis of OS will be performed in the ITT population to estimate the O'Brein-Fleming cut point by Lan-DeMets spending function;
- The OS dropout rate of 2%/12 months is assumed for both treatment groups.

Statistical Analyses

The evaluation and analysis of efficacy endpoints are based on the ITT analysis set. The ITT analysis set includes all randomized subjects; for analyses requiring baseline data, only subjects with baseline data are included (e.g., for ORR, measurable disease as defined per RECIST 1.1 is required at baseline). Safety analyses will be performed according to the Safety Analysis Set (SS), which includes all subjects who receive any amount of the study drug.

Efficacy analyses

The primary objective of this study is to compare the efficacy of the AK104 plus oxaliplatin and capecitabine (XELOX) regimen and the placebo plus XELOX regimen as first-line treatment for unresectable locally advanced or metastatic gastric adenocarcinoma or adenocarcinoma of the gastroesophageal junction, with OS in the ITT Analysis Set as primary endpoint. One interim and final analyses of OS are planned for this study in the ITT Analysis Set.

OS: OS will be compared between the study group and the control group by stratification factors at randomization using a stratified Log-rank test based on the ITT Analysis Set. The hazard ratio and 95% confidence interval will be estimated using a stratified Cox regression model, and the same event time will be handled with an exact method. The Kaplan-Meier method will be used to estimate the distribution of event time in each treatment group. The median survival time (if evaluable) and its two-sided 95% confidence intervals will be assessed every 6 months using the Kaplan-Meier method.

Safety analyses

The SS includes all subjects who receive any amount of the study drug. The analysis will be conducted according to the treatment actually received by the subjects.

Adverse events (AEs) will be coded using the Medical Dictionary for Regulatory Activities (MedDRA) and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) version 5.0. The number and percentage of subjects with TEAEs and study drug-related TEAEs will be summarized by the worst CTCAE grade, system organ class (SOC), and preferred term (PT). Laboratory abnormalities and toxicity grades will be summarized according to NCI CTCAE (Version 5.0), shifts in CTCAE grade at baseline and worst postbaseline grade will be displayed in shift tables for clinical laboratory tests, and CTCAE Grade 3 or 4 clinical laboratory abnormalities will be presented.

Pharmacokinetic Analyses

Individual AK104 concentrations will be tabulated by descriptive statistics, and a population PK model will be established. The impact of physiologically relevant subject characteristics (covariates) and disease characteristics on PK will be evaluated. The relationship between exposure and response or safety will be evaluated. Such analysis can be described in PK analysis plans for different populations, and the result report of this PK analysis can be separated from the main study report.

Immunogenicity Analyses

The immunogenic potential of AK104 will be assessed by summarizing the number and percentage of subjects

who develop detectable ADA. The impact of ADA on PK will be assessed if data is available. Samples will be collected for evaluating the neutralizing capacity of ADAs in the future. Only subjects who have received at least one dose of AK104 and provided a baseline sample and at least one post-treatment sample will be evaluated.

Interim Analyses

An interim analysis of OS will be performed once based on the ITT population. The O'Brein-Fleming cut point will be estimated using the Lan-DeMets spending function. The interim analysis will be performed when approximately 354 OS events (80% of target OS events) are observed in the ITT population, which is estimated to be 28 months after the first subject is randomized. After approximately 443 deaths are observed in the ITT population, the final analysis of OS will be performed, which is estimated to be 39 months after the first subject is randomized.

Independent Data Monitoring Committee (IDMC)

An independent data monitoring committee (IDMC) will be established for this study. The committee will include an independent oncologist and an independent statistician. The IDMC will periodically assess the on-study safety data and review the OS efficacy interim analysis data, and make corresponding recommendations to the Sponsor according to the preset OS stopping boundary. After review and discussion by the IDMC, the Sponsor will make all final decisions regarding any change in study conduct. Details are provided in the IDMC charter. The functions and membership of IDMC will be described in the IDMC Charter.

An independent data monitoring committee (IDMC) will perform an interim power data review for the results of interim analysis. After reviewing the data, considering the overall risks and benefits of subjects, IDMC will provide recommendations on whether the final analysis can be performed as planned, whether protocol modification is required, and whether a registration marketing application can be submitted to regulatory authorities in advance, based on the predefined statistical hypotheses and superiority determination rules.

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LIST OF ABBREVIATIONS

Abbreviations or Specific Terms	Definitions
AACR	American Association for Cancer Research
ADA	Anti-drug antibody
AE	Adverse event
AESI	Adverse event of special interest
ALT	Alanine aminotransferase
ANC	Absolute neutrophil count
AST	Aspartate aminotransferase
ASCO	American Society of Clinical Oncology
ASCO-GI	American Society of Clinical Oncology Gastrointestinal Cancers Symposium
AUC	Area under concentration-time curve
BID	Bis in die (twice daily)
BOR	Best overall response
CI	Confidence interval
CL	Clearance
C_{max}	Maximum concentration
CR	Complete response
c-Met	Tyrosine protein kinase Met
CRO	Contract research organization
CrCl	Creatinine clearance
CT	Computed tomography
ctDNA	Circulating tumor DNA
CTLA-4	Cytotoxic T lymphocyte-associated antigen 4
CPS	Combined positive score
vCPS	Visualized combined positive score
DCR	Disease control rate
DLT	Dose-limiting toxicity
DoR	Duration of response
DPD	Dihydropyrimidine dehydrogenase
EBV	Epstein-Barr virus
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	Electronic case report form
EDC	Electronic data capture
EGFR	Epidermal growth factor receptor
ESMO	European Society for Medical Oncology
Fc	Crystallizable fragment
FDA	Food and Drug Administration
FFPE	Formalin-fixed paraffin-embedded
FOLFOX	Oxaliplatin + fluorouracil + calcium folinate
GCP	Good Clinical Practice
GEJ	Gastroesophageal junction
GLP	Good Laboratory Practice
HBV	Hepatitis B virus
HCV	Hepatitis C virus
HER2	Human epidermal growth factor receptor 2
HIV	Human immunodeficiency virus
HGF	Hematopoietic growth factor
HNSTD	Highest non-severely toxic dose
HR	Hazard ratio
ICF	Informed consent form
ICOS	Inducible co-stimulator

Abbreviations or Specific Terms	Definitions
AACR	American Association for Cancer Research
ICH	International Conference on Harmonization
IEC	Independent ethics committee
IgG1	Immunoglobulin G1
IHC	Immunohistochemistry
irAE	Immune-related adverse event
INR	Anticoagulation parameter
IRB	Institutional review board
ITT	Intent-to-treat population
LFTs	Liver function test
MAD	Maximum administered dose
MedDRA	Medical Dictionary for Regulatory Activities
MRA	MRI angiography
MRI	Magnetic resonance imaging
mRNA	Messenger RNA
MSI	Microsatellite instability
MTD	Maximum tolerated dose
dMMR	Mismatch repair deficiency
NCI CTCAE	National Cancer Institute Common Terminology Criteria for Adverse Events
NE	Not evaluable
NSAID	Non-steroidal anti-inflammatory drug
ORR	Objective response rate
OS	Overall survival
mOS	Median overall survival
PBMC	Peripheral blood mononuclear cell
PCR	Polymerase chain reaction
PD	Disease progression
PD-1	Programmed cell death-1
PD-L1	Programmed cell death ligand-1
PD-L2	Programmed cell death ligand-2
PET	Positron emission tomography
PFS	Progression-free survival
mPFS	Median progression-free survival
PK	Pharmacokinetics
PLT	Platelets
PR	Partial response
Q3W	Every 3 weeks
QTcB	QT interval corrected for heart rate using Bazett's formula
QTcF	QT interval corrected for heart rate using Fridericia's formula
RECIST	Response evaluation criteria in solid tumors
SAE	Serious adverse event
SAP	Statistical analysis plan
SCr	Serum creatinine
SD	Stable disease
SID	Subject identification
$t_{1/2}$	Half-life
TBIL	Total bilirubin level
TRAE	Study drug-related AEs
TCGA	The Cancer Genome Atlas
TPS	Tumor proportion score
TIL	Tumor infiltrating lymphocyte

Abbreviations or Specific Terms	Definitions
AACR	American Association for Cancer Research
TSH	Thyroid stimulating hormone
TTR	Time to response
ULN	Upper limit of normal
VEGF	Vascular endothelial growth factor
WHO-DD	World Health Organization Drug Dictionary
XELOX	Oxaliplatin + capecitabine

1. STUDY BACKGROUND

1.1 Introduction

Gastric cancer is the fifth most common malignancy worldwide and the fourth leading cause of cancer death; the incidence and mortality of gastric cancer vary by regions, with 75.3% of gastric cancers incidence and 74.8% of gastric cancer mortality from Asia[1]. At present, no nationwide early screening program for gastric cancer has been conducted in China, and most patients are diagnosed with gastric cancer at advanced stages, which is similar to the United States and Europe where about 60% of new cases are diagnosed with advanced gastric cancers, while in Japan where the early gastric cancer screening is common, the number of patients diagnosed with advanced gastric cancer accounts for only about 20% of newly diagnosed cases[2].

Gastroesophageal junction cancer spans the distal esophagus and proximal stomach anatomically, and its biological behavior and clinical characteristics are more similar to the gastric cancer than esophageal cancer, but the data are scarce[3]. For this part of patients, the mode of treatment and management can refer to that of gastric cancer before optimal treatment is found. Adenocarcinoma is the main pathological type of gastric cancer and gastroesophageal junction cancer, accounting for more than 90% of all pathological types. Studies have shown that [4]-[8] Helicobacter infection, smoking, alcohol consumption, obesity, and high salt diet are all risk factors for gastric cancer, among which Helicobacter infection is the most important risk factor for gastric cancer. China is a country with a high incidence of Helicobacter infection, and the infection rate is up to 50%; patients with Helicobacter infection have a 1-fold increased risk of gastric cancer compared with those without infection.

The treatment of advanced gastric cancer is limited and the efficacy is poor. Fluorouracil, platinum, and taxane chemotherapies are the mainstay of treatment for advanced gastric cancer. First-line chemotherapy regimens are usually two- or three-drug chemotherapy regimens based on fluorouracils in combination with platinum and/or taxanes. Oxaliplatin plus capecitabine is often used as first-line standard of care for unresectable locally advanced gastric cancer and HER2-negative advanced metastatic gastric cancer. Over the past 50 years, despite advances in first-line chemotherapy regimens, the median survival still struggles to break 1 year[3]. Although the objective response rate (ORR) of first-line chemotherapy is 30%-50% [9] , it is also accompanied by higher toxicities, especially hematological adverse reactions. Studies have shown that platinum-containing two-drug chemotherapy regimens can cause about 40% of Grade 3/4 neutropenia, if taxanes are added again, this proportion will increase to 82% [10]-[14] .

Several drugs targeting anti-VEGF, EGFR, HER2, c-Met and other targets have been tried in gastric cancer, but only the trastuzumab in combination with chemotherapy (ToGA study) targeting HER2

positivity has been successful in the first-line treatment of advanced gastric cancer, none of the remaining targeted drugs has been shown to improve patient survival[15]. HER2 overexpression positivity is reported globally in 7.3%-20.2%, and the HER2 positivity in gastric cancer in China is 12%-13% [16]-[18] . Therefore, there is a huge unmet need for patients with unresectable locally advanced or metastatic gastric adenocarcinoma, and new therapies are urgently needed to improve patient survival.

Immune checkpoints are a class of inhibitory molecules in the immune system. Physiologically, they play an important role in avoiding damage to normal tissues and organs by the immune response. However, in tumors, immune checkpoints can mediate immune escape of tumors, so immune checkpoints have been developed as targets of anti-tumor therapies in recent years. Anti-PD-1/PD-L1 antibodies, acting primarily in the tumor microenvironment, enhance immune-mediated anti-tumor responses by restoring the activity of preexisting anti-tumor T cells, increasing its response to tumor cells, and promoting cytokine production [19]-[22] . At present, immune checkpoint blockades targeting the anti-PD-1/PD-L1 pathway, i.e., the anti-PD-1 monoclonal antibodies represented by nivolumab, pembrolizumab, etc., have shown clinical benefit in a variety of advanced tumor types and have obtained corresponding indications. PD-1/PD-L1 is not the only immunosuppressive molecule in the immune microenvironment, and other immunosuppressive molecules, including CTLA-4, may also cooperate with PD-1/PD-L1 to inhibit T cell killing activity. Therefore, the combination of targeted drug therapies with different immunosuppressive molecules is a potential approach to potentially increase efficacy. At present, in addition to anti-PD-1/PD-L1 targeted drugs, anti-CTLA-4 targeted drugs have also been developed well. Anti-CTLA-4 antibodies, represented by ipilimumab, mainly act in lymph nodes; it can restore T-cell activation and proliferation by contact with T-cell activation inhibitory signals, promote the generation of memory T cells, and increase the compensatory increase in tumor PD-L1 to enhance immune-mediated anti-tumor responses; therefore, simultaneous blockade of the PD-1 and CTLA-4 pathways can show synergistic anti-tumor activity [23]-[25] . At present, the anti-PD-1 antibody nivolumab in combination with the anti-CTLA-4 antibody ipilimumab has been approved for 9 indications in 6 large tumors: melanoma, renal cell carcinoma, colorectal cancer, hepatocellular carcinoma, non-small cell lung cancer, and pleural mesothelioma, and has been confirmed to bring long-term survival for patients in melanoma, kidney cancer, intestinal cancer, non-small cell lung cancer.

Chemotherapy can enhance immunotherapeutic activity by inducing immunogenic death and increasing the release of tumor antigens, which in turn increases the tumor antigen presentation[26]. In the CheckMate-568[27] and CheckMate-9LA[28] studies, nivolumab in combination with

ipilimumab followed by chemotherapy were used as first-line treatment for advanced non-small cell lung cancer. Based on the results of the study, double immunotherapy in combination with chemotherapy significantly improved the ORR, progression-free survival (PFS), and overall survival (OS) compared with chemotherapy alone, and this benefit is consistent regardless of whether the patient had squamous or non-squamous cell carcinoma and whether the PD-L1 expression level was positive or negative. In addition, double immunotherapy in combination with chemotherapy showed better efficacy in early disease control compared to immunotherapy alone (e.g., KEYNOTE-042[29]) and double immunotherapy alone (e.g. CheckMate-227[30]). This combination regimen has also been approved by U.S. Food and Drug Administration (FDA) for relevant indications. Therefore, double immunotherapy in combination with chemotherapy can bring deep and durable responses to patients, as well as long-term survival benefits to patients.

Immunotherapy combined with chemotherapy has also made breakthrough progress in gastric cancer and adenocarcinoma of the gastroesophageal junction. In the first-line Phase III KEYNOTE-062[31] study of gastric cancer, pembrolizumab in combination with chemotherapy (5-FU in combination with cisplatin) or chemotherapy alone (5-FU in combination with cisplatin) was used for advanced gastric cancer or adenocarcinoma of the gastroesophageal junction without HER2 overexpression and with PD-L1 positivity, the group of pembrolizumab in combination with chemotherapy showed higher ORR and longer median OS (mOS) compared with the group of chemotherapy alone. In the CheckMate-649 study for the first-line treatment of gastric cancer and gastroesophageal junction cancer and esophageal adenocarcinoma[32], three treatment groups were included, i.e., nivolumab in combination with Ipilimumab, nivolumab in combination with oxaliplatin and capecitabine (XELOX regimen) or nivolumab in combination with fluorouracil, calcium folinate, and oxaliplatin (FOLFOX regimen), and chemotherapy alone (XELOX regimen/FOLFOX regimen); the results reported on the 2020 ESMO showed a significant survival benefit of nivolumab in combination with chemotherapy compared to chemotherapy alone in both the CPS ≥ 5 and CPS ≥ 1 populations stratified by CPS or the full population. In the CPS ≥ 5 population, the ORR was 60% vs. 45% and the median OS was 14.4 months vs. 11.1 months in the nivolumab plus chemotherapy group compared with the chemotherapy alone group; in the CPS ≥ 1 population, the mOS was 14.0 months vs. 11.3 months and the mPFS was 7.5 months vs. 6.9 months, respectively; in the full population, the mOS was 13.8 months vs. 11.6 months and the PFS was 7.7 months vs. 6.9 months, respectively. In terms of adverse reactions, the overall incidence of TRAEs of any grade was 94%, and that of Grade 3-4 TRAEs was 59%, which did not significantly increase the toxicity compared to chemotherapy alone; the most common (≥ 25%) TRAEs of any grade were nausea, diarrhea, and peripheral neuropathy. The occurrence of adverse reactions in the

CPS ≥ 5 population was consistent with that in the full population. In addition, subgroup analyses of the Chinese population of Checkmate649 reported in the AACR in 2021 showed[33] consistent benefit between the Chinese population and the global population; regardless of in the CPS ≥ 5, CPS ≥ 1, and the full populations, nivolumab in combination with chemotherapy significantly improved PFS and OS compared with chemotherapy alone. In August 2021, the regimen of nivolumab in combination with fluorouracil- and platinum-containing chemotherapy was approved by the National Medical Products Administration (NMPA) for the first-line treatment of patients with advanced or metastatic gastric cancer, gastroesophageal junction cancer, or esophageal adenocarcinoma.

An ATTRACTION-4 study in Asian population[34] used nivolumab in combination with SOX and XELOX for the treatment of gastric cancer and adenocarcinoma of the gastroesophageal junction with primary endpoints of PFS and OS in the full population; in terms of PFS, nivolumab in combination with chemotherapy was significantly different from chemotherapy alone with 10.45 months versus 8.34 months, and the mOS was 17.45 months versus 17.15 months, respectively, although there was no significant difference in mOS between the two groups, it is the longest OS achieved by immunotherapy for gastric cancer at present, extending the median survival of unresectable locally advanced or metastatic gastric cancer and adenocarcinoma of the gastroesophageal junction to nearly one and a half years; in terms of ORR and disease control rate (DCR), the ORR was 57.5% versus 47.8% and the DCR was 71.8% versus 68.5% in the nivolumab plus chemotherapy group and the chemotherapy alone group, respectively, which also showed significant improvements. The Phase II small sample study BGB-A317-205[35] in the Chinese population included a total of 30 patients with untreated advanced GC/GEJC to be treated with tislelizumab in combination with XELOX regimen, the ORR was 46.7% and the DCR was 80%; and the results of another domestic study with the anti-PD-1 monoclonal antibody, toripalimab, in combination with chemotherapy (XELOX regimen) as first-line treatment in patients with advanced gastric adenocarcinoma showed that the ORR and DCR of toripalimab in combination with XELOX regimen could reach 66.7 and 88.9%; both showed high response rates[36]. A Phase III study (ORIENT-16) [37], which was conducted in the Asian population, used sintilimab in combination with chemotherapy as first-line treatment for advanced gastric cancer and adenocarcinoma of the gastroesophageal junction. The primary endpoint was OS in the PD-L1 CPS ≥ 5 population and the ITT population. The results of the study presented at the ESMO General Assembly in 2021 showed that sintilimab in combination with chemotherapy (XELOX regimen) showed significant survival benefit compared with chemotherapy alone (XELOX regimen) in both the PD-L1 CPS ≥ 5 population (18.4 months vs 12.9 months, HR 0.66, P = 0.0023) and ITT

population (15.2 months vs 12.3 months, HR 0.766, P = 0.009). Subgroup analysis of further OS found that the OS benefit was more pronounced in the PD-L1 CPS ≥5 population than that in the PD-L1 CPS ≥1 population (HR 0.64 vs 0.73).

In the CSCO Guidelines for the Diagnosis and Treatment of Gastric Cancer 2022 [38], for gastric adenocarcinoma patients with HER2-negative and PD-L1 CPS ≥5, FOLFOX/XELOX in combination with nivolumab and XELOX in combination with sintilimab regimens were supplemented to the first-line Grade I recommendation.

1.2 AK104

AK104 is a humanized immunoglobulin G1 (IgG1) bispecific antibody independently developed by Akeso Pharmaceuticals, Inc. (Akeso) with fragment crystallizable (Fc) mutations to eliminate Fc receptor- and complement-mediated cytotoxic effects. AK104 binds to both programmed cell death receptor-1 (PD-1) and cytotoxic T lymphocyte associated antigen-4 (CTLA-4) and blocks PD-1/programmed cell death ligand-1 (PD-L1), PD-1/PD-L2, CTLA-4/B7.1 and CTLA-4/B7.2 interactions. AK104 is a tetravalent IgG-ScFv bispecific antibody. AK104 contains a point mutation in its constant region that prevents binding of complement C1q and Fc involved in cytotoxic effects to receptor. AK104 is expressed in a Chinese hamster ovary cell line with a total molecular weight (including that of oligosaccharides) of approximately 200 kDa.

The bispecific antibody AK104, which targets both PD-1 and CTLA-4, has a quadrivalent structure, with a shorter half-life ($t_{1/2}$) and lower toxicity in monkey toxicity studies compared to anti-PD-1/L1 in combination with anti-CTLA-4 antibody. The use of AK104 in advanced cancer patients may have better efficacy and safety than that achieved by anti-PD-1/L1 in combination with anti-CTLA-4 antibody.

The level of coexpressed PD-1 and CTLA-4 in tumor infiltrating lymphocytes (TILs) is higher than that in normal tissues and peripheral blood cells. A search of The Cancer Genome Atlas (TCGA) research network public database revealed that PD-1 and CTLA-4 messenger RNA (mRNA) are coexpressed in a large number of solid tumors. It is well known that bispecific antibodies tend to have high affinity and long-term target retention due to their multivalent nature. The higher coexpression of PD-1 and CTLA-4 in tumor tissue compared to that in normal tissue has this high affinity and high retention effect more enriched in tumor tissue. In contrast, anti-PD-1 in combination with anti-CTLA-4 antibody does not have this advantage. In addition, CTLA-4 blocking antibodies have been shown to upregulate PD-1, whereas PD-1 blocking antibodies upregulate CTLA-4 in some studies. Prolongation of tumor retention by tetravalent PD-1/CTLA-4 bispecific antibodies such as AK104 can be enhanced by cross-upregulation of PD-1 and CTLA-4. Thus, the tetravalent bispecific antibodies targeting PD-1 and CTLA-4 can provide a therapeutic

advantage superior to that provided by the combination of anti-PD1/L1 and anti-CTLA-4.

1.2.1 Summary of Preclinical Pharmacokinetics

Pharmacokinetic (PK) properties of AK104 were investigated in cynomolgus monkeys. Following single intravenous administration of AK104 injection at 1, 4, and 16 mg/kg to cynomolgus monkeys, the clearance (Cl) observed was 2.2, 1.6, and 1.9 ml/h/kg, respectively, and the apparent volume of distribution (V_{ss_obs}) observed were 91, 92, and 113 mL/kg, respectively, indicating that AK104 was significantly distributed in tissues. The half-life ($t_{1/2}$) after administration of 4 mg/kg AK104 was 47.9 hours, approximately 1/3 to 1/2 that of typical antibodies. It should be noted that the volume of distribution of AK104 in monkeys was significantly greater than the blood volume, indicating significant tissue distribution. Rapid clearance from blood can reduce nonspecific (off-target) cytotoxicity. The preferential distribution of AK104 in tumor tissues may lead to better safety than the conventional anti-PD-1 and anti-CTLA-4 antibodies described above.

A nearly dose-proportional increase between the exposure (C_{max} and AUC) and dose was observed in the dose range from 1 to 16 mg kg. The increase in exposure (AUC_{last}) to AK104 increased dose proportionally with a power function model beta of 1.05, suggesting a linear PK profile of AK104.

A multiple dose PK study of AK104 was conducted in cynomolgus monkeys. After four weeks of weekly intravenous injection of 4 mg/kg AK104, the ratios of $AUC_{0\text{-}168h}$ and C_{max} after the last and first dose were 0.554 and 0.230, respectively, suggesting that anti-drug antibodies (ADA) may interfere with the detection of analytes or accelerate the clearance of AK104, which resulted in a decrease in AK104 exposure after the last dose compared to that after the first dose.

1.2.2 Summary of Preclinical Pharmacology

AK104 is a new humanized bispecific antibody drug developed by Akeso BioPharma, Inc. ("Akeso") targeting human PD-1 and CTLA-4. Preclinical primary pharmacodynamic experiments confirmed that AK104 could bind to human CTLA-4 and PD-1 with high affinity and block the binding of CTLA-4 and PD-1 to their ligands B7.1/B7.2 and PD-L1/PD-L2, thereby activating T lymphocytes, promoting secretion of cytokines IFN-γ and IL-2 by immune cells, and improving the immune clearance efficiency of tumor cells. Preclinical *in vivo* pharmacodynamic studies of AK104 explored the pharmacological activity of AK104 injection in the human PD-1 transgenic mouse line PD-1 HuGEMM mouse MC38 colon cancer cell xenograft tumor model, SCID/Beige mouse HCC827 cell xenograft tumor model, and SCID/Beige mouse Raji-PDL1 cell xenograft tumor model. AK104 injection effectively inhibited tumor growth in SCID/Beige mouse HCC827 cell xenograft tumor model and SCID/Beige mouse Raji-PDL1 cell xenograft tumor model; in the PD-1 HuGEMM mouse MC38 colon cancer cell xenograft tumor model, AK104 injection had statistically significant anti-tumor effect compared with the isotype control group, and had an anti-tumor effect

similar to that of nivolumab at similar doses, suggesting that AK104 injection has good *in vivo* pharmacodynamic activity.

1.2.3 Summary of Preclinical Toxicology

The cynomolgus monkey is considered the only pharmacologically relevant animal species in the preclinical safety study of AK104. In the GLP repeated-dose toxicity study of AK104 injection by intravenous injection to cynomolgus monkeys for 4 weeks with a 4-week recovery period and by intravenous infusion to cynomolgus monkeys for 13 weeks with a 6-week recovery period, the low, medium, and high doses of AK104 were 6, 20, and 60 mg/kg/week, respectively.

In the 4-week repeated-dose toxicity study, AK104 injection was well tolerated at intravenous infusion doses up to 60 mg/kg. Elevated ALT and/or AST and a decrease in the percentage of $CD16^+$ cells were observed at the end of the dosing period, and at the end of the recovery period, the recovery of ALT and AST parameters was observed, but a trend towards a decrease in CD16+ was still observed. Histopathological changes related to AK104 were mainly characterized by slight to mild lymphocytic infiltration in the sciatic nerve, interstitium of lacrimal glands, interstitium of salivary glands (submandibular gland) and renal interstitium. At the end of the recovery period, slight to mild lymphocytic infiltration was still observed in the interstitium of the salivary glands (submandibular glands) as well as in the renal interstitium. Therefore, the highest non-severely toxic dose (HNSTD) was considered to be 60 mg/kg, corresponding to an AUC (mean exposure in both male and female animals after repeated dosing) of 19,514 μg*hr/mL.

In the 13-week repeated-dose toxicity study, 2 males at 60 mg/kg were found dead on the day after the 5th (D29) and 6th (D36) doses, respectively, and the main cause of death was considered to be acute respiratory failure due to lung injury caused by infusion reactions. Except for the dead animals, one female at 20 mg/kg and one male at 60 mg/kg were thin and had decreased body weight during the dosing period; the male at 60 mg/kg had decreased albumin (Alb) and increased immunoglobulin IgA during the dosing period, decreased thymus volume was observed macroscopically at the end of the dosing period, and a moderate decrease in thymus cell count and moderate edema of the femoral bone marrow were observed microscopically; these pathological changes were not observed in animals euthanized and necropsied at the end of dosing and after the end of the 6-week recovery period. Test article-related changes during the dosing period included a trend towards increased high-sensitivity C-reactive protein (HsCRP) in males at 6 mg/kg and in both sexes at 20 and 60 mg/kg; decreased complement C4 on D30 in males at 6, 20, and 60 mg/kg and females at 60 mg/kg; increased IL-6 after dosing on D29 in individual males at 20, 60 mg/kg and individual females at 60 mg/kg; decreased albumin to globulin ratio (A/G) and increased globulin (Glb) and immunoglobulin IgG in males at 60 mg/kg, and decreased percentage of NK

cells in both sexes at 60 mg/kg. These changes were not observed in animals euthanized at the end of the 6-week recovery period. Therefore, HNSTD was considered to be 20 mg/kg, and the AUC after repeated dosing was 10870.75 μg * hr/mL in males and 15587.24 μg * hr/mL in females.

1.2.4 Summary of Clinical Studies

Multiple Phase I and II clinical studies of AK104 have been conducted in Australia and China, involving the treatment regimens of AK104 monotherapy and AK104 in combination with chemotherapy.

As of May 2021, AK104 has been studied as monotherapy in 457 subjects. In the dose escalation study of AK104 alone, the escalation from 0.2 mg/kg Q2W, 0.5 mg/kg Q2W, 1 mg/kg Q2W, 2 mg/kg Q2W, 4 mg/kg Q2W, 6 mg/kg Q2W, fixed dose of 450 mg Q2W, 10 mg/kg Q2W, 15 mg/kg Q3W, and 25 mg/kg Q3W has been completed in sequence, and only 1 subject in the 1 mg/kg dose group had a dose-limiting toxicity (DLT) event: Grade 3 AST increased. Maximum tolerated dose (MTD) was not reached at any dose level. Of the 457 subjects, the incidence of TRAEs was 80.5%, the incidence of ≥ Grade 3 TRAEs was 18.2%, and the incidence of ≥ Grade 3 irAEs was 9.2%. Twenty-one (21) subjects received AK104 10 mg/kg Q2W monotherapy. The incidence of TRAEs was 90.5%, the incidence of ≥ Grade 3 TRAEs was 19.0%, and the incidence of ≥ Grade 3 irAEs was 9.5%.

Two hundred and seven (207) subjects received AK104 combination regimens including AK104 + carboplatin + paclitaxel, AK104 + carboplatin + pemetrexed, AK104 + oxaliplatin + capecitabine, and AK104 + anlotinib, etc. Of the 207 subjects, the incidence of AK104-related adverse events was 82.1% for TRAEs, 45.4% for ≥ Grade 3 TRAEs , and 14.0% for ≥ Grade 3 irAEs. Twenty-five (25) subjects received AK104 10 mg/kg Q2W combination therapy. The incidence of TRAEs was 100.0%, the incidence of AK104-related ≥ Grade 3 TRAEs was 44.4%, and the incidence of ≥ Grade 3 irAEs was 20.0%. Compared with the safety data of AK104 monotherapy, preliminary results showed that AK104 combination therapy did not increase AK104-related TRAEs or increase the risk of irAEs.

AK104 monotherapy and combination therapy can not only significantly reduce immunotoxicity, but also show better clinical efficacy than the combination therapy of anti-PD-1 antibody and anti-CTLA-4 antibody.

At the 2022 ASCO-GI meeting, preliminary results of the Phase II study of AK104 in combination with mXELOX as first-line treatment for advanced gastric cancer or adenocarcinoma of the gastroesophageal junction (AK104-201 study[39]) showed that, as of August 2021, 96 patients with advanced gastric adenocarcinoma or adenocarcinoma of the gastroesophageal junction received AK104 at different doses in combination with oxaliplatin and capecitabine as first-line treatment.

Of these, 18 subjects were enrolled in the 4 mg/kg Q2W group, 40 subjects were enrolled in the 6 mg/kg Q2W group, 18 subjects were enrolled in the 10 mg/kg Q2W group, 18 subjects were enrolled in the 15 mg/kg Q3W group, and 2 subjects were enrolled in the 10 mg/kg Q3W group, with a median follow-up time of 9.95 months. Of the 88 evaluable patients, the ORR was 65.9%, including 2 complete responses (CRs) and 56 partial responses (PRs); the DCR was 92.0%. Median PFS and OS in the overall population were 7.10 months (95% CI 5.55, 10.48) and 17.41 months (95% CI 12.35, NE), respectively. Eighty-four (84) patients had been tested for PD-L1 expression, and OS was 17.08 months (95% CI 12.35, NE) in PD-L1-negative (CPS < 5) patients and not reached in PD-L1-positive (CPS ≥ 5) patients.. The safety results of the study showed that the incidence of treatment-related adverse events (TRAEs), including AK104 or chemotherapy, was 97.9%, and the most common TRAEs were platelet decreased (60.4%), white blood cell decreased (58.3%), neutrophil decreased (56.3%), anemia (47.9%), nausea (30.2%), vomiting (30.2%), and aspartate aminotransferase increased (30.2%). The incidence of Grade 3 or higher TRAEs related to AK104 or chemotherapy was 62.5%.

The staged efficacy and safety data of AK104 for recurrent or metastatic squamous carcinoma of the cervix after standard of care showed an ORR of 47.6% and a DCR of 66.7%, the median DoR was not reached (0.03+ to 10.35 months), and the maximum duration of treatment had exceeded 1 year. One (1) subject had stable disease for nearly 1 year with a maximum reduction in target lesions of -25.9%. Compared with the results of PD-1 monotherapy or PD-1 in combination with CTLA-4 in the treatment of cervical cancer, AK104 can show a more potential and sustainable tumor response in subjects who have failed platinum-based therapy. In August 2020, FDA had granted AK104 a fast track qualification for the treatment of advanced squamous carcinoma of the cervix; in October 2020, the Center for Drug Evaluation (CDE) of the National Medical Products Administration (NMPA) also included AK104 in the list of breakthrough therapies.

The results of AK104 in the treatment of malignant mesothelioma were also of concern. At the 2020 ESMO meeting, Australian investigators orally reported the results of the study with AK104 monotherapy in 18 subjects with malignant mesothelioma. Of the 15 efficacy evaluable subjects, ORR was 20.0% and DCR was 80.0%. The longest DoR exceeded 12.9 months. One (1) subject who had received anti-PD-1 monoclonal antibody therapy was treated with 6 mg/kg AK104, and the duration of stable disease was 7.2 months. For subjects at doses above 1 mg/kg, the 9-month progression-free survival rate was 54.5%.

1.3 Study Risk Assessment

1.3.1 Potential Risks Associated with AK104

Potential risks based on the mechanism of action of AK104 and related molecules include immune-mediated adverse reactions. Potential immune-related adverse events (irAEs) may be similar to those caused by anti-PD-1/L1 and/or anti-CTLA-4 drugs, which may include, but are not limited to the following types of toxicity: infusion-related reactions (pyrexia, rash, pruritus, hypotension, dyspnea, chest discomfort, wheezing, tachycardia, stiffness), skin toxicity (rash, pruritus, vitiligo), endocrine toxicity (hyperglycemia, hypothyroidism, hyperthyroidism, primary hypoadrenalism, hypophysitis), hepatotoxicity (hepatitis, aspartate aminotransferase increased, alanine aminotransferase increased, bilirubin increased), gastrointestinal toxicity (diarrhoea, colitis), pulmonary toxicity (pneumonia, pulmonary sarcoidosis), rheumatoid/boneiliac toxicity (rheumatoid arthritis, myositis, myalgia), neurotoxicity (myasthenia gravis, aseptic meningitis, encephalitis, transverse myelitis, Guillain-Barre syndrome), hematotoxicity (autoimmune hemolytic anemia, aplastic anemia, immune thrombocytopenia,acquired hemophilia), nephrotoxicity (nephritis, renal insufficiency), cardiotoxicity (myocarditis, pericarditis, arrhythmia), ocular toxicity (uveitis, scleritis). At present, the safety events found in the AK104 clinical study are consistent with the characteristics of tumor immunotherapy targeting PD-1 and/or CTLA-4 without new adverse reactions.

1.3.2 Potential Benefits Associated with AK104

Phase Ia clinical data of AK104 have preliminarily shown that AK104 is safe and well tolerated in patients with advanced tumors, and its anti-tumor activity and definite pharmacological activity are preliminarily observed. Given that the recently published data have suggested significant efficacy of PD-1 inhibitors as monotherapy, anti-PD-1 in combination with CTLA-4, and anti-PD-1 in combination with standard chemotherapy for advanced or metastatic gastric adenocarcinoma or adenocarcinoma of the gastroesophageal junction, it is concluded that AK104, a bispecific antibody targeting both anti-PD-1 and CTLA-4, may also achieve good efficacy in patients with the same disease condition, and relieve disease and prolong progression-free survival after combining with standard chemotherapy.

The level of coexpressed PD-1 and CTLA-4 receptors in tumor infiltrating lymphocytes is higher than that in lymphocytes found in T cells from normal tissues and peripheral blood. AK104 is a bispecific antibody that targets both PD-1 and CTLA-4, with a quadrivalent structure and a short half-life. It is less toxic compared to anti-PD-1/L1 in combination with anti-CTLA-4 (observed in monkey studies). It is superior to anti-PD-1/L1 antibody alone or anti-PD-1/L1 in combination with anti-CTLA-4 in terms of efficacy and/or safety. These hypotheses need to be validated in prospective clinical trials.

2. STUDY OBJECTIVES AND ENDPOINTS

2.1 Study Objectives

2.1.1 Primary Objective

- To compare OS of AK104 plus XELOX and placebo plus XELOX in the ITT population.

2.1.2 Secondary Objectives

- To compare OS of AK104 plus XELOX and placebo plus XELOX in the CPS ≥ 5 population.
- To compare the PFS, ORR, DCR, DoR, TTR of AK104 plus XELOX and placebo plus XELOX in the PD-L1 CPS ≥ 5 population and the ITT population, as assessed by the Investigator per RECIST v1.1.
- To compare the safety and tolerability of AK104 plus XELOX and placebo plus XELOX.
- To evaluate the pharmacokinetic (PK) profile of AK104 plus XELOX.
- To evaluate the immunogenicity of AK104 plus XELOX.
- To compare the health-related quality of life (HRQoL) of AK104 plus XELOX and placebo plus XELOX.

2.2 Study Endpoints

2.2.1 Primary Endpoints

- OS, based on the ITT population.

2.2.2 Secondary Endpoints

- OS, based on the CPS ≥ 5 population.
- PFS, ORR, DCR, DoR, TTR assessed by Investigator per RECIST v1.1 in the PD-L1 CPS ≥ 5 population and the ITT population.
- Safety assessment: Incidence and severity of adverse events (AEs) and serious adverse events (SAEs), and clinically significant abnormal laboratory results.
- PK characteristics: Serum drug concentrations at different time points after AK104 administration.
- Immunogenicity assessment: Number and percentage of subjects with detectable anti-drug antibodies (ADA).
- HRQoL assessment: Measurements will be performed using the EORTC QLQ-30 and EORTC QLQ-ST022 Quality of Life Questionnaire Module.

3. STUDY DESIGN

3.1 Overall Study Design

This is a Phase III, randomized, double-blind, multicenter clinical study to compare the efficacy and safety of AK104 in combination with oxaliplatin + capecitabine (XELOX) and placebo in combination with XELOX in subjects with unresectable locally advanced or metastatic gastric adenocarcinoma or gastroesophageal junction adenocarcinoma.

The study plans to include 588 eligible subjects and randomize in a ratio of 1:1 according to the stratified block randomization scheme to:

- Study group: AK104 10 mg/kg + XELOX, once every 3 weeks (Q3W) for up to 6 cycles; followed by AK104 monotherapy, Q3W;
- Control group: Placebo + XELOX, up to 6 cycles; followed by placebo monotherapy, Q3W.

Randomization will be stratified according to the following specified stratification factors:

- ECOG PS (0 vs 1);
- PD-L1 expression (vCPS ≥ 5% vs. vCPS < 5%);
- Liver metastases (without vs. with).

The study consists of a screening period (no more than 28 days after the subject signs the informed consent form up till the first dose), a treatment period (until the Investigator assesses no more clinical benefit, intolerable toxicity, or withdrawal of consent, whichever occurs first), and a follow-up period (including safety follow-up and survival follow-up).

Subjects will be screened and evaluated 28 days before the first dose to determine if they meet the study eligibility criteria. Tumor samples from subjects should be collected during screening. Eligible subjects will receive AK104 10 mg/kg in combination with XELOX chemotherapy every 3 weeks or placebo plus XELOX chemotherapy every 3 weeks based on randomization results. XELOX will be administered for up to 6 cycles. Treatment with XELOX should be discontinued for disease progression prior to the end of XELOX in Cycle 6. After 6 cycles of XELOX treatment, subjects with stable disease or responders will continue to receive AK104 (or placebo). AK104 (or placebo) will be administered once every 3 weeks until any of the following occurs: the subject is unable to continue to benefit as determined by the Investigator, intolerable toxicity, concomitant diseases that affect further treatment, Investigator decision, subject's withdrawal of consent, or other reasons as specified in the protocol (whichever occurs first). Subjects may continue to receive AK104 monotherapy between initial assessment of disease progression and confirmation of progression if performance status remains stable, and if in the opinion of the Investigator, the subject can benefit from ongoing AK104 treatment, and if other criteria listed in the protocol are

met. In this study, AK104 (or placebo) will be administered for a maximum of 24 months.

Regular tumor response assessments will be performed by the Investigator for all subjects per RECIST 1.1. Subjects will undergo tumor assessments every 6 weeks (± 7 days) within 54 weeks after enrollment, and then every 9 weeks (± 7 days). For subjects receiving the treatment, the clinical decision will be made based on the response assessed by the Investigator per RECIST version 1.1, including confirmation of progressive disease. Subjects may continue receiving treatment between the initial assessment and confirmation of progression if the performance status remains stable, and if in the opinion of the Investigator, the subject will benefit from ongoing treatment, and subjects meet other criteria listed in the protocol.

After completion of treatment, each subject will be followed-up for AE for at least 30 days and for SAE for at least 90 days or until the start of other anti-cancer treatments, whichever occurs first. Immune-related AEs (irAEs), whether serious or nonserious, will be followed up until 90 days after the end of AK104/placebo (whether other anti-tumor therapy have been initiated). After this period, the Investigator should report any SAEs that are assessed as related to study drug. A survival assessment will be conducted monthly (± 7 days) after the end of study treatment to collect survival information and anti-cancer treatment information after discontinuation of study drug.

The study design is detailed in Figure 1:

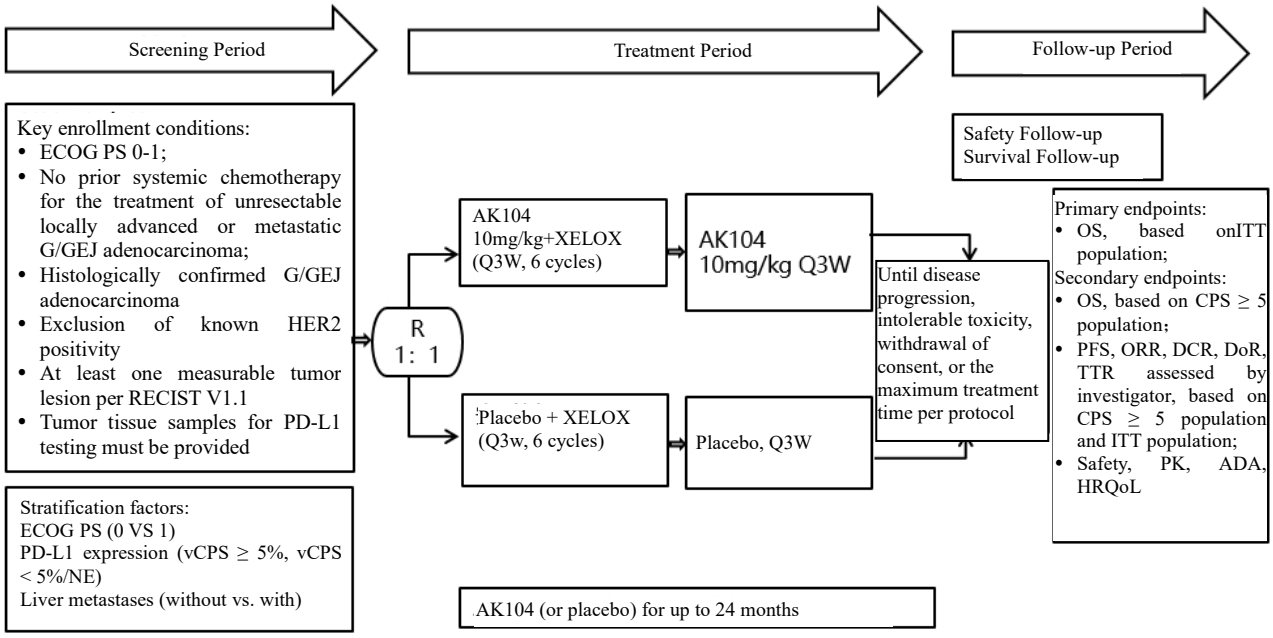

Figure 1 Study Design Diagram

3.2 Randomization

Subjects who met all of the inclusion criteria and none of the exclusion criteria will be randomized. Subjects should try to receive the first dose of study drug on the day of randomization, or no later

than Day 3 after randomization. The actual treatment given to the subject in this study will be determined by the randomization scheme in the IVRS/IWRS. The randomization scheme will be generated by a computer software program with a standard process capable of generating randomization codes. Randomization will be stratified according to the following specified stratification factors: ECOG PS (0 vs. 1), PD-L1 expression (vCPS ≥ 5% vs. vCPS < 5%), and liver metastases (without vs. with). Within each randomization stratification, randomization codes are assigned strictly in the order in which subjects are eligible for randomization.

3.3 Blinding and Unblinding

The subject's treatment assignment information will remain blinded until unblinding. Before the interim or final analysis of this study is planned to be unblinded, individual subjects from the clinical study may be unblinded only in the following circumstances:

1) Unblinding for expedited safety reporting to regulatory authorities;

Unless an unblinding waiver has been granted, the Sponsor will unblind all suspected and unexpected serious adverse reactions (SUSARs) in the clinical study report authorized by personnel of the Sponsor's pharmacovigilance department.

Blinding will be maintained for, e.g., study delivery team members and biostatisticians responsible for the analysis and interpretation of study results at the end of the clinical study.

2) In the conditions of emergency medical, the Investigator considers it necessary to know the treatment assignment information of the subjects to guide the emergency medical measures given to them;

The Investigator may unblind the subject's treatment assignment during the conduct of the clinical study only when the unblinding is meaningful for the subject's safety, i.e., he/she is aware that the study treatment assignment will affect the subject's treatment. Whenever possible, the decision to unblind should be discussed with the Sponsor's medical monitor. If unblinding occurs, the reason for unblinding must be documented in the original file.

3) Unblinding for other safety reasons.

In exceptional circumstances, it may be necessary to unblind the treatment of one or more subjects, e.g., to facilitate further assessment of potentially important safety concerns, or at the request of regulatory authorities. The decision to unblind and all unblinding procedures performed will be documented in the minutes of the meeting or in the notes to the file.

Cases requiring unblinding to facilitate the assessment of potential safety concerns are generally kept blinded in the drug safety database; cases requiring unblinding at the request of regulatory authorities are recorded as unblinded cases in the drug safety database.

All other subjects who are not unblinded during the study for the above reasons should remain

blinded until the end of the study.

3.4 Study Duration

Subjects who achieve CR, PR, or SD will receive the study treatment for up to 24 months without withdrawal of informed consent, confirmed progression, and intolerable toxicity. The number of OS events from the date of randomization of the first subject until the number of OS events required for the final analysis is expected to be 443 in this study. The specific primary study time will also depend on the time reaching the required number of OS events for the final analysis.

Subjects who are still receiving study treatment at the end of the study or beyond the longest treatment duration of 24 months may continue treatment through additional extension studies or other forms agreed upon by the Sponsor if stable or relieved disease is achieved in response evaluation, the adverse reactions are tolerable, and the approval of health authorities and ethics committees is obtained.

3.5 End of Study

During the operational phase of the study, the Sponsor may terminate the study at any time after all subjects complete the trial based on the actual clinical development.

3.6 Study Design Rationale

3.6.1 Rationale for Dose Selection of AK104

In the dose escalation study of AK104 alone, the escalation of 0.2, 0.5, 1, 2, 4, 6, 10 mg/kg Q2W, 15 mg/kg Q3W, and 25 mg/kg Q3W has completed. Only 1 subject in 1 mg/kg group had a DLT event: Grade 3 AST increased. Maximum tolerated dose (MTD) was not reached at any dose level. The results of the preliminary pharmacokinetic study showed that the escalation study from 0.2 mg/kg to 25 mg/kg Q3W indicated that the exposure of AK104 increased with the increase of dose, showing a specific linear pharmacokinetic profile. No significant differences in pharmacokinetics were found between Caucasians and Chinese. Based on factor analysis such as receptor occupancy, the recommended Phase II dose (RP2D) of AK104 monotherapy is 6 mg/kg, Q2W.

In March 2021, CDE agreed with the Sponsor to conduct a Phase III study of AK104 in combination with XELOX regimen as first-line treatment for advanced gastric cancer. As of May 2021, 87 patients with untreated advanced gastric cancer have been enrolled in the first-line gastric cancer study of AK104 in combination with mXELOX/XELOX, 75 patients are evaluable, with an overall ORR of 65.8% and a DCR of 93.2%, of which 2 patients have achieved CR. Eighteen (18) subjects were treated with 10 mg/kg in combination with chemotherapy once every other week. The incidence of Grade 3 TRAEs (TRAEs defined as adverse events related to AK104 or chemotherapy)

was 44.4%, and the incidence of Grade 3 or higher irAEs was 20.0%; of the 18 subjects who had at least one tumor assessment, the ORR had reached 83.3%; the median PFS was not reached. The study showed that tumor response may increase correspondingly with the dose increase of AK104. Different doses of AK104 (4 mg/kg Q2W, 6 mg/kg Q2W, 10 mg/kg Q2W) in combination with mXELOX and AK104 15 mg/kg Q3W in combination with XELOX regimens were investigated in the first-line treatment of gastric cancer, showing no significant differences in safety profile. Different doses of AK104 in combination with chemotherapy did not increase AK104-related TRAEs. The objective response rate of AK104 10 mg/kg Q2W combination therapy was significantly improved compared with AK104 4 mg/kg Q2W combination therapy (83.3% vs. 68.8%).

The comparison of simulated exposures showed that the steady-state exposures AUC_{ss} of 6 mg/kg Q2W and 10 mg/kg Q3W were consistent, with consistent concentration range of 5% to 95% quartiles. The C_{min} of 10 mg/kg Q3W was slightly lower than that of 6 mg/kg Q2W, which could maintain certain receptor occupancy. Therefore, AK104 10 mg/kg Q3W in combination with XELOX was selected as the study dose for the first-line treatment of advanced gastric cancer or adenocarcinoma of the gastroesophageal junction.

3.6.2 Rationale for Stratification by PD-L1 vCPS ≥ 5% or < 5%

It has been shown in studies that PD-L1 is an independent prognostic factor for gastric cancer, and the prognosis of patients with gastric cancer can be assessed by immunodetection of PD-L1[40].

The correlation between PD-L1 expression levels and efficacy was analyzed in an exploratory analysis of CheckMate-032 (GC/GEJC cohort)[41]. Different PD-L1 cut points (< 1, ≥ 1, ≥ 5, ≥ 10) were used to assess the positive rate of PD-L1 expression in gastric cancer/adenocarcinoma of the gastroesophageal junction by TPS and CPS, respectively. The results showed that at different cut points, the positive rate of PD-L1 expression was higher as assessed by CPS compared to TPS. When assessed by CPS, the ORR was better in patients with higher cut point; the correlation between CPS and OS was further assessed, and the results showed that the assessment of PD-L1 expression status using the CPS was strongly correlated with OS in patients both in both the full population and the nivolumab 1 mg/kg plus ipilimumab 3 mg/kg cohort, i.e., higher CPS values led to better OS.

Although there was a significant survival benefit for subjects regardless of the PD-L1 positive population or the full population in the results of Checkmate-649 study[32], the benefit was better in the PD-L1 positive population. The results of the KEYNOTE-061[42] study showed that the primary endpoint was not reached in the CPS ≥ 1 population, but a more significant OS benefit of pembrolizumab compared to paclitaxel was observed in the CPS ≥ 5 population.

In addition, studies have shown that the use of VENTANA PD-L1 (SP263) vCPS ≥ 5% (including PD-L1 expression in tumor cells and tumor-infiltrating immune cells) cut point and Dako 22C3 CPS ≥ 1 cut point can better identify the clinical benefit in populations with high and low PD-L1 expression in patients with gastric cancer or adenocarcinoma of the gastroesophageal junction [43]. Therefore, PD-L1 expression levels in gastric cancer and adenocarcinoma of the gastroesophageal junction were assessed using vCPS and stratified by vCPS ≥ 5% and vCPS < 5%.

3.6.3 Rationale for ECOG PS Score and Liver Metastasis as Stratification Factors

In a multivariate pooled analysis of prognostic factors, a total of 1080 patients with locally advanced and metastatic gastroesophageal junction cancer from 1992 to 2001 were enrolled in three randomized controlled trials, and the results showed that ECOG PS score ≥ 2 (P < 0.0001) and liver metastasis (P < 0.0001) were independent adverse prognostic factors for locally advanced and advanced gastric/gastroesophageal junction cancer. The subgroup analysis from study KEYNOTE-061[42] showed that patients with better performance status (ECOG PS 0) were more likely to benefit from treatment with pembrolizumab than patients with poor performance status (ECOG PS 1)[44]. Additionally, both the global data from study Checkmate-649[32] and the subgroup analysis of the Chinese population (PD-L1 CPS ≥ 5 population)[33] showed that patients with ECOG PS 0 had a longer mOS than patients with ECOG PS 1 (17.6 months vs. 12.6 months; 17.1 months vs. 14.3 months); patients without liver metastases also had a significantly better mOS than patients with liver metastases (15.5 months vs. 13.1 months; 15.6 months vs. 14.3 months); therefore, we used ECOG performance status (0, 1) and liver metastases (with, without) as stratification factors for this study.

3.6.4 Rationale for up to 2 Years of AK104 Administration

The optimal duration of immunotherapy is not yet fully defined, and the available evidence supports a more pronounced survival benefit with longer treatment duration.

The optimal duration of administration was explored in the CheckMate-153[45] study, which included a total of 1245 patients with advanced NSCLC, 220 of whom had been treated with nivolumab for 1 year, and then they were subsequently randomized to receive continued treatment with nivolumab or to discontinue nivolumab, followed by efficacy analysis of patients. After a median follow-up of 14.9 months, the analysis results showed that the median OS was not reached in patients who continued to receive nivolumab, with an OS rate of 88% at 1 year, and the median PFS was also not reached, with a PFS rate of 65% at 1 year. The efficacy was better compared to patients who discontinued treatment after 1 year of treatment (mOS of 23.2 months, 1-year OS rate of 81%; mPFS of 10.3 months, 1-year PFS rate of 40%).

In the KEYNOTE-010[46] study, a total of 1033 patients with advanced NSCLC were randomized to

receive either pembrolizumab or docetaxel. Of the 690 patients treated with pembrolizumab, 79 patients completed 35 cycles or 2 years of treatment with pembrolizumab. After a median follow-up of 43 months, among these patients, the 36-month OS rate was 99%, 75 (95%) patients had a best response of CR or PR, and 48% remained in durable remission. Of the patients who completed 35 cycles or 2 years of pembrolizumab, 91% remained alive. In addition, the safety profile was consistent with previously reported results. This indicated a better survival benefit for patients who completed 2 years of pembrolizumab.

Meanwhile, another study analyzed patient outcomes after interruption of PD-1/PD-L1 treatment, and patients who interrupted treatment within 12 months had an increased likelihood of relapse compared to patients who continued treatment for more than 12 months (P = 0.002). For these patients who restarted immunotherapy, some still benefited from retreatment. Thus, the survival benefit was greater for patients treated for more than 1 year[47]. In addition, in the pooled analysis of CheckMate-017 and CheckMate-057[48], the median duration of treatment for patients still alive at 5 years of follow-up was 37 months.

Based on these study results, this study stipulates that AK104 will be administered for up to 24 months in the absence of disease progression or intolerable toxicity.

3.6.5 Rationale for Selection of XELOX Regimen

Platinum compounds and fluorouracil are common first-line standard of care for metastatic gastric adenocarcinoma or adenocarcinoma of the gastroesophageal junction in different geographic regions[49]. In Western countries and Asia, the reference chemotherapy regimen for first-line treatment of metastatic gastric and adenocarcinoma of the gastroesophageal junction is fluorouracil (5-FU or capecitabine) in combination with a platinum agent (cisplatin or oxaliplatin) with or without a third cytotoxic drug (usually epirubicin or docetaxel). The NCCN guidelines indicate that two-drug chemotherapy is the preferred regimen for first-line therapy and that three-drug therapy is used only in patients with good performance status due to its toxicity profile. In addition, three-drug regimens are not commonly used in Asia due to their high toxicity. Oxaliplatin and capecitabine (XELOX) have shown non-inferiority to cisplatin and 5-FU and both have favorable safety profiles. The three-week regimen of XELOX is more convenient for clinical treatment, which is considered a reasonable comparator in many Phase III studies.

In a meta-analysis comparing oxaliplatin-containing regimens with cisplatin-containing regimens, oxaliplatin-containing regimens improved PFS (HR 0.88, 95% CI: 0.80-0.98) and OS (HR 0.88, 95% CI: 0.78-0.99); Compared to cisplatin-containing regimens, oxaliplatin-containing regimens had a lower incidence of neutropenia, anemia, alopecia, and thrombocytopenia and a higher incidence of neurotoxicity and diarrhea[50].

A meta-analysis combining the study data of REAL-2 and ML17032 showed that the regimen containing capecitabine significantly improved the OS (HR 0.87, 95% CI: 0.77-0.98, P = 0.02) compared to the 5-fluorouracil regimen. Moreover, capecitabine, as an oral fluorouracil drug, has good convenience in clinical use[51]. In another Phase II clinical study, a total of 54 patients with advanced gastric cancer were included and treated with capecitabine in combination with oxaliplatin (XELOX regimen) as the first-line treatment regimen, which also showed that XELOX regimen could provide a good survival benefit and was well tolerated in patients with gastric cancer[52].

In addition, the combination of immune checkpoint blockades and XELOX regimens in advanced gastric cancer has also shown an advantage in the study of immunotherapy in combination with chemotherapy. In the global data from Checkmate-649[32] and in the China subgroup[33] (PD-L1 CPS ≥ 5 population), 51% and 85% of patients received the three-week regimen of nivolumab in combination with XELOX, respectively. Compared with the two-week regimen of nivolumab in combination with FOLFOX, the efficacy was superior (15.0 months vs. 14.3 months; 15.6 months vs. 11.7 months) and the safety was good.

Therefore, we selected the three-week regimen of AK104 in combination with XELOX in this study and the XELOX three-week regimen as the treatment regimen for the control group.

3.6.6 Rationale for Treatment After Disease Progression

Studies and practices have shown[53] that, unlike conventional cytotoxic chemotherapy or radiotherapy, checkpoint inhibitor therapy may result in atypical responses in patients, which may be manifested as pseudoprogression, isolated reponses, etc. Pseudoprogression may be manifested as temporary increase in tumor volume, or the appearance of new lesions, or concurrent appearance of tumor volume increase and new lesions (as demonstrated by imaging), but without worsening of clinical symptoms, and the disease appears stable with objective response over time. This phenomenon may be due to the enrichment of inflammatory cells around the tumor or edema or necrosis produced during immunotherapy, resulting in a transient increase in tumor volume on imaging, which in fact is not a true disease progression. Isolated response is another atypical response to checkpoint inhibitor therapy, defined as concomitant regression of certain tumors and progression of other tumors (similar to a mixed response to chemotherapy and targeted therapy); if the patient is clinically stable, monitoring and management of isolated responses is similar to that of pseudoprogression.

Studies have shown that continued immunotherapy is still beneficial in patients with increased tumor size but clinically stable and tolerant to immunotherapy. A systematic review had analyzed 5588 patients (including 268 patients from the third-line ATTRACTION-2 study of gastric cancer)

from 35 immunotherapy-related clinical studies involving different tumor types who had first assessment of disease progression after immunotherapy but continued immunotherapy with stable clinical symptoms, of whom 254 (4.6%) patients had a response again after the first RECIST-defined disease progression[54]. In another study, 154 of 168 patients with advanced metastatic renal cancer were first assessed as disease progression per RECIST after treatment with nivolumab, of whom 36 patients continued treatment with nivolumab after the first assessment of disease progression and 92 patients discontinued nivolumab, the results also showed that 25 (69%) patients who continued treatment with nivolumab after the first assessment of disease progression had continued shrinkage of or stable tumor target lesions and good safety[55]. Another study in patients with advanced NSCLC also showed this effect. The study included a total of 125 patients with advanced NSCLC who progressed after immune checkpoint (PD-1/PD-L1) monotherapy or in combination with chemotherapy/anti-vascular therapy. Patients were divided into two groups according to their clinical baseline characteristics, one group continued immunotherapy for more than 6 weeks, and one group continued immunotherapy for less than 6 weeks or discontinued immunotherapy; both the median OS (mOS: 26.6 months vs. 9.5 months, HR 0.40; $P < 0.001$) and PFS (mPFS: 8.9 months vs. 4.1 months; HR 0.41; $P < 0.001$) for patients who continued immunotherapy for more than 6 weeks were longer compared to patients who discontinued immunotherapy or continued immunotherapy for less than 6 weeks[56].

The patient was prematurely discontinued from immunotherapy due to pseudoprogression, isolated responses and thus did not benefit from continued immunotherapy. Therefore, subjects will be allowed to continue AK104 monotherapy if they are assessed as clinically beneficial and tolerant to study drug after the Investigator has initially assessed disease progression as defined by RECIST v1.1. But subjects must discontinue AK104 treatment after evidence of further progression appears.

4. STUDY POPULATION

4.1 Subject Selection

4.1.1 Inclusion Criteria

Subjects must meet all of the following inclusion criteria to be enrolled in the study:

1. Ability to understand and voluntarily sign a written informed consent form (ICF), which must be signed before the specified study procedures required for the study are performed.
2. Males or females aged ≥ 18 to ≤ 75 years at the time of signing informed consent.
3. Histopathologically confirmed gastric adenocarcinoma or gastroesophageal junction (GEJ) adenocarcinoma.
4. Unresectable locally advanced or metastatic gastric adenocarcinoma or GEJ adenocarcinoma (a tumor centered within 5 cm proximal and distal to the anatomical location of the cardia as described in the Siewert Classification System).
5. Subjects have not received prior systemic therapy for locally advanced or metastatic gastric adenocarcinoma or adenocarcinoma of the gastroesophageal junction. For subjects who have received prior neoadjuvant/adjuvant chemotherapy or chemoradiotherapy for curative intent, the time between disease progression and last treatment should be at least 6 months.
6. Subjects have at least one measurable tumor lesion per RECIST v1.1; lesions that received radiotherapy are not selected as target lesions, unless the lesion is the only measurable lesion and has unequivocal progression as judged by imaging, it can be considered as a target lesion.
7. All subjects must provide archival or freshly obtained tumor tissue samples within 2 years prior to signing the ICF.
8. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.
9. Expected survival ≥ 3 months.
10. Patients who have adequate organ function:

Test items	Inclusion requirements (based on the normal range of test values in corresponding study site)
Hematology (subjects are not allowed to receive blood transfusion or growth factor support within 7 days prior to the first dose)	
Absolute neutrophil count	$\geq 1.5\times10^9$/L
Platelet count	$\geq 100\times10^9$/L
Hemoglobin	≥ 90 g/L or ≥ 5.6 mmol/L
Liver and kidney function	
Creatinine (Cr) or calculated serum creatinine clearance (CrCl)	Cr ≤ 1.5× ULN or CrCl ≥ 50 mL/min (Calculated according to Cockcroft-Gault formula, see Appendix 3)
Album in	≥ 28 g/L (albumin infusion is not allowed within 14 days prior to the first dose)
Total bilirubin (TBIL)	≤ 1.5× ULN; TBIL ≤ 3× ULN for subjects with documented/suspected Gilbert's syndrome

Test items	Inclusion requirements (based on the normal range of test values in corresponding study site)
Alanine transaminase (ALT) and aspartate aminotransferase (AST)	≤2.5×ULN; ALT and AST ≤ 5 × ULN for subjects with liver metastases
Coagulation function	
International normalized ratio (INR), Activated partial thromboplastin time (APTT), Prothrombin time (PT)	≤ 1.5×ULN;

11. Women of childbearing potential must have a negative serum pregnancy test within 7 days prior to the first dose and agree to take effective contraception measures during the study drug administration and within 120 days after the last dose. In this protocol, women of childbearing potential are defined as sexually mature women:

1) No hysterectomy or bilateral oophorectomy,

2) Spontaneous amenorrhea does not last for at least 24 consecutive months (amenorrhea caused by cancer treatment does not preclude fertility) (i.e., have menstruation at any time during the preceding 24 consecutive months).

Male patients with female partners of childbearing potential must agree to take effective contraception measures during the study drug administration and within 120 days after the last dose.

4.1.2 Exclusion Criteria

Subjects who meet any of the following criteria are not eligible to participate in this study:

1. Subjects with known HER2-positive gastric or GEJ adenocarcinoma.

2. Histopathology or cytology confirmed other pathological types, such as squamous cell carcinoma, sarcoma, or undifferentiated carcinoma.

3. Subjects who received palliative local therapy for non-target lesions within 2 weeks prior to the first dose; systemic nonspecific immunomodulatory therapy (e.g., interleukin, interferon, thymosin, etc.) within 2 weeks prior to the first dose; and Chinese herbal medicine or traditional Chinese medicinal products with anti-tumor indications within 2 weeks prior to the first dose.

4. Subjects who received any prior treatments targeting the mechanism of tumor immunity, such as immune checkpoint blockades (e.g., anti-PD-1 antibody, anti-PD-L1 antibody, anti-CTLA-4 antibody, etc.), immune checkpoint agonists (e.g., antibodies against ICOS, CD40, CD137, GITR, OX40 targets, etc.), immune cell therapy, etc.

5. History of gastrointestinal perforation or gastrointestinal fistula within 6 months prior to the first dose. If the perforation or fistula has been treated with resection or repair and the disease has recovered or resolved as judged by the Investigator, enrollment may be allowed.

6. Active or previously documented inflammatory bowel disease (e.g., Crohn's disease or ulcerative colitis). Subjects who can't swallow or have malabsorption syndrome, uncontrolled nausea, vomiting, diarrhea, or other gastrointestinal disorders that severely affect drug intake and absorption.

7. Active malignancies within the past 3 years, with the exception of tumors in this study and cured local tumors, such as basal cell carcinoma of skin, squamous cell carcinoma of skin, superficial bladder cancer, carcinoma in situ of cervix, carcinoma in situ of breast, localized prostate cancer, etc.

8. Known active or untreated brain metastases, meningeal metastases, spinal cord compression, or leptomeningeal disease. However, subjects who meet the following requirements and have measurable lesions outside the central nervous system are allowed to be enrolled: asymptomatic after treatment and radiographically stable for at least 4 weeks prior to the start of study treatment (e.g., no new or enlarged brain metastases), and the systemic glucocorticoids and anticonvulsant medications have been discontinued for at least 2 weeks.

9. Presence of clinically symptomatic pleural effusion, pericardial effusion, or ascites requiring frequent drainage (≥ 1/month).

10. Active autoimmune disease requiring systemic treatment within 2 years prior to the start of study treatment, or autoimmune diseases that may relapse or require scheduled treatment as judged by the Investigator. With the following exceptions:

a) Skin diseases not requiring systemic treatment (e.g., vitiligo, alopecia, psoriasis, or eczema);
b) Hypothyroidism due to autoimmune thyroiditis, which requires only stable doses of hormone replacement therapy;
c) Type I diabetes mellitus requiring only stable doses of insulin replacement therapy;
d) Asthma which has resolved in childhood and does not require any intervention after reaching adulthood;
e) The Investigator judges that the disease will not recur without external triggering factors.

11. Presence of any of the following cardiovascular and cerebrovascular diseases or cardiovascular and cerebrovascular risk factors:

a) Myocardial infarction, unstable angina, cerebrovascular accident, transient ischemic attack, acute or persistent myocardial ischemia, symptomatic heart failure (Grade 2 or higher according to New York Heart Association functional classification), symptomatic or poorly controlled arrhythmia, or any arterial thromboembolism within

6 months prior to the first dose.

b) History of deep vein thrombosis, pulmonary embolism, or other serious thromboembolism within 3 months prior to the first dose.

c) Presence of major vascular diseases such as aortic aneurysm, dissecting aortic aneurysm, internal carotid artery stenosis that may be life-threatening or require surgery within 6 months.

d) Previous history of myocarditis and cardiomyopathy.

e) Left ventricular ejection fraction (LVEF) < 50%.

12. Unresolved toxicities during prior anti-tumor therapy are defined as the toxicities that do not resolved to National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) (NCI CTCAE v5.0) Grade 0 or 1, or to the levels specified in the inclusion/exclusion criteria, with the exception of alopecia/pigmentation. Subjects with irreversible toxicity that are not expected to worsen after study drug administration (e.g., hearing loss) may be included in the study after consultation with the medical monitor. Subjects with long-term toxicity due to radiotherapy that cannot be recovered at the discretion of the Investigator may be included in the study after consultation with the medical monitor.

13. Presence of ≥ Grade 2 peripheral neuropathy as defined by NCI CTCAE v5.0.

14. Known presence of interstitial lung disease or noninfectious pneumonitis that is currently symptomatic or requires prior systemic glucocorticoid therapy that, in the judgment of the Investigator, may affect the assessment or management of toxicity related to study treatment.

15. Known active pulmonary tuberculosis. Subjects with suspected active pulmonary tuberculosis should be examined through tests of chest x-ray, sputum, clinical symptoms and signs to exclude this condition.

16. Systemic anti-infective therapy (excluding antiviral therapy for hepatitis B or C) within 2 weeks prior to the first dose.

17. Known medical history of allogeneic organ transplantation and allogeneic hematopoietic stem cell transplantation.

18. Presence of clinically active hemoptysis, active diverticulitis, abdominal abscess, and gastrointestinal obstruction.

19. History of clinically significant hemorrhage symptom or clear hemorrhagic diathesis within 1 month prior to the first dose, such as digestive tract hemorrhage, gastirc ulcer hemorrhage or vasculitis.

20. Endoscopic signs of active bleeding of lesions are known.

21. Major surgery other than the diagnosis of gastric cancer within 28 days prior to the first dose or

major surgery is expected during the study.

22. Subjects with active hepatitis B or active hepatitis C:

a) Subjects with hepatitis B surface antigen positive (HBsAg +) and/or hepatitis B core antibody positive (HBcAb +) requiring hepatitis B virus deoxyribonucleic acid (HBV DNA) testing;

b) Those who are stable with HBV DNA copy number < 1000 cps/mL (200 IU/ml) or below the lower limit of detectable value at the study site after antiviral therapy may be enrolled, and subjects are required to continue anti-hepatitis B virus therapy during the study treatment;

c) Patients who are hepatitis C antibody positive (HCV Ab+) require HCV RNA testing and are eligible to participate in the study only if they are HCV RNA negative (defined as below the lower limit of detectable value at the study site);

23. Known medical history of immunodeficiency or positive HIV test.

24. Known active syphilis infection.

25. Concurrent participation in another clinical study, unless it is an observational, non-interventional clinical study or the follow-up period of an interventional study.

26. Subjects who require systemic treatment with glucocorticoids (> 10 mg/day prednisone or equivalent) or other immunosuppressive drugs within 14 days prior to the first dose. With the following exceptions:

a) Inhaled, ophthalmic, or topical glucocorticoids or other glucocorticoids at doses ≤ 10 mg/day prednisone or equivalent are permitted if there is no active autoimmune disease.

b) Physiologic doses of systemic glucocorticoids ≤ 10 mg/day prednisone, or equivalent doses of other glucocorticoids.

c) Glucocorticoids are used as pretreatment for infusion-related reactions or allergic reactions (e.g., before CT examination, etc.).

27. Presence of severe neurological or psychiatric disorders, including dementia and epileptic seizures.

28. Live vaccines inoculated within 30 days prior to first dose or planned to be inoculated during the study.

29. Subjects with known medical history of severe hypersensitivity reactions to other monoclonal antibodies.

30. Subjects with known contraindications to oxaliplatin and capecitabine chemotherapy (see instructions for oxaliplatin and capecitabine).

31. Subjects with a known history of allergy or hypersensitivity to AK104, oxaliplatin, capecitabine,

or any of its ingredients.

32. Subjects with known dihydropyrimidine dehydrogenase (DPD) deficiency.

33. Pregnant or lactating women.

34. Any condition that, in the opinion of the Investigator, may result in a risk when receiving the study drug, or would interfere with the evaluation of the study drug or the safety of subjects, or the interpretation of the study results.

4.2 End of Study for Subjects

4.2.1 End of Study Treatment

All subjects may discontinue study treatment at any time, including early withdrawal from the study. If a subject experiences end of study treatment or end of study, the reason for any end must be recorded in the electronic case report form (eCRF). The Investigator will complete the end-of-study visit and report the results in as much detail as possible until the end of the study, including the date of the last treatment and the reason for the end of treatment.

Reasons for the end of study treatment mainly include, but are not limited to, the following:

- Disease progression: Radiographic disease progression and/or clinical disease progression.
- Adverse events (e.g., any adverse event that meets the protocol-recommended criteria for permanent discontinuation of study treatment, or contraindications that, in the opinion of the Investigator or Sponsor, preclude further dosing), laboratory abnormalities, or concomitant diseases, indicating that continuation of the study is not the greatest benefit for the subject as judged by the Investigator.
- Subject's completion of anti-tumor therapy (up to 24 months of treatment)
- Major protocol deviations (e.g., subject did not meet the inclusion criteria, pregnancy or planned pregnancy, concurrent initiation of other anti-tumor therapy, etc.).
- Withdrawal of informed consent by the subject (subjects who have ended study treatment will generally continue to participate in the study unless they expressly disagree with further participation in any assessment of the study).
- Subject's refusal (refused to continue study treatment, but agreed to continue with subsequent follow-up).
- Subjects are lost to follow-up.
- Subjects die.
- The Sponsor terminates the study.
- Others (e.g., at the discretion of the Investigator, etc.).

Subjects who end study treatment must continue to be followed up and the safety follow-up and/or

survival follow-up data must be collected as required until the subject dies or the end of the study. Except for subjects who discontinue treatment due to disease progression based on imaging assessments, imaging assessments will be continued until disease progression, withdrawal of consent, death, or end of the study, whichever occurs first.

4.2.2 Continued Treatment After Disease Progression

AK104 is a bispecific antibody that can target both PD-1 and CTLA-4, and studies have shown that pseudoprogression may occur when tumors are treated with checkpoint inhibitors, which is due to immune cell infiltration or other mechanisms that leads to a significant increase in existing tumor lesions or the appearance of new tumor lesions. Study evidence suggests that this subset of subjects may still have clinical benefit despite preliminary evidence of PD. If the Investigator suspects that the disease progression is pseudoprogression, the disease progression should be confirmed by re-imaging at least 4 weeks later or at the next scheduled imaging assessment (it should be noted that the time point of re-imaging should not be later than 9 weeks after initial confirmation of disease progression). Between initial documentation of PD and tumor reassessment, the following criteria must be met if the subject is considered to continue to receive study drug AK104/placebo:

a) Clinical benefit as assessed by the Investigator, such as absence of clinical symptoms and signs of significant disease progression (including worsening of laboratory results);
b) Tolerance to study drug;
c) Stable Eastern Cooperative Oncology Group (ECOG) performance status;
d) Absence of rapid disease progression or tumor progression at critical anatomical sites (e.g., spinal cord compression) that necessitates urgent alternative medical intervention.

In rare cases, if, in the opinion of the Investigator, treatment is beneficial to the subject, the treatment may be continued after the re-imaging assessment of disease progression of the tumor. It is the responsibility of the Investigator to determine whether a subject may benefit from continued treatment beyond progression. Decisions should be made with careful consideration to enable subjects who may benefit to continue treatment as far as possible, while avoiding prolonged use of ineffective therapies. Any decision to continue treatment beyond progression must be documented in the study file.

4.2.3 End of Study

Withdrawal from the study includes discontinuation of all follow-up procedures. If a subject decides to withdraw from the study, a complete final evaluation should be performed as far as possible and no further data will be collected after withdrawal from the study. The Investigator should make every effort to contact the subject or his/her guardian to determine the reason for withdrawal from the study as completely as possible and to document it in the original medical record and eCRF.

Reasons for withdrawal include:

- Death
- Lost to follow up
- Subject's withdrawal of consent
- Sponsor's termination of study

4.3 Subject Management

4.3.1 Subject Number

Each subject will be assigned a unique identification number. The SID number will be used to identify subjects throughout the study and will be entered in all study documents. The SID will consist of 8 digits, with the first 4 digits being the study site number and the last 4 digits being the sequential number assigned to the study site at screening. If a subject discontinues from the study at any time, their study number cannot be reassigned to another subject.

4.3.2 Subject Replacement

Subjects randomized into the study will not be replaced.

4.3.3 Rescreening of Subjects

Re-screening (only once) will be allowed for subjects who failed to meet eligibility criteria at initial screening (basic information and reasons for screening failure should also be recorded in the eCRF). At the time of re-screening, subjects must re-sign the informed consent form and obtain a new subject identification number (SID).

4.3.4 Procedures for Handling Subjects Who Do Not Meet Study Criteria

Subjects who do not meet the study population criteria should not be eligible for study drug administration under any circumstances. If a subject does not meet all study population criteria but unexpectedly starts study treatment, the Investigator should inform the Sponsor immediately upon discovery and discuss with the Sponsor whether to continue or discontinue the subject's study treatment. The Sponsor must ensure that all decisions are accurately documented.

5. STUDY TREATMENT AND MANAGEMENT

5.1 Study Drug

Akeso will provide sufficient quantities of study drugs directly or through designated distribution party to the Investigator: AK104 injection, oxaliplatin injection, and capecitabine tablets.

AK104 is of solvent type with a strength of 125 mg/10 mL/vial. AK104 does not contain any antimicrobial preservatives. Each vial is designed for single use only and should not to be used to administer the study drug to more than one subject. AK104 should be stored at 2-8℃ and protected from light. Freezing and vigorous shaking are prohibited.

For chemotherapeutic drugs, marketed products will be purchased uniformly by the Sponsor, and their management (e.g., preparation, administration, storage, and disposal) will follow the drug labeling provided by the manufacturer and site requirements.

Medications used to treat infusion-related reactions (e.g., diphenhydramine, cimetidine, glucocorticoids) are not investigational medicinal products and are not provided by the Sponsor. These drugs are marketed products that will be purchased by the study site.

All drugs involved in the study should be used before the expiry date indicated on the package. All instructions for drug shipments must be followed to ensure that the drug is in good condition prior to use.

5.2 Study Drug Preparation

This study drug can only be provided by the Investigator to subjects participating in this study. The dose of investigational product for administration must be prepared by the Investigator or the sites' authorized personnel using aseptic techniques. The preparer should inspect each vial selected to contain the study drug. If there are any defects noted with the investigational product, the Investigator and site monitor should be notified immediately. Please refer to the Drug Complaints section for further instructions.

AK104 (or placebo): The dose of AK104 (or placebo) will be calculated based on the body weight within the time required by the protocol, and the corresponding volume of AK104 (or placebo) is withdrawn from the vial to obtain the actual administered dose (deviation within ± 5% due to non-calculation error is allowed). It is formulated in an IV bag containing normal saline (0.9% [w/v] sodium chloride for injection). The vial cannot be shaken during preparation, and the infusion bag should be gently inverted after preparation to thoroughly mix the solution. Do NOT shake the bag vigorously. Visually inspect the final solution. If the infusion is not clear or the contents appear to contain precipitate, the solution should be discarded and documented on the Drug Accountability Log. If the AK104 (or placebo) diluent cannot be used immediately and requires storage, the total

storage time from the prepared diluent to completion of infusion should not exceed 24 hours in the refrigerator (2℃ to 8℃), which includes a maximum of 6 hours at room temperature.

For the preparation of oxaliplatin, please refer to the product labeling of the drug.

Capecitabine is an oral formulation and does not require preparation.

5.3 Study Drug Treatment

5.3.1 Therapeutic Methods

Both AK104 (or placebo) and oxaliplatin are intravenous injections and must be administered in an outpatient clinic or ward of the study site, and the drug administration process must be performed by qualified or experienced investigators, and the drugs must not be taken out of the study site for use. Capecitabine is an oral formulation and subjects are required to record the dosage and time of capecitabine administered daily in writing.

The AK104 (or placebo) intravenous infusion should be kept on the same day as the oxaliplatin intravenous infusion. AK104 (or placebo) will be administered intravenously at least 30 minutes before oxaliplatin intravenous injection and capecitabine administration. The administration time window is ± 3 days (except for C1D1), but subjects must complete various examinations including vital signs, physical examination, laboratory examination, and physical status score to evaluate the safety and tolerability of continued treatment within 3 days before each administration.

The study personnel will record the time of dosing in the source documents. Treatment compliance will be monitored by drug dispensing records, subject's medical records. The subjects will take capecitabine orally twice daily, and be required to record the dosage and administration timing of capecitabine administered daily in writing and submit it to the study coordinator of the study site at the next visit, and return all used and unused drug containers at the end of treatment for compliance evaluation.

5.3.1.1 AK104 Injection (or Placebo)

AK104 (or placebo) will be administered at a dose of 10 mg/kg Q3W. The actual dose to be administered should be determined by the body weight at each dose administration. The dose to be administered will be calculated based on the body weight up to 7 days (the most recent one will prevail) prior to the first dose. For subsequent cycles, if the patient's weight change is ≤ ± 10%, the baseline weight can be used for dose calculation; otherwise, the actual body weight should be used.

AK104 (or placebo) will be administered via infusion over 60 minutes (± 10 minutes). Ongoing monitoring of potential infusion reactions will be performed and secondary prophylaxis of infusion-related reactions and/or adjustment of infusion rate will be allowed as per the protocol guidelines. For subjects who cannot tolerate a 60-minute infusion or may have infusion-related risks,

the infusion time may be extended to a maximum of 120 minutes (±15 minutes). Dose modification is not allowed for AK104 (or placebo), the administration time window is ± 3 days, dose delay is allowed for a maximum of 12 weeks, calculated from the last administration time, and otherwise treatment should be discontinued. AK104 may be held for more than 12 weeks from the last dose until the steroid reduction is completed or the dose of prednisone is reduced to ≤ 10 mg/day (or equivalent) if the patient needs to be tapered gradually after receiving steroids due to an AE. The acceptable duration of dose interruption must be agreed upon by the Investigator and the medical monitor. If a major public health emergency or an AE unrelated to AK104/placebo results in the interruption of AK104/placebo for more than 12 weeks after the last dose, the subject may continue on treatment after agreement with the medical monitor provided that the subject has not experienced disease progression, no other intolerable AE, and no other anti-tumor therapy is initiated, and the subject can still benefit from treatment with AK104/placebo in the judgment of the investigator.

Following dilution in the IV bag, the entire contents of the IV bag should be administered as an IV infusion, using an in-line filter of 0.2 μm or 0.22 μm. After completion of the infusion, the IV line should be flushed with a volume of normal saline equal to (approximately 10 mL or the routine volume of the study site) the priming volume of the infusion set to ensure that the contents of the IV bag are fully administered to the subject. Since the compatibility of AK104 with other IV medications and solutions other than normal saline (0.9% [w/v] sodium chloride for injection) is not known, the AK104 (placebo) solution should not be infused through an IV line in which other solutions or medications are being administered. The date, start time, interruption time, and completion time of AK104 administration must be recorded in the source documents.

5.3.1.2 XELOX Regiments

Oxaliplatin: 130 mg/m^2 intravenous infusion for 2-6 hours on Day 1 in every 3-week cycle. Oxaliplatin is dissolved in 250 mL to 500 mL of 5% dextrose solution, intravenously infusion for 2-6 hours on Day 1 in a 3-week cycle. Deviation within ± 5% due to non-calculation error is allowed. Avoid exposure to cold air and cold water during oxaliplatin treatment (3-5 days after infusion). Please refer to the product labeling of the drug for detailed usage and precautions.

Capecitabine: 1000 mg/m^2, p.o., twice daily (for each course of treatment, the administration time is from the morning of Day 1 to the evening of Day 14 or from the evening of Day 1 to the morning of Day 15), swallowed with water within 30 minutes after a meal; the doses will be administered for 14 days every 3 weeks. If a subject experiences vomiting within 30 minutes after taking capecitabine, he/she will be considered to take no drugs, which will be recorded in the subject diary card and the eCRF. Subjects who experience vomiting are not re-administered. However, capecitabine may be re-administered if vomiting occurs within a short period of time after

administration and the capecitabine tablet taken is clearly visible. Under no circumstances should capecitabine be administered more than twice daily. The dose of capecitabine will be adjusted according to the strength of the drug, and the dose should be as close as possible to a complete tablet. Please refer to the product labeling of the drug for detailed usage and precautions.

Treatment with capecitabine can be initiated on the same day after oxaliplatin administration.

Treatment of the toxicities associated with the listed drugs, including dose delay, dose reductions, and dose modifications, should be based on the instructions in oxaliplatin and capecitabine's prescription information and standard clinical practice.

5.3.2 Monitoring during Treatment

The Investigator will monitor subjects during the treatment period and assess vital signs at the following times: 20 minutes (± 10 minutes) before the start of AK104 (or placebo) infusion and 20 minutes (± 10 minutes) after the end of infusion. Additional monitoring and vital sign assessments will be performed during the infusion at the discretion of the Investigator based on the subject's clinical indications and standard clinical practice. In Cycles 1 and 2, additional vital sign measurements will be performed 3 hours (± 30 minutes) after the end of all study drug infusions. For subsequent treatment cycle, a 3-hour observation period is not required unless the subject experiences an infusion-related reaction.

As with any antibody, allergic reactions to dose administration are possible. Therefore, appropriate drugs and medical equipment to treat acute anaphylactic reactions must be immediately available, and study personnel must be trained to recognize and treat anaphylaxis. Subjects will be monitored for signs and symptoms of infusion-related reactions and serious systemic allergic reactions.

5.3.3 Management of AK104-Related Toxicities

5.3.3.1 Infusion-Related Reactions

The study personnel will monitor subjects for infusion-related reactions, including but not limited to, fever, chills, stiffness, rash, pruritus, flushing, headache, dizziness, syncope, sweating, shortness of breath, cough, wheezing, angioedema, hypertension, hypotension, hypoxemia, arthralgia, myalgia and other local or systemic symptoms or signs.

In this study, to avoid confusion of potential safety signals regarding the assessment of AK104 infusion-related reactions:

1. Primary prophylaxis for infusion-related reactions (prophylaxis in subjects without infusion-related reactions) is not permitted in this study.
2. Appropriate secondary prophylaxis (i.e., prevention of infusion-related reactions after the initial episode) should be performed at the discretion of the Investigator:

- Acetaminophen and/or antihistamines (e.g. diphenhydramine) may be administered within approximately 30 min prior to the start of subsequent infusions, and/or glucocorticoids or equivalents may be administered according to the routine diagnosis and treatment of each site.
- Non-sedating antihistamines (e.g., cetirizine) may be considered for subjects with recurrent infusion-related reactions.
- After these treatments above, glucocorticoid therapy should be considered if symptoms persist.
- In accordance with the routine of diagnosis and treatment of each site, pethidine/morphine sulfate and promethazine or their equivalents may be administered intravenously at the discretion of the Investigator prior to the start of subsequent infusions.

In the event of ≤ Grade 2 infusion-related reactions, the infusion rate of AK104 may be reduced by 50%, or the infusion may be interrupted until resolution of the event, and the infusion can be resumed at 50% of the initial infusion rate until the infusion is completed. Once the infusion rate of AK104 is reduced by 50% or the infusion is interrupted due to an infusion-related reaction, it is recommended that all subsequent infusions must be continued at a reduced infusion rate. If a subject experiences a second Grade 2 infusion-related reaction under proper preventive medication and/or at a slower infusion rate, the Investigator is advised to consider whether to discontinue AK104 based on the subject's performance of infusion-related reactions and disease status.

If a subject experiences a Grade 3 or 4 infusion-related reaction at any time, treatment with AK104 must be discontinued and no further AK104 treatment will be administered to the subject. Supportive care should be provided according to standard medical practice. If a subject experiences symptoms or signs of systemic allergic reactions or severe type I hypersensitivity reactions during administration of AK104, the subject will be treated with appropriate medications and medical equipment, which should be readily available at all study sites. Glucocorticoids, epinephrine, anti-allergic agents, antihistamines or equivalents should be readily available. Subjects should be instructed to report any delayed reactions immediately to the Investigator.

Treatment modifications and recommended toxicity management guidelines for AK104 infusion-related reactions are shown in Table 1. At the discretion of Investigators, treatment may be discontinued in the presence of a combination of lower-grade AEs that would not mandate discontinuation individually.

Table 1 Recommendations on treatment modifications for infusion-related reactions caused by AK104

NCI-CTC AE Grading	Recommendations on AK104 Treatment Modifications
Grade 1-Mild Mild transient reaction; infusion interruption not indicated; intervention not indicated.	Reduce the infusion rate by 50% and closely monitor for any worsening.

NCI-CTC AE Grading	Recommendations on AK104 Treatment Modifications
Grade 2-Moderate Therapy or infusion interruption indicated but responds promptly to symptomatic treatment (e.g., antihistamines, NSAIDs, narcotics); prophylactic medications indicated for < 24 hours.	Interrupt the infusion. Resume infusion at 50% of previous rate once IRR has resolved or decreased to Grade ≤ 1 in severity, and monitor closely for any worsening. If a subject experiences a second Grade 2 infusion-related reaction under proper preventive medication and/or at a slower infusion rate, the Investigator is advised to consider whether to discontinue the treatment based on the subject's performance of infusion-related reactions and disease status.
Grade 3 or Grade 4 – Severe or life-threatening Grade 3: Prolonged (e.g., not rapidly responsive to symptomatic medication and/or infusion interruption); recurrence of symptoms following improvement; hospitalization indicated for clinical sequelae. Grade 4: Life-threatening consequences; urgent intervention indicated.	Stop the infusion immediately and disconnect the infusion tube from the subject. Subjects must be withdrawn from the treatment immediately and must not receive any further treatment.

If hypersensitivity reactions occur, patients must be treated according to the routine of diagnosis and treatment, and the guidelines for the emergency treatment of anaphylactic reactions developed by the Working Group of the Resuscitation Council (UK) are recommended [57]. Subjects should be instructed to report any delayed reactions to the Investigator immediately.

If systemic anaphylactic/anaphylactoid reactions occur (usually within minutes following administration of the drug/antigen and characterized by: respiratory distress, laryngeal edema, and/or severe bronchospasm, followed by circulatory failure or shock in the absence of dyspnea; cutaneous manifestations such as pruritus and urticaria with/without edema; gastrointestinal manifestations such as nausea, vomiting, crampy abdominal pain, and diarrhea), the infusion must be immediately discontinued and the subject discontinued the study drug. If hypersensitivity reactions are observed, the subjects will be infused with epinephrine and dexamethasone, followed by immediate monitoring, and the ICU (intensive care unit) will be notified of the possible transfer if needed.

For prophylaxis of flu-like symptoms, a dose of 25 mg indomethacin or an equivalent dose of nonsteroidal anti-inflammatory drugs (NSAIDs; e.g., 600 mg ibuprofen, 500 mg naproxen sodium) may be administered 2 hours before and 8 hours after the start of each dose of study drug infusion. Alternative antipyretic therapy (e.g., acetaminophen) may be given to subjects at the discretion of the Investigator.

5.3.3.2 Immune-Related Adverse Events

An immune-related adverse event (irAE) is defined as an ADR of all grades judged to be causally related to the immune mechanism in clinical trials of anti-tumor drugs/therapies. Based on the mechanism of action of AK104 targeting PD-1 and CTLA-4 leading to T cell activation and

proliferation, there is the possibility of observing irAEs during the conduct of this study. Potential irAEs may be similar to those seen with anti-CTLA-4 and anti-PD-1/L1 drugs, which may include but are not limited to the following types of toxicities: skin toxicity (rash, pruritus and vitiligo), endocrine toxicity (hyperglycemia, hypothyroidism, hyperthyroidism, primary hypoadrenalism and hypophysitis), hepatotoxicity (hepatitis, increased AST, increased ALT, increased bilirubin), gastrointestinal toxicity (diarrhea and colitis), pulmonary toxicity (pneumonia and pulmonary sarcoidosis), rheumatoid/boneiliac toxicity (rheumatoid arthritis, myositis and myalgia), neurotoxicity (myasthenia gravis, aseptic meningitis, encephalitis, transverse myelitis and Guillain-Barre syndrome), hematotoxicity (autoimmune hemolytic anemia, aplastic anemia, immune thrombocytopenia and acquired hemophilia), nephrotoxicity (nephritis and renal insufficiency), cardiotoxicity (myocarditis, pericarditis and arrhythmia), and ocular toxicity (uveitis and scleritis). Subjects should be monitored by the Investigator for signs and symptoms of irAEs. In the absence of an alternate etiology (e.g., infection or PD), an immune-related etiology should be considered for signs or symptoms.

To determine whether it is an irAE, the most important thing is to identify the cause and characteristics of the event. The Investigator should thoroughly evaluate the subject (e.g., symptoms, signs, previous diseases, concomitant medications, and imaging characteristics) and support the diagnosis of irAEs using serological, immunological and histological examinations, as appropriate, to exclude any alternative etiology (e.g., PD, concomitant medications, and infectious disease). In general, the recommended management and handling of irAEs are as follows:

- In the absence of a clear alternative etiology, all events may be considered inconclusive or possibly immune-related. With the occurrence, development and management of the event, whether there is an alternative etiology is determined as clearly as possible.
- Symptomatic and topical therapy should be considered for low-grade (Grade 1 or 2, unless otherwise specified) events.
- Systemic corticosteroids (e.g., prednisone or IV equivalent) should be considered for persistent low-grade or severe (Grade ≥ 3) events.
- If symptoms recur or worsen during the gradual reduction of corticosteroids, increase the dose of corticosteroids until symptoms stabilize or improve and then the gradual reduction of corticosteroids will be given again at a slower rate.
- For events that do not respond to systemic steroids, after discussion with the Medical Monitor, consideration should be given to the use of more potent immunosuppressants, i.e. TNF antagonists (e.g., infliximab, and mycophenolate mofetil).

- Discontinuation of investigational product is not mandated for Grade 3 or Grade 4 inflammatory reactions attributed to local tumor response (e.g., inflammatory reaction at sites of metastatic disease, lymph nodes).
- If additional specialist consults are needed, e.g., immunologist, these should be obtained in time to make appropriate treatment decisions.
- At the discretion of the Investigator, AK104 may be discontinued in the presence of a combination of lower-grade AEs that would not mandate discontinuation individually.

The Investigator will assess the severity of immunotoxicity that occurs during treatment according to NCI CTCAE v5.0, and conduct treatment modifications for AK104 to manage irAEs according to the subject's medical history, symptoms or signs. The general recommendations are as follows:

- For Grade 1 irAE, no treatment modification is needed.
- For Grade 2 irAE, the treatment of AK104 will be interrupted until Grade 2 resolves to Grade ≤ 1 or baseline:
  a) If toxicity worsens or suggests worsening, it is recommended to treat it as Grade 3 or 4.
  b) If toxicity improves, resumption of AK104 treatment at an appropriate scheduled treatment visit may be considered.
  c) In the event of persistent Grade 2 irAEs that does not resolve to Grade ≤1 or baseline within 12 weeks after symptomatic treatment, the Investigator will consider whether to permanently discontinue AK104 based on the subject's clinical conditions (the communication with the Sponsor's Medical Monitor, if necessary).
- For Grade 3 irAEs, the AK104 treatment will be interrupted until Grade 3 resolves to ≤ Grade 1 or baseline:
  a) If toxicity improves, the Investigator may consider whether to permanently discontinue AK104 treatment based on the type of individual toxicity, occurrence, and outcome process of the toxicity.
  b) If a Grade 3 irAE does not resolve to ≤ Grade 1 or baseline within 12 weeks after symptomatic treatment, permanent discontinuation of AK104 treatment is recommended.
- For Grade 4 irAEs, treatment with AK104 must be discontinued permanently.

AK104 treatment modification recommendations and toxicity management guidelines for irAEs are provided in Table 2. For irAEs not covered in Table 2 or under special circumstances for certain specific toxicities, refer to the toxicity management guidelines for immunotherapy at home and abroad. For sites with extensive experience in immune research or treatment management, where there are discrepancies in the specific procedures of toxicity management guidelines from immune

checkpoint blockades, they may be performed in accordance with site standard clinical practice, provided that the Investigator monitors and processes as promptly as possible to maximize the safety of subjects; if necessary, the Investigator may discuss with the Sponsor and manage the toxicity accordingly.

Table 2 AK104 treatment modification recommendations and recommended toxicity management guidelines for irAEs

irAEs	Toxicity Grade	Actions Taken with AK104	Management of irAEs (corticosteroids and other treatments)	Monitoring and Follow-up
Pneumonitis[1]	Grade 2	Interruption	Corticosteroids (initial dose of 1-2 mg/kg prednisone or equivalent), followed by tapering	Monitor subjects for pneumonitis-related symptoms and signs Assess and confirm by imaging follow-up, pulmonary function test, arterial oxygen saturation test, etc., and exclude infection (e.g., sputum/blood culture and drug sensitivity test) and underlying disease-related etiology If infection has not been completely ruled out, empiric anti-infective therapy is indicated
	Grade 3-4 or recurrent Grade 2	Permanent discontinuation		
Diarrhea/Enteritis	Grade 2-3	Interruption	Corticosteroids (initial dose of 1-2 mg/kg prednisone or equivalent), followed by tapering	Monitor subjects for gastroenteritis-related symptoms and signs (e.g., diarrhea, abdominal pain, hematology, feces with or without fever) or intestinal perforation-related symptoms and signs (signs of peritoneal irritation or intestinal obstruction) Consult with a gastroenterologist and undergo endoscopy to rule out enteritis for subjects with suspected enteritis with ≥ Grade 2 diarrhea Subjects with diarrhea/enteritis should receive fluid replacement (if not orally, IV fluid replacement and electrolytes)
	Grade 4 or recurrent Grade 3	Permanent discontinuation		
AST/ALT Increased or bilirubin increased	Grade 2	Interruption	Corticosteroids (initial dose of 0.5-1 mg/kg prednisone or equivalent), followed by tapering	Monitor liver functions (weekly or more frequently until liver enzyme level return to baseline or remain stable)
	Grade 3	Interrupt, or	Corticosteroids (initial	

irAEs	Toxicity Grade	Actions Taken with AK104	Management of irAEs (corticosteroids and other treatments)	Monitoring and Follow-up
		permanently discontinue drug	dose of 1-2 mg/kg prednisone or equivalent), followed by tapering	
	Grade 4	Permanent discontinuation	Corticosteroids (initial dose of 1-2 mg/kg prednisone or equivalent), followed by tapering	
Type 1 diabetes mellitus (T1DM) or hyperglycemia	New-onset T1DM or hyperglycemia (associated with cell failure)	Interruption	Initiate insulin replacement therapy for subjects with T1DM; initiate hypoglycemic agents for subjects with hyperglycemia	Monitor the subject's blood glucose level, or other symptoms and signs of diabetes
Hypophysitis	Grade 2	Interruption	Use corticosteroids and initiate hormone replacement therapy if clinically indicated	Monitor for symptoms and signs of hypophysitis (including hypopituitarism and adrenal insufficiency)
	Grade 3-4	Interrupt, or permanently discontinue drug		
Hyperthyroidism	Grade 2	Continuation	Use β-blockers (e.g., propranolol) or other antithyroid drugs (e.g., iodine, iodides, thioureas) as appropriate	Monitor symptoms and signs of thyroid diseases
	Grade 3-4	Interrupt, or permanently discontinue drug	Continue the treatment if clinical symptoms are improved to ≤ Grade 2 and can be controlled by the antithyroid drug therapy, continued treatment is recommended	
Hypothyroidism	Grade 2	Continuation	Thyroid hormone replacement therapy (e.g., levothyroxine or triiodothyronine) as appropriate	Monitor symptoms and signs of thyroid diseases
	Grade 3-4	Interrupt, or permanently discontinue drug	Continue the treatment if clinical symptoms are improved to ≤ Grade 2 and can be controlled by the hormone replacement therapy, continued treatment is recommended	
Nephrotoxicity[2]	Grade 2	Interruption	Corticosteroids (initial dose of 1-2 mg/kg prednisone or equivalent), followed by tapering	Discontinue nephrotoxicity-related drugs [proton pump inhibitors (PPIs) or nonsteroidal anti-inflammatory drugs (NSAIDS)] and monitor changes in renal function
	Grade 3-4	Permanent discontinuation		

irAEs	Toxicity Grade	Actions Taken with AK104	Management of irAEs (corticosteroids and other treatments)	Monitoring and Follow-up
Myocarditis	Grade 1	Interruption	Use corticosteroids based on the type and seriousness of AE	Ensure adequate assessment to confirm etiology and exclude other causes
	Grade 2	Interruption or permanent discontinuation		
	Grade 3-4	Permanent discontinuation		
All other irAEs	Intolerable or persistent Grade 2	Interruption	Use corticosteroids based on the type and seriousness of AE	Ensure adequate assessment to confirm etiology and exclude other causes
	Grade 3	Interrupt or permanently discontinue the treatment based on the type of AE (AEs requiring permanent discontinuation include, but are not limited to, Guillain-Barre syndrome and encephalitis)		
	Grade 4 or recurrent Grade 3	Permanent discontinuation		

Notes:
1: Dose interruption is recommended for Grade 1 pneumonia, and chest CT and pulmonary function test should be repeated 3 to 4 weeks later, and treatment may be continued if imaging is improved or unchanged.
2: refer to the 2021 CSCO guidelines for nephrotoxicity. Grade 1 nephrotoxicity is characterized by asymptomatic or mild symptoms with only clinical observations or diagnostic increases in creatinine level > 0.3 mg/dL and creatinine 1.5-2×ULN; it is recommended to consider drug interruption, find out possible causes, discontinue nephrotoxicity-related drugs (PPIs or NSAIDS), and repeat creatinine and urine protein every 3 to 7 days.

5.3.3.3 Non-immune-related Adverse Events

All non-irAEs will be graded according to NCI CTCAE version 5.0. For AK104-related Grade 3 or higher non-irAEs assessed by the Investigator, AK104 treatment may be interrupted and clinical management will be given. The Investigator may determine whether to continue the study drug until the AE resolves or improves. For laboratory abnormalities that are clearly not attributable to AK104 or are not deemed to be clinically significant will not require a dose omission. Dose reduction will not be allowed. Non-IRRs will be managed according to the clinical diagnosis and treatment practice at each site.

5.3.4 Management of Chemotherapy-Related Toxicity

The dose of chemotherapy will be adjusted according to the Investigator's clinical judgment, guided by local standards and labeling, and with the information in this section as a reference. The Investigator should monitor and observe as closely as possible to ensure the safety of subjects to the greatest extent possible.

5.3.4.1 General Precautions for Dose Modifications

- Baseline body weight will be used to calculate the required chemotherapy dose. Dose modifications are required if the subject's body weight changes by ± 10% from baseline. The dose of chemotherapy can not be changed by less than ± 10% body weight change;
- If a subject experiences concurrent toxicities with varying grades or severity, dose modification should be conducted based on the highest grade or severity of toxicities observed;
- Absolute neutrophil count should be ≥ 1.5×10^9/L and platelet count ≥ 75×10^9/L prior to the start of each treatment cycle, otherwise dosing should be delayed; based on the lowest platelet and neutrophil counts in the previous treatment cycle, dose modifications will be made at the start of the next treatment cycle.
- If, in the judgment of the Investigator, some toxic reactions are unlikely to develop into serious or life-threatening AEs, such as alopecia, taste alteration, or nail alteration, treatment may be continued at the original dose level without dose reduction or delay;
- If anemia (non-hemolytic anemia) is well controlled by transfusion, dose reduction or delay may be not performed;
- If any individual study drug must be delayed by 1 day or longer, all medications should be delayed by the same time. However, the time of tumor assessment will not be changed due to the delay of chemotherapy;
- If a subject requires an administration delay of XELOX for > 3 weeks (i.e. > 6 weeks after the start of the previous treatment cycle) due to toxicity in the study treatment, the XELOX chemotherapy part of the regimen will be permanently discontinued due to unacceptable toxicity;
- If a major public health emergency results in discontinuation of XELOX for more than 6 weeks after the last dose, the subject may continue on treatment after agreement with the medical monitor provided that the subject has not experienced disease progression, no other intolerable AEs, and no other anti-tumor therapy is initiated, and the subject can still benefit from the treatment with XELOX chemotherapy in the judgment of the investigator;
- If, in the opinion of the Investigator, toxicity is due to only one drug under study treatment, the administration of the drug should be delayed or dose adjusted; if the dose of chemotherapy is required to be reduced, the reduced dose will be continued in subsequent treatment cycles; if there is no contraindication, the administration of other chemotherapeutic agents should be continued;

- After completion or discontinuation of chemotherapy, subjects should continue receiving AK104/placebo until disease progression, intolerable toxicity, other anti-tumor therapy, loss to follow-up, death, or study termination, whichever occurs first;
- If continuation of treatment is not in the best interest of the subject, the Investigator may decide to discontinue any of the chemotherapeutic agents or all chemotherapeutic agents.
- Reasons for dose modifications or delays, supportive actions taken, and results will be documented in the patient's record and recorded on the eCRFs.

5.3.4.2 Dose Modifications for Hematotoxicity

Table 3 Therapeutic dose modifications for hematotoxicity

Febrile neutropenia	Grade 3 (ANC < 1.0 × 10^9/L and single temperature > 38.3°C or sustained temperature ≥ 38°C for more than 1 hour)	1st occurrence	Capecitabine 75% of original dose + oxaliplatin 85 mg/m^2
		2nd occurrence	Permanently discontinue[a]
	Grade 4 (ANC < 1.0 × 10^9/L, single temperature > 38.3°C or sustained ≥ 38 °C for more than 1 hour, and life-threatening consequences)	1st occurrence	Permanently discontinue[a]
		2nd occurrence	Permanent discontinuation
Neutropenia	Grade 2 (1.0 ≤ ANC < 1.5 × 10^9/L)	1st/2nd/3rd occurrence	No modification
	Grade 3 (0.5 ≤ ANC < 1.0×10^9/L)	1st occurrence	Capecitabine 75% of original dose + oxaliplatin 100 mg/m^2
		2nd occurrence	Capecitabine 75% of original dose + oxaliplatin 85 mg/m^2
		3rd occurrence	Permanently discontinue[b]
	Grade 4 (ANC ≤ 0.5×10^9/L)	1st occurrence	Capecitabine 50% of original dose + oxaliplatin 85 mg/m^2
		2nd occurrence	Permanent discontinuation
		3rd occurrence	Not applicable
Thrombocytopenia	Grade 2 (50 ≤ PLT < 75 × 10^9/L)	1st/2nd/3rd occurrence	No modification
	Grade 3 (25 ≤ PLT < 50 × 10^9/L)	1st occurrence	Capecitabine 75% of original dose + oxaliplatin 100 mg/m 2
		2nd occurrence	Capecitabine 75% of original dose + oxaliplatin 100 mg/m^2
		3rd occurrence	Capecitabine 50% of original dose + oxaliplatin 100 mg/m^2
	Grade 4 (PLT < 25 × 10^9/L)	1st occurrence	Capecitabine 50% of original dose + oxaliplatin 85 mg/m^2
		2nd occurrence	Permanently discontinue[c]
Non-hemolytic anemia	Hemoglobin (8.0 ≤ Hemoglobin < 10.0 g/dL; 6.5 ≤ Hemoglobin < 8.0 g/dL Hemoglobin < 6.5 g/dL)	Any occurrence	No modification (transfusion control)

Comment:

a: Capecitabine 50% of original dose + oxaliplatin 85 mg/m^2 may be administered unless, in the opinion of the Investigator, it is in the best interest of the subject to continue treatment after the hematotoxicity has recovered to Grade 0-1.
b: Capecitabine 75% of original dose alone may be administered unless, in the opinion of the Investigator, it is in the best interest of the subject to continue treatment after the hematotoxicity has recovered to Grade 0-1.
c: Capecitabine 50% of original dose alone may be administered unless, in the opinion of the Investigator, it is in the best interest of the subject to continue treatment after the hematotoxicity has recovered to Grade 0-1.

5.3.4.3 Dose Modifications for Non-hematologic Toxicities

Capecitabine

Recommendations for dose modification of capecitabine apply to toxicities related to capecitabine and not to toxicities associated with combination therapy.

Toxicities caused by administration of capecitabine may be managed with symptomatic treatment and/or treatment interruption or dose reduction. Once the dose has been reduced, it should not be increased later. Capecitabine treatment interrupted due to toxicity should be considered as missing of treatment time, and missing doses will not be compensated. The specific modification protocol of capecitabine is shown in Table 4:

Table 4 Dose modification recommendations for capecitabine

Toxicity*	**During Treatment**	**Dose Modification at Next Treatment (% of Initial Dose)**
Grade 1	Maintain dose level	Maintain dose level
Grade 2		
1st occurrence	Interrupt treatment until toxicity resolves to Grade 0-1	100% （1000 mg/m^2）
2nd occurrence		75%（750mg/m^2）
3rd occurrence		50%（500mg/m^2）
4th occurrence	Permanent discontinuation	-
Grade 3		
1st occurrence	Interrupt treatment until toxicity resolves to Grade 0-1	75%（750mg/m^2）
2nd occurrence		50%（500mg/m^2）
3rd occurrence	Permanent discontinuation	-
Grade 4		
1st occurrence	Permanently discontinue the treatment; or interrupt the treatment until the toxicity resolves to Grade 0-1, if, in the opinion of physician, it is in the best interest of the patient to continue treatment.	50%（500mg/m^2）

Oxaliplatin

Recommendations for dose modification of oxaliplatin apply to toxicities related to oxaliplatin and not to toxicities associated with combination therapy. Modification of oxaliplatin for non-hematologic toxicities are presented in Table 5 and for neurotoxicities in Table 6.

Table 5 Dose modifications of oxaliplatin for non-hematologic toxicities

Toxicity	Grade	Dose Modification
Allergic reaction*	Grade 3 or 4	Permanently discontinue the treatment
Pulmonary fibrosis indicated by respiratory symptoms*	Any	Interrupt the treatment and investigate the clear reason
New interstitial lung disease*	Any	Permanently discontinue the treatment
Nausea and vomiting after prophylactic use of antiemetics	Grade 3	100 mg/m^2
Nausea and/or vomiting	Grade 4	100 mg/m^2
Diarrhoea	Grade 3 or 4	100 mg/m^2
Stomatitis	Grade 3 or 4	100 mg/m^2
Skin toxicity	Grade 3 or 4	Prolong the rest of chemotherapy until recovery to Grade 0-1, without dose modification

*: Dose modification of capecitabine may not be performed if the patient's best benefit is taken into account

Table 6 Dose modifications of oxaliplatin for neurotoxicities

Toxicity	Grade	Duration		
		1-7 days	> 7 days	Ongoing between cycles
Paresthesia/Dysesthesia without affecting function	1	Dose unchanged	Dose unchanged	Dose unchanged
Paresthesia/Dysesthesia, affecting function but not affecting self-care activities of daily living	2	Dose unchanged	Dose unchanged	100 mg/m^2
Paresthesia/Dysesthesia with pain, affecting function and self-care activities of daily living	3	Dose unchanged	100 mg/m^2	Permanently discontinue the treatment
Paresthesia/Dysesthesia, disabling or life threatening	4	Permanently discontinue the treatment	Permanently discontinue the treatment	Permanently discontinue the treatment
Acute (during infusion or within 2 hours after the end of infusion) pharyngolaryngeal paresthesia		It should be distinguished from allergic reactions caused by oxaliplatin. Once it reoccurs, the infusion rate should be reduced to 1/3 of the original rate of oxaliplatin infusion	NA	NA

5.4 Study Drug Management

5.4.1 Labeling and Packaging

Akeso Pharmaceuticals, Ltd. should uniformly provide labeling and packaging for clinical study drugs for clinical trials.

The labeling of the study drug will be performed in accordance with the relevant regulatory

guidelines of Good Clinical Practice (GCP). It should include, but not be limited to clinical approval number, protocol number, drug name, drug number, packaging specification, manufacturing batch number, expiry date, dosage and administration, storage conditions, and Sponsor name, and should be marked with "only for clinical study use".

5.4.2 Storage and Records

The study drug should be stored in their original container as indicated on the label. If the storage conditions on the label and in other materials of the study drug (such as the Pharmacy Manual) are inconsistent, the storage conditions on the label should prevail.

AK104 must be stored at 2°C to 8°C and protected from light. Exposure to temperatures outside the recommended range and vigorous shaking should be avoided. Temperature exposure warmer than 8°C or colder than 2°C should be reported promptly to the Sponsor as a temperature excursion and the impact on the study drugs must be evaluated by the Sponsor.

Records of the actual storage conditions during the study (e.g., date and time, continuous temperature records of the freezer used to store study drugs, or maintenance records of instruments such as temperature alarm systems used in conjunction with temperature recorders) must be maintained.

5.4.3 Accountability and Destruction

The Clinical Research Associate (CRA) or study authorized personnel will be responsible for performing drug accountability. The study drug provided for this study should be administered according to the mode of administration and frequency specified in the protocol. It is the Investigator/site's responsibility to establish a system for handling study treatments, so as to ensure that:

- Deliveries of the investigational product are correctly received by a responsible person.
- Such deliveries are truthfully recorded by the authorized personnel or CRA.
- Study treatments are handled and stored safely and properly as stated on the label.
- Study treatments are only dispensed to study subjects in accordance with the protocol.
- Any unused study drug should be destroyed in liaison with the CRA. Unused drug supplies may be returned to the Sponsor for destruction, or may be destroyed at the site, in accordance with drug management manual.
- Used study drug and its packaging will be disposed of and destroyed per site requirements.
- At the end of the study, it must be possible to reconcile delivery records with records of usage and destroyed stock. Records of usage should include the SID of the subject to whom the study treatment will be dispensed and the quantity and date of dispensing. This record is in addition to any drug accountability information recorded in the eCRF. Any

discrepancies must be accounted for on the appropriate forms. All used study drugs should be destroyed when appropriate drug accountability procedures are performed by the CRA.

5.4.4 Drug Complaints

Any defects with the investigational product must be reported by the site immediately to the Sponsor with further notification to the site monitor. During the investigation of the product complaint, all investigational products must be stored at labeled conditions unless otherwise instructed.

Table 7 Contact information for study drug complaints

E-mail	productcomplaints@akesobio.com
Tel	+86-760-89873999
Fax	+86-760-89873900
Sponsor	Akeso Pharmaceuticals, Inc.
Department	Product Complaint Department
Address	Building A1, 6 Shennong Road, Torch Development Zone, Zhongshan, Guangdong, China

5.5 Concomitant Medications and Treatments

All concomitant medications taken from 30 days prior to the first dose of study drug to the treatment discontinuation visit or 30 days after the last dose, whichever occurs first, will be recorded in the eCRF. Investigators may prescribe concomitant medications or treatments deemed necessary to provide adequate supportive care.

After these periods, AEs-related (see Section 9.2.2 for AE collection timelines) concomitant therapies should also be recorded in the eCRF.

5.5.1 Permitted Concomitant Medications

1. Supportive or symptomatic treatment: antibiotics, analgesics, nutritional infusion, psychotherapy, any other supportive treatments for control of non-tumor-related essential symptoms, or any symptomatic treatment for AEs (e.g., granulocyte colony-stimulating factor [G-CSF] and other hematopoietic growth factors may be given for the treatment of acute toxic reactions such as neutropenic fever if clinically indicated or at the discretion of the Investigator).
2. Pretreatment medications for IRRs and hypersensitivity reactions during the study treatment.
3. Concurrent use of hormones for noncancer-related conditions (e.g., insulin for diabetes and HRT).
4. Palliative radiotherapy for known metastatic sites (e.g., radiotherapy for pain control of bone metastases) is permitted, provided that target lesions are not included in the radiation field, and the

palliative radiotherapy does not affect tumor response assessment or interrupt study treatment for more than 2 weeks.

5. Supportive cares that relieve tumor-related symptoms, such as bisphosphonates and RANK L inhibitors for bone metastases, are allowed.
6. Immunosuppressive medications for the management of investigational product-related AEs or in subjects with contrast allergies.
7. Inhaled, topical and intranasal corticosteroids are permitted.
8. Temporary courses of corticosteroids for treatment of underlying or concurrent illness or in the setting of palliative radiotherapy, upon discussion with the Medical Monitor.
9. Oral contraceptives.

Permitted Medications, Precautions, and Other Restrictions for XELOX Chemotherapy Regimen:

1. If nausea, vomiting, or diarrhea occurs, effective symptomatic treatment must be conducted.
2. Erythropoietin may be used in accordance with professional guidelines for chemotherapy-related anemia.
3. Subjects taking concomitant capecitabine and oral coumarin anticoagulants should undergo frequent coagulation test (INR/PT) for monitoring to modify the dose of anticoagulants. Alterations in coagulation test results or bleeding, including death, have been reported in subjects taking concomitant capecitabine and coumarin anticoagulants such as warfarin and phenprocoumarin; drug-drug interactions in pharmacokinetics have also been found. Low-molecular-weight heparin (LMWH) will be used instead of coumarin at the discretion of the Investigator.
4. Use of prophylactic medications such as magnesium/calcium infusion, or other agents to prevent oxaliplatin-induced neuropathy, as determined by the Investigator; however, these agents are not recommended in this protocol because their efficacy has not been clearly demonstrated.
5. The neurological safety of oxaliplatin should be closely monitored when coadministered with drugs with potential neurotoxicity.

5.5.2 Prohibited Concomitant Medications

Subjects must be instructed not to take any medications, including over-the-counter products, without first consulting with the Investigator.

The following medications are considered exclusionary during the study: if a subject receives any of the following medications during the study, the Sponsor must be notified.

1. Any other study drugs.
2. Any other anticancer therapy: including but not limited to surgery (for tumor removal), chemotherapy, antibody-based therapy, targeted therapy, endocrine therapy, cell therapy,

radiotherapy (except for palliative radiation, as defined in Section 5.5.1), and traditional Chinese medicinal products or Chinese herbal medicines with anti-tumor indications.

3. Immunosuppressive medications including, but not limited to systemic corticosteroids at >10 mg/day of prednisone or equivalent, methotrexate, azathioprine, and TNF-alpha blockers (except for immunosuppressive medications, defined in Section 5.5.1).
4. Live vaccines.
5. Concomitant use of dietary supplements, non-investigator-prescribed medications, and alternative/complementary therapies is discouraged but not prohibited, and should be discussed with the Medical Monitor.
6. Hepatotoxic drugs (i.e., drugs that are warned of hepatotoxicity in the product labeling) should be avoided during study treatment. Investigators are encouraged to review each potentially hepatotoxic drug by searching at www.livertox.nih.gov.
7. Prohibited treatments as specified in other inclusion and exclusion criteria, such as nonspecific immunomodulatory therapy, etc.

5.5.3 Concomitant Medications Used with Caution

Considering that AK104 is a therapeutic bispecific antibody, the potential for drug-drug interactions between the study drug AK104 and small molecule drugs is considered low. Since AK104 is expected to be degraded to amino acids and recycled to other proteins, it is unlikely to have an effect on drug metabolizing enzymes or transporters. For concomitant medications used with the chemotherapeutic agents oxaliplatin and capecitabine with caution, refer to their product labeling.

Coumarin Anticoagulants Used with Caution

Changes in coagulation parameters and/or bleeding have been reported in subjects treated with capecitabine with concomitant anticoagulants of coumarin derivatives such as warfarin and phenprocoumon. These conditions occur within days to months of capecitabine treatment, with some subjects occurring within 1 month of capecitabine discontinuation. Anticoagulation parameters (INR or PT) should be monitored routinely in subjects receiving capecitabine and oral coumarin derivative anticoagulants, and the dose of the anticoagulants should be adjusted accordingly.

Substrates of Cytochrome P-450 2C9

Substrates of cytochrome P-450 2C9 (diclofenac, phenytoin sodium, piroxicam, S-warfarin, and tolbutamide) should be used with caution during treatment.

Prohibited Medications for XELOX Chemotherapy Regimen:

1. Combinations of drugs with potential ototoxic or nephrotoxic effects (e.g., aminoglycosides, cephalosporins, furosemide, amphotericin B) should be avoided or adequately monitored.

2. Capecitabine should not be administered concomitantly with the antiviral drug sorivudine or its chemically related analogues such as brivudine. Sorivudine could inhibit dihydropyrimidine dehydrogenase (DPD) and has a significant clinical interaction with 5-fluorouracil drugs. This interaction leads to increased toxicity of fluorouracil with the potential for lethality. There must be a waiting period of at least 4 weeks between the end of treatment with sorivudine and its analogues (e.g., bromovudine) and the start of treatment with capecitabine.
3. The use of oxaliplatin with basic drugs or media, chlorine compounds, basic preparations, etc., as well as the use of aluminum-containing intravenous appliances should not allowed.

6. STUDY PROCEDURES

6.1 Screening Period

The subject's consent and ICF must be obtained before any study-specified screening tests or assessments. Eligibility of subjects to participate in the study will be assessed according to the inclusion and exclusion criteria, and all eligibility screening assessments must be completed within 28 days prior to the start of the first study treatment. Table 8 shows all procedures performed during screening.

Table 8 Schedule of planned procedures for screening

Study Periods	**Screening Period**
Procedure	**D -28 —D-1**
Written informed consent/assignment of SID number[a]	X
Demographics[b]	X
Past medical history, drug allergy history, smoking history, alcohol consumption[c]	X
Prior tumor history and prior anti-tumor therapy[d]	X
Clinical Examinations and Assessments	
Height, weight[e]	X
Physical examination[f]	X
Vital signs[g]	X
ECOG performance status	X
12-lead ECG[h]	X
Echocardiogram[h]	X
Concomitant medications and treatments[i]	X
AE/SAE assessment[i]	X
Laboratory Tests/Assessments	
Hematology[j]	X
Blood chemistry[j]	X
Urinalysis[j]	X
Coagulation[j]	X
Myocardial markers[k]	X
Thyroid function test (TSH, free T3, free T4)[l]	X
HBV, HCV, HIV virus screening[m]	X
Pregnancy test[n]	X
Sampling for Biomarker Testing	
Archived/Fresh tumor tissue samples	X
Tumor imaging and assessment[o]	X
Enrollment/Exclusion criteria assessment	X

Comment:
In addition to the test items and time points listed in the table, the Investigator may add items that are necessary for unscheduled tests based on the subject's clinical conditions.

a. Written ICF must be obtained prior to any clinical study procedures, except for tumor imaging, tumor tissue biopsy, and serum virology available within the specified time frame prior to the first dose. Screening tests could not be performed several times for purpose of enrollment in this project; if necessary, screening assessments could be repeated during the screening period, and the investigator will assess the subject's eligibility based on the latest screening assessment results; Subjects who have previously failed screening are allowed to be re-screened (only once) in this study. At re-screening, the ICF must be re-signed and a new subject number must be reassigned.
b. Demographic data: including date of birth, sex and ethnicity.
c. Past medical history: All past medical history (except for this indication), including prior Helicobacter pylori (Hp) infection, that occur prior to signing of the ICF will be collected.
d. Past tumor history: including histological or cytological diagnosis of the tumor, pathological type, histological classification, HER2 expression, EBV status (EBER in situ hybridization), MSI/MMR status, clinical TNM stage, date of first diagnosis, date of metastasis/relapse, etc.; prior anti-tumor therapy history: including surgical history, chemotherapy history, radiotherapy history, targeted therapy history, etc. The specific regimen, start and end date, best response evaluation and reason for discontinuation of all treatments should be recorded as far as possible.
e. Height will be collected during the screening period; body weight will be collected within 7 days prior to the first dose; and the most recent body weight prior to the first dose will be used for the calculation of the first dose.
f. A complete physical examination, including general condition, skin, mucosa, superficial lymph nodes throughout the body, head and organs, neck, chest, abdomen, spine and extremities, nervous system, and others, will be performed at screening.
g. Vital signs: including sitting blood pressure (after at least 5 minutes rest), pulse, respiration, and body temperature.
h. ECG that has been performed within 7 days prior to the first dose does not need to be repeated. Echocardiography that has been performed within 28 days prior to the first dose does not need to be repeated.
i. Concomitant medications and therapies: Record medications and therapies for diseases other than those for the study indication within 30 days prior to the first dosing. AE/SAE assessment: AEs and SAEs will be recorded from the time of signing the ICF to the first dose.
j. Hematology, blood chemistry, urinalysis, and coagulation tests that have been performed within 7 days prior to the first dose does not need to be repeated.
k. Myocardial marker testing includes creatine phosphokinase and its isoenzymes and, if any abnormal findings with clinical significance, includes cardiac troponin I (cTnI), and/or cardiac troponin T (cTnT), brain natriuretic peptide (BNP), and/or N-terminal pro-brain natriuretic peptide (NT-proBNP). Myocardial marker testing that have been performed within 7 days prior to the first dose do not need to be repeated.
l. Thyroid function tests that have been performed within 28 days prior to the first dose do not need to be repeated.
m. Hepatitis B virus screening includes testing of hepatitis B surface antigen (HBsAg), hepatitis B surface antibody (HBsAb), and hepatitis B core antibody (HBcAb); if HBsAg or HBcAb is positive, HBV DNA testing is required. Hepatitis C virus screening includes HCVAb testing; if HCVAb testing is positive, HCV RNA testing is required. HIV virus screening includes HIVAb. All viral screening test results within 28 days prior to the first dose are acceptable.
n. For women of childbearing potential, a serum pregnancy test will be performed within 7 days prior to the first dose.
o. Baseline assessments at screening should be completed within 28 days prior to the first dose of study treatment and should include head, neck, chest, abdomen, pelvis, and bones. Ideally, it should be performed as close as possible to first dose. Tumor assessments (other than bone) performed prior to obtaining informed consent and within 28 days prior to the first dose may be used for screening assessments and do not need to be repeated; bone scans are not required to be repeated if bone scans have been performed within 42 days prior to dosing on Day 1;If PET-CT is available within 42 days prior to the first dose, no bone scan is required if PET-CT shows no bone metastases, and if PET-CT shows bone metastases, additional bone scan is required. For lesions detected at baseline, all subsequent tumor assessment visits should performed using the same imaging method and equipment, and examination modality. Contrast-enhanced CT or contrast-enhanced MRI (except for bone CT/MRI) is the preferred method for tumor assessment in this study.

6.2 Treatment Period

The study procedures during the treatment period is presented in Table 9. The treatment period

starts after the randomization using the IVRS/IWRS. Subjects will be randomized at a ratio of 1:1 to the AK104+chemotherapy (XELOX) group and the placebo+chemotherapy (XELOX) group. Study treatment will begin no more than 3 days after randomization.

The treatment period ends when the subject discontinues study treatment (e.g., PD, intolerable toxicity, withdrawal of consent or other reasons by the subject, treatment with AK104/placebo for up to 2 years). On the day of infusion, AK104 (or placebo) should be administered first.

AK104 (or placebo) + XELOX will be administered every 3-week cycle. For AK104 (or placebo) + XELOX, chemotherapy (XELOX) should be discontinued until PD occurs prior to the end of Cycle 6 (3-week cycle). After 6 cycles of XELOX treatment, subjects with stable disease or responders will continue to receive AK104 or placebo. The study drug AK104 (or placebo) will be administered once every 3 weeks until any of the following occurs: the subject is unable to continue to benefit as determined by the Investigator, intolerable toxicity, concomitant diseases that affect further treatment, Investigator decision, subject's withdrawal of consent, or other reasons as specified in the protocol (whichever occurs first). Treatment with AK104 (or placebo) can be given over a maximum of 24 months.

AK104 (or placebo) alone will be allowed if chemotherapy have to be discontinued due to toxicity. If discontinuation is mandatory due to intolerable toxicity of AK104 (or placebo), a single chemotherapeutic agent is allowed in doublet combination or monotherapy. Subjects will be allowed to continue treatment with AK104 (or placebo) if the Investigator suspects that the subject's initial PD is pseudoprogression and evaluates that the subject is clinically benefiting from PD and tolerates treatment. Subjects will undergo tumor assessments every 6 weeks (± 7 days) within 54 weeks after enrollment, and then every 9 weeks (± 7 days). Biomarker, pharmacokinetics and immunogenicity collection will be performed according to the schedule specified in the protocol. Report questionnaires for quality-of-life assessment such as EORTC QLQ-30 and EORTC QLQ-ST022 will be collected every 2 cycles.

At unscheduled visits, the Investigator should perform a clinical assessment based on the subject's symptoms. These assessments may include a symptom-specific physical examination, ECOG performance status score, clinical laboratory tests, 12-lead ECG, echocardiography, and imaging for tumor assessment. Abnormal value change from baseline should be recorded in the subject's medical history. New or worsening findings should be recorded as AEs on the adverse event page of the eCRF.

Table 9 Schedule of planned procedures for treatment period

Procedures/Study Days[a]	Treatment Period (a 21-day treatment cycle)							End of Treatment Visit[q] -
	C1	C2	C3	C4	C5	C6	C7 and beyond[c]	

	D1[b]	±3	±3	±3	±3	±3	±3	±7
Clinical Procedures and Assessments								
Randomization	X							
Height, weight[d]	X	X	X	X	X	X	X	X
Vital signs[e]	X	X	X	X	X	X	X	X
Physical examination[f]	X	X	X	X	X	X	X	X
ECOG Performance Status	X	X	X	X	X	X	X	X
12-lead ECG[g]	X		X		X		X	X
Echocardiogram	As clinically indicated or as judged by the Investigator							
Concomitant medications and treatments	X	X	X	X	X	X	X	X
AE/SAE assessment	X	X	X	X	X	X	X	X
Laboratory Tests and Assessments								
Hematology	X	X	X	X	X	X	X	X
Blood chemistry	X	X	X	X	X	X	X	X
Urinalysis[h]	X	X	X	X	X	X	X	X
Coagulation function	X		X		X		X (every 2 cycles)	X
Thyroid function	X		X		X		X (every 2 cycles)	X
Myocardial markers[i]	As clinically indicated or as judged by the Investigator							
HBV, HCV, HIV [j]	As clinically indicated or as judged by the Investigator							
Pregnancy test[k]	X	X	X	X	X	X	X	X
Pharmacokinetic and Immunogenicity Testing[l]								
PK sample collection	X		X		X		X	
Immunogenicity sample collection	X		X		X		X	
Tumor imaging and assessment[m]								
Tumour imaging	X							X
Tumor assessment	X							X
Administration of study drug[n]								
AK104 (or placebo)	X	X	X	X	X	X	X	
Oxaliplatin	X	X	X	X	X	X		
Capecitabine O	X	X	X	X	X	X		
Quality of Life Assessment [p]								
EORTC QLQ-30	X		X		X		X	X
QLQ-ST022	X		X		X		X	X

Comment:

In addition to the test items and time points listed in the table, the Investigator may add items that are necessary for unscheduled tests based on the subject's clinical conditions.

a. Examinations and evaluations at scheduled visits need to be completed prior to each dose unless otherwise noted. Dosing on the nearest feasible date is allowed if scheduled dosing is not possible due to procedural reasons such as holidays, weekends. Every cycle of treatment is 21 days long. Each treatment visit window is ± 3 days long; pre-dose examinations and quality of life assessments will be performed within 3 days prior to the actual dosing date.

b. Subjects received the first study treatment as soon as possible on the day of randomization (no later than Day 3 after randomization). If the following assessments have been completed within 7 days prior to the first study treatment, repeated assessments are not required on Day 1: hematology, blood chemistry, urinalysis, 12-lead ECG, coagulation, and pregnancy tests; if the following assessments have been completed within 4 week prior to the first study treatment, repeated assessments are not required on Day 1: thyroid function and echocardiography.

c. Subject's C7 visit and subsequent scheduled visits will be performed according to the requirements of previous treatment cycles, including examination content, frequency, dosing, etc.

d. Body weights must be measured within 3 days prior to each dose for all subjects, except for C1D1. Subjects' body surface area may be calculated using the height measured at baseline during the treatment period unless, in the opinion of the Investigator, there is a significant change in height. In that case, the height will be re-measured.

e. The Investigator will monitor subjects during the treatment period and assess vital signs at the following times: 20 minutes (± 10 minutes) before the start of AK104 (or placebo) infusion and 20 minutes (± 10 minutes) after the end of infusion. Additional monitoring and vital sign assessments will be performed during the infusion at the discretion of the Investigator based on the subject's clinical indications and standard clinical practice. In Cycles 1 and 2, additional vital sign measurements will be performed 3 hours (± 30 minutes) after the end of all study drug infusions. For subsequent treatment cycle, a 3-hour observation period is not required unless the subject experiences an infusion-related reaction.
f. A targeted physical examination should be performed based on symptoms and signs at screening and newly identified symptoms and signs within 3 days prior to treatment in each cycle and should be performed prior to dosing.
g. A 12-lead ECG will be performed within 3 days prior to every 2 treatment cycles. Subjects with a cardiovascular medical history, or at risk of cardiac function as assessed by the Investigator, may be tested repeatedly. The Investigator will confirm appropriate pre-dose results of ECGs. Additional ECG monitoring may also be performed if clinically needed.
h. Urinalysis (e.g., urine dipstick examination or microscopic examination of urine sediment) will be performed within 3 days prior to treatment in each cycle. If the subject's urinalysis results show proteinuria ≥ 2+, a 24-hour urine protein quantitative test should be performed prior to dosing. The study drug may be administered without waiting for the 24-hour urine protein test results during the study.
i. During the treatment, cardiac markers (including at least troponin T and/or troponin I) are recommended to be tested at the Investigator's discretion as clinically indicated (e.g., fatigue, chest tightness, chest pain, and/or ECG, echocardiographic abnormalities).
j. HBsAg-positive or HCVAb-positive subjects at enrollment will be monitored and assessed as clinically indicated or as judged by the Investigator.
k. It is only applicable for women of childbearing potential. Urine pregnancy tests will be performed at each cycle during treatment (samples must be obtained within 3 days prior to the infusion, except for C1D1). If the urine pregnancy test is positive, it must be confirmed by a serum pregnancy test.
l. PK and immunogenicity sample collection: PK blood samples will be collected within 30 minutes prior to AK104 (or placebo) infusion at Cycle 1, 3, 5, and 7, and every 6 cycles for the 10th dose and subsequent doses. Immunogenicity blood samples will be collected within 30 minutes prior to AK104 (or placebo) infusion at Cycle 1, 3, 5, and 7, every 6 cycles for the 10th and subsequent doses, with safety follow-up. Sampling time points and requirements are detailed in Section 8.1.
m. Tumor imaging examinations and assessments will be performed every 6 weeks (± 7 days) within 54 weeks after enrollment (relative to the date of the first dose of AK104/placebo), and every 9 weeks (± 7 days) after 54 weeks after enrollment until subsequent anti-tumor therapy, disease progression, withdrawal of consent, death, or study closure, whichever occurs first. Additional tests or unscheduled assessments may be performed by the Investigator if clinically indicated. Unscheduled radiographic tests may be performed when disease progression is suspected (e.g., worsening symptoms). If an unscheduled assessment is performed and the subject shows no disease progression, subsequent tumor assessments will need to follow the next scheduled examination; if the most recent scheduled date is < 4 weeks apart from the date of the unscheduled examination performed, the original scheduled assessment will not be performed. In the event of treatment delay, tumor assessments are performed as originally planned. The same radiographic procedures must be followed throughout the study for each subject. Assessments should be performed by the same assessor, if possible, to ensure internal consistency across visits. Bone scans should be repeated if bone metastases are present at screening but are not seen on CT or MRI scans, or if clinically indicated, when complete response is identified in target lesions or when progression of bone metastases is suspected. For a subject with brain metastases identified at baseline, the assessment should be repeated at each subsequent tumor assessment. If no brain metastases are found at baseline, CT or MRI of the head may be added if brain metastases are suspected during subsequent treatment periods.
n. All tests should be completed within 3 days prior to the actual dose date (except for C1D1), and the drug may not be continued after safety assessment by the Investigator. For the first dose of AK104 (or placebo), the dose should be calculated based on the weight within 7 days prior to dosing, and in subsequent cycles, the dose should be calculated based on the weight data obtained within the first three days before each dose. If the patient's weight change is ≤ ± 10%, the baseline weight can be used for dose calculation; otherwise, the actual body weight should be used. The intravenous infusion time of AK104 (or placebo) should be 60 minutes (± 15 minutes). For subjects who cannot tolerate a 60-minute infusion or may have infusion-related risks, the infusion time may be extended to a maximum of 120 minutes (± 15 minutes).
o. Capecitabine will be administered for up to 6 cycles. Refer to Sections 5.3.1.2 and 5.3.4 for specific medications and dose modifications.

p. Quality of life assessments will be performed within 3 days prior to the actual dosing date, and EORTC QLQ-30 and EORTC QLQ-ST022 quality of life assessment scales should be completed before any clinical activities are performed on the day of completion.
q. The End of Treatment (EOT) date is the day on which the Investigator determines that dosing will no longer be administered. If routine laboratory tests (hematology, blood chemistry, urinalysis, etc.) are performed within 7 days prior to the EOT visit date, they do not have to be repeated at the End of Treatment visit. Tumor imaging tests and assessments should be performed at the end of treatment. Tumor assessments need not be repeated at the EOT visit if the interval from the last examination is < 4 weeks. If the EOT visit is performed at least 30 days (± 7 days) after the last dose for any reason, the safety follow-up visit is not necessary and the EOT examinations may be considered as a part of the safety follow-up visit.

6.3 Follow-up Period

Table 10 shows the planned procedures and requirements of subjects during the follow-up period. Subjects will be followed until death, withdrawal of consent, loss to follow-up, or study closure, whichever occurs first.

Table 10 Schedule of planned procedures for follow-up period

Procedure	Follow-up Period		
	Safety Follow-up[a] (D30 ± 7 days after last treatment)	**PD Follow-Up Visit**	**Survival Follow-up[g] (Every month ± 7 days after the End of Treatment Visit)**
Clinical Procedures and Assessments			
Physical examinations[b]	X		
ECOG performance status	X		
Vital Signs	X		
12-Lead ECGs	X		
AE/SAE Assessment[c]	X	X	X
Concomitant medications and treatments[d]	X	X	
Laboratory Tests and Assessments			
Hematology	X		
Blood chemistry	X		
Coagulation function	X		
Urinalysis[e]	X		
Thyroid function	X		
Pregnancy test	X		
Tumor imaging and assessment f	X		
Pharmacokinetics and Immunogenicity			
PK sample collection			
Immunogenicity sample collection	X		
Subsequent Anti-tumor Therapy	X	X	X

Comment:
a. The safety follow-up will be conducted on Day 30 (± 7 days) after the last dose or before the initiation of new anti-tumor therapy, whichever occurs first. If the EOT visit is more than 30 days (± 7 days) from the last dose, the safety follow-up visit is not necessary. Tumor imaging assessment is required at the Safety Follow-up Visit if it is not performed at the EOT and the Safety Follow-up Visit is more than 4 weeks from the last imaging examination.
b. A targeted physical examination will be performed as clinically indicated or as judged by the Investigator.

c. After the subject signs the ICF, all AEs and SAEs will be reported (regardless of causality to the study drug) until 30 days (AE) and 90 days (SAE) after the last dose of study treatment or initiation of other anti-tumor therapy, whichever occurs first (whether observed by the Investigator or reported by the subject) in the original records and eCRFs. Immune-related AEs (irAEs) , regardless of serious or non-serious irAEs, will be reported until 90 days after the last dose of AK104/placebo (whether other anti-tumor therapy have been initiated). After this period, the Investigator should report any SAEs that are assessed as related to AK104/placebo.
d. All concomitant medications taken from 30 days prior to the first dose of study drug to the treatment discontinuation visit or 30 days after the last dose, whichever occurs first, will be recorded in the eCRF. After these periods, AEs-related (see Section 9.2.2 for AE collection timelines) concomitant medications and therapies should also be recorded in the eCRF.
e. If the subject's urinalysis results show proteinuria ≥ 2+, a 24-hour urine protein quantification test should be performed.
f. For subjects who discontinue study drug due to intolerable toxicity or for reasons other than confirmed disease progression, tumor assessments will continue to be performed as originally planned until the subject begins a new anti-tumor therapy, experiences disease progression, withdraws consent, is lost to follow-up, or dies, or until the study is completed or the study is terminated, whichever occurs first.
g. In the survival follow-up period, information on subject survival status and subsequent anti-tumor therapy will be collected via telephone calls or clinical visits approximately once monthly (± 7 days) after the End of Treatment Visit (after the last dose if no End of Treatment Visit is available) until death, loss to follow-up, withdrawal of informed consent.

7. EFFICACY ASSESSMENT

7.1 Tumor Assessment

The assessment of tumor response will be based on RECIST v1.1. The assessment schedule also applies to those subjects who continue to receive the study drug through progression. After disease progression is confirmed as per RECIST 1.1, if the investigator determines that the subject may still benefit from the continuous treatment, the iRECIST criteria are recommended for subsequent tumor assessments. The RECIST v1.1 and iRECIST are described in detail in Appendix 1 and 2.

Baseline assessments should be completed within 28 days prior to the first dose and ideally as close as possible to the time of first dose. Results of tumor assessments performed prior to the signing of the informed consent and within 28 days of the first administration of the drug might be used for screening assessments rather than repeating such assessments. For lesions detected at baseline, all subsequent tumor assessment visits should performed using the same imaging method and equipment, and examination modality.

Contrast-enhanced CT or contrast-enhanced MRI (except for bone CT/MRI) is the preferred method for tumor assessment in this study. Prior to any imaging, study sites are advised to assess subjects for potential contrast contraindications, such as potential allergic reactions to contrast media or risk of renal impairment. If the subject is allergic or contraindicated to contrast agents of both enhanced CT or MRI, CT or MRI is preferred based on the site of imaging examination. Subjects will undergo one tumor assessment every 6 weeks (± 7 days) within 54 weeks after the enrollment (relative to the date of first dose) and every 9 weeks (± 7 days) 54 weeks after the enrollment until disease progression, new anti-tumor therapy, withdrawal of consent, death, or study closure, whichever occurs first.

Tumor assessments must include neck, chest, abdomen, and pelvis throughout the study. Contrast-enhanced MRI or CT of the head is required at baseline assessment to determine brain metastases; re-examination of the head will be performed if abnormalities are suspected by the Investigator or clinically indicated during the course of the study. The presence of bone metastases is also required at baseline assessments. Skeletal scans (e.g., bone scan) are not required to be repeated if they have been performed within 42 days prior to the first dose. If PET-CT is available within 42 days prior to the first dose, no bone scan is required if PET-CT shows no bone metastases; otherwise, additional bone scan is required. If bone metastases are present at screening or if clinically indicated during the study treatment phase, skeletal imaging should be repeated at least when progression of bone metastases is suspected or when a complete response is identified in target lesion. If there is evidence of metastases to any other site at screening, imaging examinations of these sites should be performed based on the clinical symptoms and signs of the individual

subject and followed throughout the study. For subsequent tumor assessments, the same test method as used to assess disease sites at screening should be used throughout the study (using the same contrast regimen as possible for contrast-enhanced MRI or CT). All known lesion sites must be recorded at screening and reassessed at each subsequent tumor assessment.

Table 11 Tumor response assessment process

	Screening Period	Treatment Period	Continuation Phase after PD (if any)[a]	Follow-up Period
Head	X	Perform as indicated at baseline or as clinically indicated		
Neck	X	X	X	X
Chest	X	X	X	X
Abdominal	X	X	X	X
Pelvis	X	X	X	X
Bone[b]	X	Perform as judged by the Investigator or as clinically indicated		
Other sites	Perform as judged by the Investigator or as clinically indicated			

a. During the continuation phase after PD, tumor assessment time points and imaging requirements will follow the original plan, and recorded and evaluated accordingly.
b. Bone scans are not required to be repeated if bone scans have been performed within 42 days prior to dosing on Day 1;If PET-CT is available within 42 days prior to the first dose, no bone scan is required if PET-CT shows no bone metastases, and if PET-CT shows bone metastases, additional bone scan is required.

7.2 Central Review

Not applicable.

7.3 Quality of Life Assessment

Quality of life (QoL) will be assessed using two scales: EORTC QLQ-30 and EORTC QLQ-ST022. Quality of life assessments will be performed within 3 days prior to the actual dosing date, and EORTC QLQ-30 and EORTC QLQ-ST022 quality of life assessment scales should be completed before any clinical activities are performed on the day of completion. Refer to the Quality of Life Assessment Manual provided by the Sponsor for detailed assessment scales and requirements.

The EORTC QLQ-30 is a core scale for all cancer patients with a total of 30 items, which can be divided into 15 dimensions with 5 functional dimensions (physical, role, cognitive, emotional, and social functioning), 3 symptom dimensions (fatigue, pain, nausea and vomiting), 1 global health status/QoL dimension and 6 single items.

EORTC QLQ-ST022 is the primary tool for QoL assessment in patients with gastric cancer and is often used in conjunction with EORTC QLQ-30. It includes 5 symptom dimensions: dysphagia, pain, reflux, eating restriction, and anxiety; and 4 individual measures: dry mouth, taste, body image, and alopecia. Items are graded in four grades: none, a little, more, and many, and scored 1-4 points.

7.4 Survival Assessment

Subjects must have a survival assessment every 1 months (± 7 days) after the End of Treatment Visit (after the last dose if no End of Treatment Visit is available). To obtain survival information, the Investigator or study authorized personnel may contact the subject or the subject's family via telephone, by contacting the subject's current physician, or reviewing the subject's clinical records. Meanwhile, details of the first and subsequent anti-tumor treatment after discontinuation will also be collected. In addition, subjects may be contacted at the time of treatment or survival follow-up to obtain complete survival data after the data cutoff for the primary analysis and all subsequent survival analyses.

8. PHARMACOKINETIC, IMMUNOGENICITY AND BIOMARKER ASSESSMENTS

8.1 Pharmacokinetic Assessment

Sparse PK samples will be collected by the study personnel at the time points shown in Table 12. Sparse PK samples will be continuously assessed for PK parameters and their relevance to demographic variables, biomarkers, and clinical endpoints by a population pharmacokinetic method.

The actual PK collection day may be adjusted accordingly with the actual medication of the subject.

Table 12 PK sample collection schedule

	Scheduled Day	Time Relative to AK104 (or Placebo) Infusion
Dose 1 (Cycle 1)	Day 1	Pre-infusion (within 30 minutes prior to infusion)
Dose 3 (Cycle 3)	Day 43	Pre-infusion (within 30 minutes prior to infusion)
Dose 5 (Cycle 5)	Day 85	Pre-infusion (within 30 minutes prior to infusion)
Dose 7 (Cycle 7)	Day 127	Pre-infusion (within 30 minutes prior to infusion)
10th dose or subsequent doses (every 6 treatment cycles)	Day 190 and beyond	Pre-infusion (within 30 minutes prior to infusion)

Subject samples will be analyzed for AK104 concentrations at a central laboratory using a validated bioanalytical method. Specific instructions on serum PK sample collection and transfer will be provided in the Laboratory Manual.

8.2 Immunogenicity Assessment

Blood samples will be collected to evaluate the response of ADA to AK104 in serum. The following bridging immunoassays will be developed and validated: screening, qualification, and titer. Once a test sample is confirmed positive for the presence of ADAs, the specificity of the ADA will be assessed to determine if the ADAs have neutralizing potential. Samples will be collected and stored for the ADA specificity assessments, which will occur at a later time when the assays have been developed and validated. The scheduled immunogenicity sampling day may be adjusted according to the actual medication of the subject.

Table 13 Immunogenicity sample collection schedule

	Scheduled Day	Time Relative to AK104 (or Placebo) Infusion
Dose 1 (Cycle 1)	Day 1	Pre-infusion (within 30 minutes prior to infusion)
Dose 3 (Cycle 3)	Day 43	Pre-infusion (within 30 minutes prior to infusion)
Dose 5 (Cycle 5)	Day 85	Pre-infusion (within 30 minutes prior to infusion)
Dose 7 (Cycle 7)	Day 127	Pre-infusion (within 30 minutes prior to infusion)
10th dose and subsequent doses (every 6 treatment cycles)	Day 190 and beyond	Pre-infusion (within 30 minutes prior to infusion)
Day 30 (± 7 days) after last dose [a]		

a. ADA sample collection is not required if a new anti-tumor therapy is initiated.

Details for collection, aliquoting, storage, and shipment of serum samples for ADA evaluations are

presented in a separate Laboratory Manual.

8.3 Biomarker Assessment

According to the inclusion criteria specified in Section 4.1.1, all subjects must provide archival or freshly obtained tumor tissue samples (formalin-fixed paraffin-embedded [FFPE] block or 9 unstained tumor tissue slides, preferably freshly obtained tumor tissue samples; if the subject is unable to provide 9 unstained tumor tissue slides, at least 7 unstained tumor tissue slides may be provided after discussion with the Medical Monitor) within 2 years prior to signing the ICF.

vCPS is an indicator to detect PD-L1 expression levels in tumor tissues using Ventana SP263. vCPS ≥ 5% is defined as that the area of tumor cells and tumor-infiltrating immune cells expressed by PD-L1 is ≥ 5% in the tumor area; vCPS < 5% is defined as that the area of tumor cells and tumor-infiltrating immune cells expressed by PD-L1 is < 5% in the tumor area, as well as that not evaluable (NE).

CPS is the percentage of PD-L1-stained positive viable tumor cells (any intensity of partial or complete membrane staining) as well as positive lymphocytes and macrophages (any intensity of membrane or cytoplasmic staining) detected using Dako 22C3 to all viable tumor cells multiplied by 100.

If the subject does not have archival tumor tissue samples or the tumor tissue sample is not suitable for use, a biopsy must be performed to collect fresh tumor tissues. Operating requirements and recommendations for collection of fresh tumor tissue samples:

- Fresh tumor biopsies should be obtained preferentially from tumor tissues that are safely collected as determined by the Investigator, rather than from the sites that are obtained by major-risk surgery (e.g., brain biopsy) . Tumor lesions for fresh biopsy should not be used as RECIST v1.1 target lesions unless no other lesions are suitable for biopsy and the longest length of lesion used for biopsy is ≥ 2 cm. If target lesions are used for biopsy, they must be performed prior to imaging at screening.
- Fine needle aspirate biopsy or burr biopsy samples are often insufficient for biomarker testing, and this method should be carefully selected to ensure that tumor tissue is adequate for biomarker testing.
- Cell smears centrifuged from pleural/abdominal/pelvic/pericardial effusion drainage, bone lesions without soft tissue components or decalcified bone tumor samples are not acceptable.

One of the study objectives of this study is to evaluate study endpoints, i.e., PD-L1 positive subjects. Tumor tissue samples from all subjects will be sent to the central laboratory for determination of PD-L1 expression level at screening. The shipping, storage, and handling of archival tumor samples,

fresh tumor samples, and leftover tumor tissues for biomarker assessment will be managed by a central laboratory. Refer to the Central Laboratory Manual for details of sample handling.

9. SAFETY ASSESSMENTS

9.1 Definition

9.1.1 Adverse Event

An adverse event (AE) is any untoward medical occurrence or worsening of an existing medical event in a subject who has signed the ICF and which does not necessarily have to have a causality with the study drug. An AE can therefore, be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease associated with the use of a/an (investigational) drug, whether or not considered related to the (investigational) drug (refer to the ICH Clinical Safety Data Management: Definitions and Standards for Expedited Reporting).

It is the responsibility of Investigators, based on their knowledge and experiences, to determine which circumstances or abnormal laboratory findings should be considered AEs. AEs include, but are not limited to, the following:

- Worsening (including worsening of symptoms, signs and laboratory abnormalities) of an existing (before the signing of ICF) medical condition/disease;
- Any new adverse medical condition (including symptoms, signs and newly diagnosed diseases);
- Abnormal and clinically significant laboratory values or findings.

9.1.2 Serious Adverse Event

A serious adverse event (SAE) is defined as any untoward medical occurrence that meets one or more of the following severity criteria:

- Results in death;
- Is life-threatening (Note: The term "life-threatening" in the definition of "serious" refers to an event in which the subject will be at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it are more severe.);
- Hospitalization or prolongation of hospitalization (see Section 9.2.4 for exceptions);
- Results in persistent or significant disability/incapacity;
- Is a congenital anomaly/birth defect;
- Is an important medical event (Note: Medical and scientific judgment should be exercised in deciding whether expedited reporting is appropriate in other situations, such as important medical events that may not be immediately life-threatening or result in death or hospitalization but may jeopardize the subject or may require intervention to prevent one of the other outcomes listed in the definition above. Examples include allergic bronchospasm, and development of drug dependency or drug abuse.)
- It should be noted that:

- A procedure is not an AE or an SAE, but the reason for the procedure may be an AE;
- Pre-planned surgery or hospitalization for an existing condition that did not worsen significantly is not an SAE.

9.1.3 Adverse Events of Special Interest

Adverse events of special interest (AESIs) are events of scientific and medical interest specific to understanding of the safety profile of the study drug AK104 injection and may require close monitoring by the Investigator. An AESI may be serious or nonserious.

AK104, a humanized IgG1 BsAb antibody, is capable of simultaneously binding PD-1 and CTLA-4 with high affinity. Potential risks based on mechanism of action are associated with AK104, which activates the immune system. AESIs for this study include, but are not limited to, IRRs and events with a potential inflammatory or immune-mediated mechanism and that may require more frequent monitoring and/or interventions such as steroids, immunosuppressants, and/or HRT.

Based on similar drugs and AK104 data, ≥ Grade 3 immune-related AEs (including but not limited to colitis, dermatitis, pneumonia, myocarditis, myositis, hepatitis, encephalitis, nephritis, endocrine disorders, and immune thrombocytopenia), ≥ Grade 3 infusion-related reactions and systemic allergic reactions/serious type I hypersensitivity are considered as AESIs.

9.1.4 Immune-related Adverse Events

An irAE is defined as an ADR of all grades judged to be causally related to the immune mechanism in clinical trials of anti-tumor drugs/therapies.

Based on the mechanism of action of AK104 leading to T cell activation and proliferation, it is possible to observe irAEs during the conduct of this study. Overlapping or more severe irAEs may occur when anti-PD-1/L1 is used in combination with anti-CTLA-4 drugs. Potential irAEs may be similar to those seen with anti-PD-1/L1 or anti-CTLA-4 drugs, which may include immune-mediated colitis, dermatitis, pneumonitis, hepatitis, encephalitis, nephritis, endocrine disorders, etc.

Subjects should be monitored for signs and symptoms of irAEs. Serologic, immunologic, and histological (biopsy) data should be used to support the diagnosis of irAEs, as appropriate. Appropriate measures should be taken to rule out tumors, infections, metabolism, toxins, or other causes that cause suspected irAEs. In the absence of an alternative etiology (e.g., infection or PD), an immune-related etiology associated with clinical signs or symptoms of colitis, dermatitis, pneumonia, hepatitis, encephalitis, nephritis, and endocrine disorders should be considered.

9.1.5 Suspected and Unexpected Serious Adverse Reaction (SUSAR)

A suspected unexpected serious adverse reaction (SUSAR) is a serious adverse reaction that is both unexpected (i.e., not present in the product's Reference Safety Information [RSI] or labeling of

marketed product) and meets the definition of a serious adverse drug reaction (SADR), the specificity or severity of which is not consistent with those noted in the Investigator's Brochure.

9.2 Handling of Adverse Events

9.2.1 Investigator's Responsibilities

The Investigator will record all directly observed AEs and all AEs reported spontaneously by the subject/legal representative in the original medical record. In addition, subjects/legal representatives will be asked about the occurrence of AEs in a non-guided manner.

The Investigators are required to fully record any case of AE, including diagnosis (if there is no diagnosis, symptoms, signs, laboratory abnormalities and other abnormalities should be recorded), dates of start and end, CTCAE grade and change reported, whether the event is an SAE or not, whether the event is an AESI or not, actions taken with study drug, treatment given due to AEs, and outcomes of the event, the causal relationship between the AE and the study drug judged by the Investigator, and whether the AE leads to withdrawal of the subject from study treatment.

The Investigators shall immediately take appropriate treatment measures to ensure the safety of the subjects if an SAE occurs during the study, regardless of relationship to study drug. Meanwhile, the Investigator must complete the SAE form and report it promptly to the Sponsor within 24 hours of awareness.

9.2.2 Time Limit for Collection of Adverse Events/Serious Adverse Events

After the subject signs the ICF, all AEs and SAEs will be reported (regardless of causality to the study drug) until 30 days (AE) and 90 days (SAE) after the last dose of study treatment or initiation of other anti-tumor therapy, whichever occurs first in the original records and eCRFs. Immune-related AEs (irAEs) , regardless of serious or non-serious irAEs, will be reported until 90 days after the last dose of AK104/placebo (whether other anti-tumor therapy have been initiated). After this period, the Investigator should report any SAEs that are assessed as related to AK104/placebo.

9.2.3 Follow-up for Adverse Events/Serious Adverse Events

Follow-up for AEs/SAEs should be continued not only during the treatment, but also for AEs/SAEs that persist in subjects at the end of the study or early withdrawal. Follow-up may be discontinued after as much information as possible has been obtained if any of the following is achieved:

- Event resolves; or
- Event becomes stable; or
- Event returns to baseline level (if baseline records are available); or

- When more information is not possible (When no more information is available (the subject refuses to provide further information, or the subject is lost to follow-up or withdraws consent); or
- The Investigator agrees with the Sponsor on the acceptable level.

Note: Subjects who discontinue treatment or withdraw early but still have AEs/SAEs related to the study drug should be followed up until the above condition or causality becomes unrelated.

Event resolution means that the subject has returned to baseline or that AE/SAE is not expected to improve or worsen as anticipated by the Investigator. For AEs/SAEs that remain unresolved at the last AE/SAE assessment, the Investigator will follow up the subjects with these AEs/SAEs as per medical practice. During the post-study follow-up phase, the Investigator is not responsible for actively collecting new AEs/SAEs.

9.2.4 Recording of Adverse Events

Diagnosis, Symptoms and Signs

If a diagnosis has been made, the diagnosis should be recorded on the eCRF rather than the individual symptoms and signs (for example, liver failure should be recorded, rather than jaundice, elevated transaminase, and flapping tremor). If it is not determined that symptoms and signs are caused by the diagnosis at the time of reporting, they will be recorded as individual AE/SAE. If it is later determined that symptoms, signs, and laboratory abnormalities are caused by the diagnosis, only the diagnosis will be reported separately, symptoms, signs, laboratory abnormalities will be included in the diagnosis, records of symptoms and signs are required to be deleted for AEs, and follow-up update reports are required to be sent for SAEs.

Adverse Event Secondary to Other Event

In general, AEs that are secondary to other events (e.g., cascade events or clinical sequelae) should be identified by their primary cause, unless it is severe or serious adverse events. However, a secondary event with clinical significance should be recorded as an independent AE in the eCRF if its time of occurrence is different from the primary event. If the association between events is unclear, the event should be recorded separately in the eCRF.

Persistent or Recurrent Adverse Events

A persistent AE is one that extends continuously, without resolution, between patient evaluation time points. Such AEs should only be recorded once on the eCRF. The initial severity of the event should be recorded and then updated in case the event is aggravated to record the highest severity of the event.

A recurrent AE is one that completely resolves between patient evaluation time points and subsequently recurs. Such two events should be recorded separately in the eCRF.

Abnormalities in Laboratory Tests or Other Assessments

Not all abnormalities in vital signs or laboratory tests are AEs. The events with values not considered clinically significant by the Investigator should not be considered AEs. Only those considered clinically significant by the Investigator (the Investigator is advised to repeat laboratory tests) need to be reported as AEs, and vital signs or laboratory findings that meet any of the following criteria are considered clinically significant:

Vital signs or laboratory findings are accompanied by relevant clinical symptoms;

Vital signs or laboratory findings result in changes in study treatment (e.g., leading to dose modification, treatment interruption, or treatment discontinuation);

Vital signs or laboratory findings result in a need for medical intervention or changes in concomitant therapy.

The Investigator is responsible for reviewing all vital signs and laboratory findings. Medical and scientific assessments should be performed in determining whether an individual vital sign or laboratory abnormality should be classified as an AE. If it is determined that symptoms, signs, and laboratory values are caused by the diagnosis, only the diagnosis is reported individually, and symptoms and signs are included in the diagnosis.

Death

Throughout the study, all deaths occurring during the period, whether related to study drug or not, including 90 days after the last dose or before the initiation of other anti-tumor therapy, whichever occurs first, should be recorded in the death report form and SAE report form of the eCRF and reported to the Sponsor in a timely manner.

When recording an event of death, the cause of death will be recorded as an SAE if the cause of death is clear, and the outcome of the SAE will be recorded as death and the event will be reported as an SAE; if the death is clearly due to disease progression, it may not be reported as an SAE unless assessed by the investigator as related to the study drug; if the cause of death is unknown at the time of reporting, it should be recorded as “unexplained death” in the AE form of the eCRF, and be reported as an SAE of “unexplained death” before further investigation of its exact cause.

Hospitalization and Prolongation of Existing Hospitalization or Surgery

Any AEs that result in hospitalization or prolongation of existing hospitalization should be recorded and reported as SAEs, with the following exceptions:

- Hospitalization or prolongation of existing hospitalization scheduled according to protocol (e.g., for drug administration, efficacy assessment);
- Hospitalization due to unchanged pre-existing medical conditions prior to study participation, e.g., elective surgery/treatment arranged prior to study participation.

- Time of visit to an emergency department or other departments in the hospital is < 24 hours and the subject is not hospitalized (unless it is a significant medical or life-threatening event);
- Hospitalization for scheduled medical procedures/surgeries as per the study protocol;
- Routine medical examinations (e.g., routine colonoscopy) that require hospitalization to evaluate the baseline level/trend of health status;
- Medical/surgical hospitalizations that have been predefined prior to study entry, which are not intended for salvage therapy. However, these conditions need to be documented;
- Hospitalization due to other conditions which are unrelated to health status and need no medical/surgical interventions (e.g., no housing, financial difficulty, temporary lack of caregivers, family environment, management reasons, hospital's administrative requirements, sanatorium);
- Hospitalization for anti-tumor therapy without any other SAEs.

However, if the pre-existing disease deteriorates during the study (e.g., requiring surgery/treatment earlier than originally planned), the elective surgery/treatment that becomes necessary due to the deterioration will be considered as an AE/SAE.

Disease progression

Progression of a potential malignant tumor (excluding the outcome of death) will not be reported as an AE/SAE if it is clearly consistent with potential tumor progression as defined by RECIST v1.1 criteria or other criteria specified in the protocol. If it is not possible to determine whether these symptoms or signs are caused by the progression of underlying malignant tumor, or do not meet the expected pattern of PD, the clinical symptoms or signs of progression may be reported as AEs/SAEs.

Deaths that are clearly due to PD during the study should not be reported as SAEs unless assessed by the investigator as related to study drug.

New Tumor

The occurrence of a new tumor should be considered an SAE. A new primary tumor refers to a tumor that is not the main reason for the use of study drug and is found after the subject is enrolled in the study.

Pre-existing Medical Conditions

The pre-existing symptoms/signs at screening by subjects should be recorded and reported as AEs only in the event of worsening in severity, frequency, and nature (except for worsening of the disease under study) after enrollment into the study. Changes relative to pre-existing conditions, such as "increase in headache frequency", should be reflected in the record.

9.2.5 Information Collected for Adverse Events/Serious Adverse Events

Information to be collected includes, but is not limited to:

1. AE name;
2. AE duration (start date, end date);
3. The causality of AEs to the study drug (definitely related, probably related, possibly related, unlikely related, unrelated);
4. AE severity: The Investigator will make an assessment of severity for each AE and SAE reported during the study. All AEs and SAEs will be evaluated and graded according to the NCI-CTCAE v5.0. Toxicities that are not specified in the NCI-CTCAE v5.0 will be graded according to the following definitions:

- Grade 1, mild: asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated;
- Grade 2, moderate: smaller, non-invasive or local treatment is required; age-related instrumental activities of daily living are limited;
- Grade 3, severe: serious or medically significant but not immediately life-threatening; leading to hospitalization or prolongation of existing hospitalization; disabling; limitations on self-care activities of daily living;
- Grade 4: life-threatening; urgent intervention indicated;
- Grade 5, AE-related death.

Note: Instrumental activities of daily living refer to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.;

Self-care activities of daily living refer to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden, etc.

5. Severity criteria for AE (fatal, life-threatening, hospitalization or prolongation of existing hospitalization, persistent or significant incapacity/loss of function, congenital anomaly/birth defect, and other important medical events);
6. Action taken with study drug due to AE;

   1 = Continuation: No change in study drug, and administration as specified;

   2 = Discontinuation: The prescribed dosing regimen is discontinued;

   3 = Interruption: The prescribed dosing regimen is interrupted and modified;

   4 = Dose reduction: including frequency, intensity, or dose reduction

   5 = Other (specify).
7. AE outcome

   1 = Resolved: The subject fully recovers from the AE and no sequelaes are observed;

2 = Resolved with sequelae: The residual effects of the AE are still present and observable; sequelae/residual effects are identified.

3 = Not recovered: The AE itself is in remission but not fully recovered.

4 = Not resolved: The AE itself is still present and observable;

5 = Fetal;

6 = Unknown.

8. Other action taken for AE

1 = None: No treatment is required.

2 = Medication required: Prescription and/or over-the-counter medication is required to treat the AE.

3 = Hospitalization or prolongation of existing hospitalization required: Hospitalization is required or prolonged due to the AE, whether or not medication is required.

4 = Other.

9.2.6 Causality Assessment

During the clinical trial, the Investigator should determine the causality between AEs and study drug, explain the basis for the judgment as much as possible, and record in detail in original medical record, eCRF and SAE report form, and organize consultation and judgment by relevant medical personnel if necessary. When an SAE occurs, the Investigator's initial report to the Sponsor/contract research organization (CRO)'s pharmacovigilance team may contain very little information. However, it is important for the Investigator to assess the causality of each event before sending the SAE report form to the Sponsor's pharmacovigilance team, may modify his/her opinion on causality based on subsequent information and may modify the SAE eCRF and SAE report forms accordingly. The causality assessment is one of the criteria used when determining regulatory reporting requirements.

The relationship between an AE and the study drug will be classified into five categories based on the criteria for determination of causality of study drug and AE: definitely related, probably related, possibly related, unlikely related, and unrelated. According to the ICH E2A guideline, "related" refers to a reasonable relationship between a drug and an AE. If there is insufficient information for causality assessment, no reasonable association is considered. Refer to the table below for specific judgments. The relationship between an AE and the study drug will be determined by the Investigator on the basis of their clinical judgment and the definition in the following table:

Table 14 Determination criteria for causality of adverse events

Grade	Assessment Criteria*
Definitely related	• The AE follows a reasonable temporal sequence from the administration of study drug;

	• The AE cannot be explained by the effects of disease or other drugs; • The response to the dechallenge test is justified; • The AE can be considered to be caused by the pharmacological effects of the study treatment; • Positive rechallenge test (if necessary)
Probably related	• The AE follows a reasonable temporal sequence from the administration of study drug; • The AE is not attributable to the effects of disease or other drugs; • The response to the dechallenge test is justified; • Rechallenge information is not required.
Possibly related	• The AE follows a reasonable temporal sequence from the administration of study drug; • The AE can be explained by the study drug or can be explained by the effects of disease or other drugs; • Information on the dechallenge test may be missing or unclear
Unlikely related	• The AE does not follow a reasonable sequence from the administration of study drug (but cannot be completely excluded) • The AE can be reasonably explained by the effects of disease or other drugs; • Or lack of sufficient information for causality assessment
Not related	• The AE occurs prior to study treatment • Or events not possibly related to study treatment (e.g., injury as a passenger in a car accident) • Or there is other more reasonable explanation (e.g., typical adverse reaction to concomitant medications and/or typical disease-related AE).

*: All key points should reasonably meet the assessment criteria.

9.3 Serious Adverse Event Reporting

9.3.1 Initial Report

The Investigator should complete the Sponsor's Serious Adverse Event Collection Form, sign and date it, and report it to the Sponsor or its representative (refer to the Safety Management Plan for reporting) within 24 hours of becoming aware of the SAE.

The Serious Adverse Event Report Form and the Sponsor’s original acknowledgement email should be kept at the study site.

If the Investigator does not have all information regarding an SAE, they should not wait to receive additional information before notifying the Sponsor or designee of the event and completing the form. The form will be updated when additional information is received.

The Investigator should provide other information (autopsy report, medical records of termination phase, and other necessary information) on SAEs (including reported deaths or adverse drug reactions) as required by the Sponsor, medical institution, and ethics committee.

9.3.2 Follow-up Report

Follow-up information should be filled in a new Serious Adverse Event Report Form, and be identified as a follow-up report.

Upon receipt of new information on the reported SAE, the follow-up report is uploaded within the specified time-frame and the subsequent reporting process is the same as the initial reporting process specified in Section 9.3.1.

9.3.3 Expedited Reporting to Drug Regulatory Authorities, Health Authorities, Investigators, Clinical Trial Institutions, Ethics Committees, and Other Relevant Organization

In accordance with national/regional regulations, the Sponsor will promptly notify Investigators, clinical trial facilities, ethics committees, drug regulatory authorities, and health authorities of any SUSARs.

In accordance with all applicable regulations, the Investigator reports SUSARs to the Institutional Review Board (IRB)/Independent Ethics Committee (IEC).

9.4 Reporting of Adverse Events of Special Interest

After the occurrence of AESIs, the eCRF module report should be completed within 7 days of the Investigator's awareness.

9.5 Pregnancy Reporting

Any female subject or the spouse of a male subject who becomes pregnant while on study drug or within 120 days of discontinuation of study drug must notify the Sponsor or Drug Safety Associate of the CRO.

When the Investigator becomes aware that a female subject is pregnant, the Investigator must ask the subject to discontinue the study drug and follow up the subject until termination of pregnancy or delivery. When the Investigator becomes aware that the spouse of a male subject is pregnant, the subject is not required to withdraw from the trial, but the Investigator must follow up the pregnancy as far as possible. The Investigator should promptly report all pregnancy using the pregnancy report to the ethics committee and Sponsor or its representative within 24 hours of learning of pregnancy. Although pregnancy is not an adverse event, close follow-up and confirmation of the outcome about the pregnancy event are required. The pregnancy follow-up report form should be completed with the outcome of pregnancy, and reported to the IEC and Sponsor or its representative. Other AEs during pregnancy which are identified as SAEs should be reported in accordance with the SAE reporting procedure. Ectopic pregnancy, spontaneous abortion, induced abortion for medical and health reasons, dead fetus, and neonatal SAEs (not limited to neonatal deaths, congenital anomaly [including congenital anomaly of aborted infants]) should be recorded and reported as SAEs.

9.6 Other Events Requiring Immediate Reporting

9.6.1 Overdose

An overdose is defined as a subject receiving a dose of study drug higher than the prescribed dose in the protocol.

- An AE-related overdose will be recorded and reported in the form of AE diagnosis/symptom on the respective AE module in the eCRF.
- The event of overdose without relevant symptoms will be reported to the Sponsor within 7 days of documentation.

Overdose of AK104 injection/placebo with and without AEs must be reported electronically or in writing to the Sponsor/CRO by the Investigator or other personnel in the study site (when this person becomes aware of the event) via EDC within the specified time. Events of overdose related to SAEs should be reported to the Sponsor within 24 hours.

9.6.2 Combined Elevation of Aminotransferase and Bilirubin

Combined elevations of aminotransferases and bilirubin, either serious or non-serious and whether or not causally related to the study drug, meeting the criteria of a potential Hy's law case (ALT or AST ≥3 × ULN together with TBiL ≥2 × ULN) should always be reported to the Sponsor/CRO through EDC or SAE paper form, with the Investigator's assessment of seriousness, causality, and a detailed narrative, according to the SAE reporting procedure with the specified period. The site will promptly initiate subsequent investigations and will conduct detailed medical history collection, inquiries for laboratory tests, as well as further tests (including viral hepatitis serology and liver imaging) to determine if the results are reproducible and/or if there is objective evidence that clearly supports the results are caused by disease (e.g., gallstones and bile duct obstruction with gallbladder dilatation) or other medications other than study drug.

9.7 Clinical Laboratory Safety Assessments

All laboratory values must be appraised by the Investigator as to clinical significance. The events with values not considered clinically significant by the Investigator should not be considered AEs. Only those considered by the Investigator to be clinically significant (the Investigator is advised to repeat laboratory tests) need to be reported as AEs. All abnormal laboratory values considered clinically significant by the Investigator should be recorded on the AE page of the eCRF. If the abnormal laboratory value constitutes an SAE, the SAE should be reported through EDC or SAE paper form and other relevant procedures must be followed.

Table 15 Laboratory test safety assessments

Hematology	Red blood cell (RBC) count, hemoglobin (Hb), hematocrit (HCT), mean corpuscular hemoglobin concentration (MCHC), mean corpuscular volume (MCV), white blood cell (WBC) count, absolute neutrophil count (ANC), neutrophil percentage (NEUT%), lymphocyte (LYM) count and percentage, monocyte (MONO) count and percentage, eosinophil (EOS) count and percentage, basophil (BASO) count and percentage, and platelet count (PLT).
Urinalysis	Acidity or alkalinity (PH), urinary protein (UPRO), specific gravity, urinary glucose (UGLU), bilirubin, ketones, occult blood, and microscopic examination (including: urine white blood cell (UWBC), urine red blood cell (URBC), if required). If the

	subject's urinalysis results show proteinuria ≥ 2+, a 24-hour urine protein test should be performed.
Blood chemistry	Aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), lactate dehydrogenase (LDH), total bilirubin (TBIL), direct bilirubin (DBIL), glutamyltransferase (GGT), total protein (TP), albumin (ALB), fasting blood glucose (FBG), triglycerides (TG), total cholesterol (TC), creatinine (Cr), uric acid (UA), urea nitrogen (BUN) or urea, sodium (Na^+), potassium (K^+), chloride (Cl^-), calcium ($Ca2^+$), magnesium ($Mg2^+$), phosphorus (P), serum C-reactive protein (CRP) and amylase.
Coagulation function	Prothrombin time (PT), international normalized ratio (INR), and activated partial thromboplastin time (APTT)
Pregnancy testing	Serum β-hCG pregnancy test or urine hCG pregnancy test
Myocardial marker	Creatine kinase (CK), creatine kinase isoenzyme (CK-MB); and further detect cardiac troponin I (cTnI) and/or cardiac troponin T (cTnT), brain natriuretic peptide (BNP), and/or N-terminal pro-brain natriuretic peptide (NT-proBNP) if any clinically significant abnormalities. Additional cardiac function tests may be performed at the discretion of the Investigator, depending on the subject's clinical condition and site conditions
Thyroid function	Free triiodothyronine (FT3), free thyroxine (FT4), and thyrotropin (TSH)
HBV	Hepatitis B surface antigen (HBsAg), hepatitis B surface antibody (HBsAb), and hepatitis B core antibody (HBcAb); if HBsAg or HBcAb is positive, HBV-DNA testing is required.
HCV	Hepatitis C antibody (HCVAb); if HCVAb is positive, HCV-RNA testing is required.
HIV	Human immunodeficiency virus antibody (HIVAb).

9.8 Vital Signs

Vital sign measurements include blood pressure, pulse rate, respiratory rate, and body temperature. It is recommended that each vital sign test should be performed after at least 5 minutes of rest.

Additional monitoring with assessment of vital signs is at the discretion of the Investigator per standard clinical practice or as clinically indicated.

9.9 Physical Examination

A complete physical examination, including general condition, skin, mucosa, superficial lymph nodes throughout the body, head and organs, neck, chest, abdomen, spine, extremities, nervous system and others, will be performed at screening. A brief focused physical examination during the treatment period should be symptom-directed and performed prior to administration of investigative product.

9.10 12-Lead ECG

A 12-lead electrocardiogram (ECG) will be performed as planned in the study flowchart. All ECGs will be reviewed by the Investigator. ECG results should be explained by a qualified physician and clinically significant abnormalities that occur when the subject signs the ICF for screening should

be recorded in the eCRF. New or worsening clinically significant findings after the informed consent form for screening was signed (ECGs are recommended to be repeated at appropriate intervals) must be recorded as AEs in the eCRF.

Original ECG tracings of each of the timed 12-lead ECGs will be retained with the subject's source documents. At a minimum the heart rate, and QRS, QT, QTc, and PR intervals will be recorded in the eCRFs. Digital copies of ECGs will be retained in the subjects' source documents (if applicable) and made available upon request.

9.11 Echocardiography

Echocardiography is required at Screening and as clinically indicated to assess cardiac function such as left ventricular ejection fraction (LVEF). For the purpose of safety monitoring, the Investigator must review all echocardiogram reports. Echocardiography reports will be retained in the subject's original files at the study site and any echocardiography abnormalities must be recorded in the eCRF.

9.12 ECOG Performance Status

The ECOG PS will be determined at Screening, and then at each scheduled visit throughout the study. ECOG performance status will be assessed at the time specified in the Assessment Schedule based on the following content.

Table 16 ECOG PS score

Performance Status Score	Performance Status
0	Fully active, able to carry on all pre-disease performance without restriction.
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work.
2	Ambulatory and capable of all self-care but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	Self-care ability is limited, bed or chair time in >50% of daytime.
4	Completely disabled; cannot carry on any self-care; totally confined to bed or chair.
5	Death

10. STATISTICAL METHODS

10.1 Analysis Set

10.1.1 ITT Analysis Set

The ITT Analysis Set includes all randomized subjects. The FAS includes all randomized subjects; for analyses requiring baseline data, only subjects with baseline data are included (e.g., for ORR, measurable disease as defined per RECIST 1.1 is required at baseline).

Subjects will be analyzed according to the treatment group to which they are randomized, regardless of the drug treatment they actually receive.

10.1.2 Safety Analysis Set

The Safety Analysis Set (SS) includes all subjects who receive any amount of the study drug. The analysis will be conducted according to the treatment actually received by the subjects.

10.1.3 Immunogenicity Analysis Set

The Immunogenicity Analysis Set will include all subjects who have been randomized, received at least one dose of study drug, and had pre-dose and at least one post-dose immunogenicity assessment data.

10.1.4 Pharmacokinetic Analysis Set

The PK Analysis Set will include all subjects who have been randomized, received at least one dose of study drug, and had at least one valid post-dose plasma concentration data.

10.2 Study Population Data

For subjects in the ITT Analysis Set, the reason for discontinuation of treatment and study will be summarized and listed.

Demographic and baseline characteristics such as age, sex, baseline ECOG performance status, prior therapy methods, and best response to prior therapy will be summarized for the ITT Analysis Set.

Study drug exposure, treatment duration, and medication compliance will be summarized for the safety analysis set using descriptive statistics.

10.3 General Statistical Considerations

Appropriate descriptive statistics will be performed for the selected population, safety, PK, immunogenicity and biomarkers. Descriptive statistics on continuous data will include the number of subjects, means, medians, standard deviations, minimum, and maximum, while categorical data will be summarized using frequency counts and percentages. Graphical summaries of the data may be presented.

Assessments of change from baseline to post-treatment or the ratio of post-treatment to baseline will

include only those subjects with both baseline and post-baseline measurements. The last non-missing value of a variable taken before the first dose of study drug will be used as the baseline value unless otherwise specified.

Safety analysis will be performed based on the Safety Analysis Set; efficacy analysis will be performed based on the ITT Analysis Set.

Detailed methodology to be used in the analyses will be described in the Statistical Analysis Plan (SAP) and finalized before database lock. All statistical analyses will be performed using SAS 9.4 (or higher).

10.4 Efficacy Analysis

The following efficacy endpoints will be analyzed. Please refer to SAP for details.

Primary endpoint:

- Overall survival (OS) is defined as the period from the date of randomization until death due to any cause. For subjects who are alive at the time of data cutoff, OS will be censored on the last date when subjects are known to be alive. For subjects without any follow-up information, OS will be censored on the day of randomization.

Secondary endpoints:

- Progression-free survival (PFS) is defined as the period from the date of randomization until documentation of progressive disease (per RECIST v1.1 criteria) or death due to any cause, whichever occurs first. Subjects who are alive and have no progressive disease at the time of analysis data cutoff are censored on the date of last evaluable tumor assessment. Subjects without post-baseline assessment will be censored on the date of randomization.
- Objective response rate (ORR) is defined as the proportion (%) of subjects with a response of CR or PR according to RECIST v1.1 at least 1 visit prior to radiographic progression (PD). The ORR assessment will include data obtained until disease progression or, in the absence of progression, until the last evaluable assessment. In the ORR analysis, subjects who discontinue treatment without disease progression, receive subsequent therapy then achieve response will not be included as responders. Patients who do not have a post-baseline response assessment (for any reason) will be considered non-responders in the best overall response assessment.
- Duration of response is defined as the duration from the first documentation of confirmed CR or PR to the first documented PD according to RECIST v1.1, or death due to any cause, whichever occurs first. For subjects who are alive and have not progressed at the time of data cut-off for analysis, DoR will be censored at the last evaluable tumor scan date

with assessment available. The DoR will be evaluated only for the subjects with CR or PR.

- Disease control rate (DCR) is defined as the proportion (%) of subjects considered CR, PR, or SD at least 1 visit per RECIST v1.1. The DCR assessment will include data obtained until disease progression or, in the absence of progression, until the last evaluable assessment. In the DCR analysis, subjects who discontinue treatment without progression, receive subsequent treatment, and then achieve response will not be included as responders or subjects with stable disease.
- Time to response (TTR) is defined as the time from the date of randomization to the first objective tumor response.

10.4.1 Analysis of Primary Efficacy Endpoints

The primary objective of this study is to compare the efficacy of the AK104 in combination with oxaliplatin and capecitabine (XELOX) regimen and the placebo in combination with XELOX regimen as first-line treatment for unresectable locally advanced or metastatic gastric adenocarcinoma or adenocarcinoma of the gastroesophageal junction, with OS in the ITT Analysis Set as primary endpoint. One interim and final analyses of OS are planned for this study in the ITT Analysis Set. The OS interim analysis will be performed when the target number of OS events is reached. The timing and stopping boundaries for the interim and final analyses are presented in Section 10.8.

OS: OS will be compared between the study group and the control group by stratification factors at randomization using a stratified Log-rank test based on and ITT Analysis Set. The hazard ratio and 95% confidence interval will be estimated using a stratified Cox regression model, and the same event time will be handled with an exact method. The Kaplan-Meier method will be used to estimate the distribution of event time in each treatment group. The median survival time (if evaluable) and its two-sided 95% confidence intervals will be assessed every 6 months using the Kaplan-Meier method.

10.4.2 Analysis of Secondary Efficacy Endpoints

The secondary efficacy endpoints analyzed in this study include OS in the CPS ≥5 population in the ITT population, PFS, ORR, DoR, DCR, and TTR assessed by investigator per RECIST v1.1(based on ITT population and CPS ≥5 population in the ITT population).

OS: based on the CPS ≥5 population in the ITT population, OS will be analyzed using the same statistical method as the OS analysis in the ITT Analysis Set.

PFS: PFS will be compared between the study group and the control group by stratification factor at randomization using a stratified Log-rank test based on the ITT Analysis Set and CPS ≥ 5 population in the ITT population. The hazard ratio and 95% CI will be estimated using a stratified

Cox regression model. The Kaplan-Meier method will be used to estimate the distribution of event time in each treatment group. The median PFS (if evaluable) and its 95% confidence interval will be assessed every 3 months using the Kaplan-Meier method.

The statistical significance of the difference in ORR between the two treatment groups will be assessed using the Cochran-Mantel-Haenszel (CMH) chi-square test stratified by the actual value of the randomization stratification factors. The difference in ORR between groups and its 95% CI will be calculated, as well as the Clopper-Pearson 95% CI for ORR in each group. DCR will be summarized similarly as ORR.

For the endpoint DOR, median DOR is estimated using the Kaplan-Meier method, with the survival curves provided.

For the endpoint TTR, the median, maximum, and minimum are estimated separately for each treatment group.

10.5 Safety Analyses

Safety analysis should be performed using the safety analysis set according to the treatment actually received by the subjects.

10.5.1 Analysis of Adverse Events

A TEAE is defined as an AE that: has been absent at pretreatment and emerges during treatment (from the date of first dose to 30 days [for non-SAEs]/90 days [for SAEs] after the date of last dose); or has been present at baseline but stopped prior to treatment and re-emerges during treatment; or worsens in severity relative to the pretreatment state if the AE is continuous.

AEs will be coded using the Medical Dictionary for Regulatory Activities (MedDRA), and assigned grades according to the NCI CTCAE v5.0. The number and percentage of subjects with TEAEs will be summarized by the worst CTCAE grade, SOC, and PT. The number and percentage of subjects with treatment-emergent SAEs and TEAEs/SAEs considered related to study drug will be summarized in the same manner.

AEs (including TEAEs) of subjects should be listed by verbatim term, PT, SOC, grade, causality to study drug, etc. Deaths, other SAEs, and other significant AEs including those that lead to permanent discontinuation of AK104, oxaliplatin or capecitabine will be listed.

10.5.2 Analysis of Laboratory Values

Descriptive statistics will be provided for the clinical laboratory results and changes from baseline by scheduled time of evaluation (including the EOT visit) and treatment group as well as for the maximum and minimum post-baseline values.

Abnormal laboratory results will be graded according to NCI CTCAE v5.0. Shift tables will be used to present CTCAE grade at baseline and the change of worst post-baseline grade for laboratory

variables. Abnormal clinical laboratory test results of Grade 3 or 4 will be listed.

10.5.3 Analysis of Vital Signs

Vital sign measurements and changes from baseline will be summarized using descriptive statistics by scheduled evaluation time, including follow-up at the end of treatment and maximum and minimum values after baseline.

10.5.4 Electrocardiogram Analysis

ECG parameters (PR, RR, QRS, QT, QT interval corrected for heart rate using the Bazetts formula [QTcB], and QTcF) will be summarized using descriptive statistics by planned evaluation time for actual values and changes from baseline, including end-of-treatment follow-up and maximum value after baseline. QTcF will be considered as the primary correction method to assess subject cardiac safety.

The incidence of outliers in maximum absolute QTcF and QTcB intervals (new > 450 msec, new > 480 msec, new > 500 msec) and the maximum absolute uncorrected QT intervals (new > 500 msec) over all post-baseline evaluations, as well as in QTcF and QTcB maximum changes from baseline (> 30 and > 60 msec) over all post-treatment evaluations will be summarized by treatment. “New” means the category of the QTc abnormality was not present at baseline and became present at least 1 post-baseline ECG assessment. A listing of ECG data will be provided.

10.6 Pharmacokinetic, Immunogenicity and Biomarker Analysis

10.6.1 Pharmacokinetic Analysis

Individual AK104 concentrations will be tabulated by descriptive statistics. A population PK model will be established in this trial. The impact of physiologically relevant subject characteristics (covariates) and disease characteristics on PK will be evaluated. The relationship between exposure and response or safety will be evaluated. Such analysis can be described in PK analysis plans for different populations, and the result report of this PK analysis can be separated from the main study report.

10.6.2 Immunogenicity Analysis

The immunogenic potential of AK104 will be assessed by summarizing the number and percentage of subjects who develop detectable ADA. The impact of ADA on PK will be assessed if data is available. Samples will be collected for evaluating the neutralizing capacity of ADAs in the future. Only subjects who have received at least one dose of AK104 and provided a baseline sample and at least one post-treatment sample will be evaluated.

10.6.3 Biomarker Analysis

The relationship between PD-L1 expression and efficacy as a predictive biomarker in the archived tumor tissue samples or baseline fresh biopsy samples will be analyzed. PD-L1 expression will be

assessed by the central laboratory for the determination of IHC of PD-L1.

10.7 Sample Size Calculation

OS in the ITT population will be used as primary endpoint in this study. The number of events required for OS in the ITT Analysis Setis estimated based on the following assumptions (assuming an exponential distribution of event time):

- The one-sided significance level is 0.025. A test will be performed in the ITT population.
- The power to test an OS HR of 0.75 (equivalent to a median OS increase from 11 to 14.7 months) in the ITT population is at least 85%.
- An interim analysis of OS will be performed in the ITT populationto estimate the O'Brein-Fleming cut point by Lan-DeMets spending function.
- Assuming the OS dropout rate of 2%/12 month

A total of 588 subjects will be enrolled. The subjects will be randomized at a ratio of 1:1. It is assumed that a total of 588 subjects will be enrolled over 18 months.

10.8 Interim Analysis

An interim analysis of OS will be performed once based on the ITT population. The O'Brein-Fleming cut point will be estimated using the Lan-DeMets spending function. The interim analysis will be performed when approximately 354 OS events (80% of target OS events) are observed in the ITT population, which is estimated to be 28 months after the first subject is randomized.

After approximately 443 deaths are observed in the ITT population, the final analysis of OS will be performed, which is estimated to be 39 months after the first subject is randomized.

The interim analysis will be performed by an external independent statistician. IDMC will perform an interim power data review for the results of the interim analysis. After reviewing the data, considering the overall risks and benefits of subjects, IDMC will provide recommendations on whether the final analysis can be performed as planned, whether protocol modification is required, and whether a registration marketing application can be submitted to regulatory authorities in advance, based on the predefined statistical hypotheses and superiority determination rules.

The stopping boundaries (p-value and Z-score) of the superiority test for OS at the interim and final analyses are shown in Table 17. The boundary values will be updated based on the actual death events observed during the interim analysis and the final planned number of death events using a spending function.

Table 17 Stopping boundaries (p-value and Z-score) for main analyses of OS

	Time (Months)	Number of events	One-sided P-value for Efficacy (Z-score)	Approximate HR Threshold for Efficacy
ITT Analysis Set (one-sided **α** = 0.025)				
OS Interim Analysis	28	354	0.012（-2.250）	0.787
Final analysis of OS	39	443	0.021（-2.025）	0.825

10.9 Independent Data Monitoring Committee (IDMC)

An independent data monitoring committee (IDMC) will be established for this study. The committee will include independent oncologists and an independent statistician. The IDMC will periodically (approximately every 6 months) assess the on-study safety data and review the OS efficacy interim analysis data, and make corresponding recommendations to the Sponsor according to the preset OS stopping boundaries. After review and discussion by the IDMC, the Sponsor will make all final decisions regarding any change in study conduct. Details are provided in the IDMC charter. The functions and membership of IDMC will be described in the IDMC Charter.

An independent data monitoring committee (IDMC) will perform an interim power data review for the results of interim analysis. After reviewing the data, considering the overall risks and benefits of subjects, IDMC will provide recommendations on whether the final analysis can be performed as planned, whether protocol modification is required, and whether a registration marketing application can be submitted to regulatory authorities in advance, based on the predefined statistical hypotheses and superiority determination rules.

10.10 Subgroup Analyses

Subjects will be divided into multiple subgroups by demographics (e.g. age, sex), baseline characteristics (e.g. ECOG PS score, prior therapy, tumor location, presence or absence of liver metastases, presence or absence of peritoneal metastases) and PD-L1 expression to further assess the consistency of study results across subgroups.

11. DATA MANAGEMENT

11.1 Data Collection

All data collected during the study will be recorded in the individual, subject-specific eCRF. The Sponsor or its designee will supply EDC training and login accounts and provide instructions for the completion of the case report form. Each subject was assigned a unique subject number in the EDC for identification. Any data modifications made by the data entry personnel in the EDC system will be automatically recorded through the "Audit Trace" function of the system.

The data entry personnel should complete the case report form as soon as possible after follow-up, answer queries as soon as possible, and keep the data of the EDC current to enable the monitor and data reviewer to review the subject's status throughout the course of the study. The CRF should be completed, reviewed and signed by the Investigator as soon as possible after the last subject follow-up.

An eCRF must be completed for each subject who signs an ICF and undergoes any screening procedures. If a subject is not treated, the reason must be recorded in the eCRF.

All data other than the EDC (or external data, e.g., central laboratory, central imaging) will be integrated with the subject's CRF data in accordance with the data transfer agreement or analysis plan.

The subject's number and date of entry into the study, along with the study code, should be recorded in the subject's medical/study records by the Investigator. The Investigator should also record the following information in the medical/study records: confirmation of written and oral consent, the subject's clinical status, date of every study visit, date of administration of study drug, concomitant medications, copies of all relevant reports and laboratory tests, comments on results, and reference to any AEs.

The Investigator will complete the electronic signature via the EDC system, which will indicate that the Investigator inspected or reviewed the data in the CRF, the data queries, and the site notifications, and agrees with the content.

All information and other material to be used by subjects and investigative staff must use vocabulary and language that are clearly understood.

11.2 Data Management

The data management department will create the EDC database for data collection, and check and review the data during the study to ensure the timeliness, accuracy, reliability and consistency of the data.

The data management personnel will check the data of the case report form electronically and

manually, the Investigator or designee will reply to the queries generated, and the data management personnel will close the queries according to responses or resend the queries until the problem is resolved. The electronic method is to use the corresponding modules of the EDC system for program and logical settings, then use the system after testing. The data will be checked at the time of data entry and queries will be triggered according to pre-set conditions. The manual method is a supplement to the electronic method, in which the data management personnel will compile the verification rule conditions outside the system, and check data problems by SAS or manually. The resulting queries will be manually added to the EDC by the data management personnel. In addition, data management personnel and medical data reviewers will also review adherence to the protocol based on EDC data. To resolve any questions arising from the Clinical Data Management review process, CRF queries will be raised and resolved within the EDC application.

Data received from external sources such as central labs will be reconciled to the clinical database. SAEs in the clinical database will be reconciled with the safety database.

11.3 Medical Coding

AEs and medical history collected during the study will be coded using MedDRA. Prior and concomitant medications will be coded as per the World Health Organization Drug Dictionary (WHO-DD).

Terms that cannot be confirmed by the medical coder should be accurately coded by communicating with the Investigator through queries to obtain more detailed information.

11.4 Database Lock

Database lock is an important milestone in the clinical trial process, and the process and time of database lock should be clearly documented. Before the data is locked, a detailed list of database to be locked should be prepared and approved by all relevant personnel.

11.5 Study Documentation and Storage

The Investigator will maintain a signature list of qualified persons who have delegated study duties. All persons authorized to make entries and/or corrections on the eCRFs will be included in the signature list.

Source documents refer to the original records, documents and data generated in the clinical study, such as hospital medical records, medical images, laboratory records, memos, subject diaries or assessment forms, drug dispensing records, data automatically recorded by instruments, microfilm, photographic negatives, magnetic media, X-ray film, subject files, documents and records related to the clinical study, including certified copies, kept by the pharmacy, laboratory, and medical technology departments. Source documents, containing source data, may be available in paper or

electronic form. The Investigator and study staff are responsible for maintaining a comprehensive and centralized filing system (Trial Master File) of all study-related (essential) documentation, suitable for inspection at any time by representatives from the Sponsor and/or applicable regulatory authorities. Essential documents include:

- Subject files, containing completed original records, informed consents and supporting copies of source documentation (if kept). For studies using EDC, a Portable Document Format (PDF) file will be generated for each subject's data (original records and queries) at the end of the study, which will be stored in the Trial Master File;
- Study files, containing the protocol with all amendments, the Investigator Brochure, copies of all relevant clinical documents required before initialization of study, and all correspondences to and from the IEC/IRB and the Sponsor;
- Study drug-related records, including site receipt, accountability records, final reconciliation, and corresponding correspondence;
- In addition, all original source documents supporting entries in the original records must be maintained and be readily available. All essential documentation will be retained by the institution for at least 5 years after completion of the study or for a longer period, where so required by other applicable regulations or requirements. It is the responsibility of the Sponsor to inform the Investigator/institution as to when these documents no longer need to be retained. No study document should be destroyed without prior written agreement between the Sponsor and the Investigator. Should the Investigator wish to assign the study records to another party or move them to another location, he/she must notify the Sponsor in writing of the new responsible person and/or the new location.

11.6 Record Storage

Records of subjects, source documents, monitoring visit logs, data correction forms, original records, inventory of study product, regulatory documents (e.g., protocol and amendments, IRB/IEC correspondence and approvals, approved and signed ICFs, Investigator's Agreement, clinical supplies receipts, distribution and return records), and other Sponsor correspondence pertaining to the study must be kept in appropriate study files at the site. Source documents include all recordings and observations or notations of clinical activities and all reports and records necessary for the evaluation and reconstruction of the clinical study. These records will be retained in a secure file for the period required by the institution or site policy. Prior to transfer or destruction of these records, the Sponsor must be notified in writing and be given the opportunity to further store such records.

12. STUDY MANAGEMENT

12.1 Quality Management and Quality Control

Each study-related institution or organization will implement internal quality management for the study process, data and biological sample collection, document archiving and completion. Standard operation procedures shall be established for all study processes.

Qualification of study institutions: Clinical study institutions must be drug clinical study centers with clinical study qualification as determined by the National Medical Products Administration.

Qualifications of study personnel: Study personnel must be investigators, pharmacists, nurses or other technician trained for clinical trials and guided by senior professionals.

Laboratory quality control measures: The laboratories should establish the standard operating procedure and the quality control procedure for the experimental indicators.

Quality control measures for clinical ward trials:

- Before a trial, the clinical ward must meet the requirements of standardization and ensure that rescue equipment is complete.
- Investigators (including caregivers) should be trained for the protocol before the clinical trial begins.
- The operators should check and ensure instruments are functional and in good condition, and conduct test runs.
- The medication should be administered to the subjects by professional nursing staff.

The study drugs should be locked in cabinets, kept by trained pharmacy personnel or nurses, and stored in accordance with the drug storage requirements. The remaining study drugs should be stored separately, recorded, and returned to the Sponsor or destroyed at the end of the clinical trial.

All observations and abnormal findings in the clinical study should be carefully verified and documented in a timely manner to ensure the reliability of the data. The instruments, equipment, reagents and reference standards etc., used for various examinations in the clinical trial should follow rigorous specifications and work normally. The recording and transfer of clinical data must be in the charge of an experienced Investigator and supervised or checked by a specially-assigned personnel to ensure the scientificity and accuracy of the data. Conclusions of clinical trials must be based on raw data.

The Investigators responsible for trials should fill in the eCRFs completely, accurately and in detail. All trial-related data should be centrally managed and analyzed.

Procedures for data storage, transfer, and query should be established. The data kept includes: Subject's medical records, imaging data, drug use registration form, subject code form, SAE report form, GCP form to be filled in by hospitals, and relevant original medical documents, etc. The data

transferred includes: Subject enrollment information, SAE report forms and data required by summaries. The drug regulatory authorities, the Sponsor, its entrusted supervisors and the Ethics Committee have the right to access relevant study data and original records with the consent of the principal Investigator.

When the results of clinical trials are summarized and analyzed, standard statistical analysis methods must be used and analysis should be performed by professional biostatisticians.

12.2 Clinical Monitoring

In order to ensure the reliability and completeness of the study data, the Sponsor and its entrusted monitors should regularly conduct systematic monitoring on the clinical hospital to determine whether the trial is conducted in accordance with the study protocol, whether the reported data are consistent with the records provided by the institutions participating in the clinical trial, and a visit report should be written for each monitoring and visit.

The specific contents of visits and monitoring include:

- Provide information and support to Investigator and study-site personnel.
- Verify that the study equipment and facilities remain compliant with relevant requirements.
- Confirm that the study team adheres to the clinical study protocol, records study data in the eCRF accurately and timely, processes biological samples according to the laboratory manual, and completes checking and counting of study drugs.
- Verification of raw data (reconciliation of data in the eCRF against the subject's medical records at the hospital or actual medical records and other study-related records), including verification of informed consent forms of participating subjects, requires direct access to all original records (e.g., medical records) of each subject.
- Review raw data, i.e. review original documents to check their quality, check protocol compliance, and ensure critical step data and original documents are properly maintained.
- Ensure that withdrawal of informed consent for the use of biological samples from subjects is reported, biological samples are identified and disposed of/destroyed accordingly, actions taken are documented and reported to subjects.
- Sponsor representatives may be consulted between visits if the Investigator or other study-site personnel need to obtain information and advice on the conduct of the study.

13. ETHICAL AND REGULATORY REQUIREMENTS

13.1 Ethical Requirements

This study will be conducted in accordance with GCP, all applicable regulatory requirements, and, where applicable, the current version of the Declaration of Helsinki. The Investigator (or Sponsor, as appropriate) is responsible for ensuring that the protocol, the study site's ICF, and any other information to be presented to potential subjects (e.g., advertisements or information to support or supplement the ICF) are reviewed and approved by the appropriate IEC/IRB. The Investigator will allow the IEC/IRB direct access to all relevant documents. The IEC/IRB must be constituted in accordance with all applicable regulatory requirements. The Sponsor will provide the Investigator with relevant document(s)/data that are needed for IEC/IRB review and approval of the study. Before the study drug can be shipped to the study site, the Sponsor must receive copies of the IEC/IRB approval letter, the approved ICF, and any other information that the IEC/IRB has approved for presentation to potential subjects.

If the protocol, the ICF of the study site, or any other information that the IEC/IRB has approved for presentation to potential subjects is amended during the study, the Investigator is responsible for ensuring the IEC/IRB reviews and approves, where applicable, these amended documents. The Investigator must comply with all applicable regulatory requirements regarding the use of the amended ICF, including that subjects enrolled after the new version of the ICF has been approved by the IEC/IRB should use the new version of the ICF. Copies of the IRB/IEC approval of the amended ICF/other information and the approved amended ICF/other information must be forwarded to the Sponsor promptly.

13.2 Confidentiality Agreement and Subject Privacy

The Investigators and the Sponsor will preserve the confidentiality of all subjects taking part in the study in accordance with GCP and local regulations. The Investigator must ensure that the subject's anonymity is maintained. In the original records or other documents submitted to the Sponsor and the CRO, subjects should be identified by a unique subject identification (SID) number as designated by the Sponsor. Documents that are not for submission to the Sponsor and CRO (e.g., signed ICFs) should be kept in strict confidence by the Investigator.

In compliance with Federal regulations/ICH GCP Guidelines, it is required that the Investigator and institution permit authorized representatives of the company, of the regulatory agency(s), and the IEC or IRB direct access to review the subject's original medical records for verification of study-related procedures and data. The Investigator is obligated to inform the subject that his/her study-related records will be reviewed by the above named representatives without violating the

confidentiality of the subject.

13.3 Process of Informed Consent

Before a subject's participation in the study, it is the Investigator's responsibility to obtain freely given consent, in writing, from the subject or legally acceptable representative, as allowed by local regulations, after adequate explanation of the aims, methods, anticipated benefits, and potential hazards of the study. A legally acceptable representative is an individual or other body authorized by the subject or under applicable law to consent, on behalf of a prospective subject, to the subject's participation in the clinical study. The written ICF should be prepared in the local language(s) of the potential subject population.

In obtaining and documenting informed consent, the Investigator should comply with the applicable regulatory requirements and should adhere to GCP and the ethical principles that have their origin in the Declaration of Helsinki. The consent form and any revision(s) should be approved by the IEC or IRB prior to being provided to potential subjects. The Investigator will ensure that each subject is notified that they are free to discontinue from the study at any time and that each subject is given the opportunity to ask questions and allowed time to consider the information provided.

The subject's written ICF should be obtained prior to their participation in the study, and should be kept on file at the study site. The ICF should be signed and dated by the subject or a legally acceptable representative, as allowed by local regulations, and the Investigator. The original signed ICF should be retained at the site, and a copy of the signed consent form should be provided to the subject or legal representative. The date that informed consent was given should be recorded in the medical records and original records.

If the subject or legally acceptable representative, as allowed by local regulations, cannot read, then an impartial witness should be present during the entire informed consent discussion. This witness should sign the ICF after the subject or the legally acceptable representative has orally consented to the subject's participation and, if possible, signed the ICF. By signing the ICF, the witness attests that the information in the ICF and any other written information was adequately explained to and apparently understood by the subject or the legally acceptable representative and that the subject or the legally acceptable representative agreed to sign the ICF.

Any incentives for subjects who participate in the study as well as any provisions for subjects harmed as a consequence of study participation need to be described in the ICF that is approved by an IRB/IEC.

13.4 Amendments to the Protocol and Informed Consent Form

Any amendments to the study protocol will be communicated to the Investigator by the Sponsor,

and all of the amendments require approval from the IRB/IEC. A protocol amendment may be implemented after it has been approved by the IRB/IEC and if applicable, the national regulatory authority, unless immediate implementation of the change is necessary for subject safety. The Sponsor will ensure the timely submission of amendments to regulatory authorities.

If a protocol amendment requires a change to a site's ICF, the Sponsor and the site's IRB/IEC are to approve the revised ICF before the revised form is used.

14. FINANCE AND INSURANCE

14.1 Finance

Prior to initiation of the study, the principal Investigator and/or the study facility will enter into a clinical study agreement with the Sponsor. This agreement will include the financial information agreed upon by the parties.

14.2 Reimbursement, Compensation and Insurance

Reimbursement, indemnity, and insurance shall be addressed in a separate agreement on terms agreed upon by the parties.

15. PUBLICATION POLICY

The Investigator should keep the information and data related to this trial confidential and consult with the Sponsor in advance and obtain the written consent of the Sponsor in order to publish any conclusions drawn from the trial. In order to protect its rights and interests, the Sponsor may require the Investigator not to publish relevant information until the investigational product is approved for marketing. The Sponsor has the right to issue or publish information or data related to this study, or submit them to the drug regulatory authority.

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17. APPENDICES

APPENDIX 1 RESPONSE EVALUATION CRITERIA IN SOLID TUMORS (RECIST) v1.1

1. Measurability of Tumor at Baseline

1.1. Definitions

At baseline, tumor lesions/lymph nodes will be categorized measurable or non-measurable as described below.

1.1.1. Measurable

Tumor lesions: Must be accurately measured in at least 1 dimension (the longest diameter in the plane of measurement is to be recorded) with a minimum size of:

- 10 mm by CT scan (CT scan slice thickness no greater than 5 mm).
- 10 mm caliper measurement by clinical examination (lesions that cannot be accurately measured with calipers should be recorded as non-measurable).
- 20 mm by chest X-ray.

Measurable malignant lymph nodes: To be considered pathologically enlarged and measurable, a lymph node must be ≥ 15 mm in short axis when assessed by CT scan (CT scan slice thickness recommended to be no greater than 5 mm). At baseline (i.e., screening for this study) and in follow-up (i.e., all measurements past Screening for this study), only the short axis will be measured and followed. See also notes below on "Baseline documentation of target and non-target lesions" for information on lymph node measurement.

1.1.2. Nonmeasurable

All other lesions include small lesions (longest diameter < 10 mm or pathological lymph nodes with ≥ 10 mm to < 15 mm short axis) as well as truly non-measurable lesions. Lesions considered truly non-measurable include: leptomeningeal disease, ascites, pleural or pericardial effusion, inflammatory breast disease, lymphangitic involvement of skin or lung, and abdominal masses/abdominal organomegaly identified by physical examination that is not measurable by reproducible imaging techniques.

1.1.3. Special Considerations Regarding Lesion Measurability

Bone lesions, cystic lesions, and lesions previously treated with local therapy require particular consideration:

1.1.4. Bone Lesions

- Bone scan, positron emission tomography scan, or plain films are not considered adequate imaging techniques to measure bone lesions. However, these techniques can be used to

confirm the presence or disappearance of bone lesions.

- Lytic bone lesions or mixed lytic-blastic lesions, with identifiable soft tissue components, that can be evaluated by cross sectional imaging techniques such as CT or MRI can be considered as measurable lesions if the soft tissue component meets the definition of measurability described above.
- Blastic bone lesions are nonmeasurable.

1.1.5. Cystic Lesions

- Lesions that meet the criteria for radiographically defined simple cysts should not be considered as malignant lesions (neither measurable nor nonmeasurable) since they are, by definition, simple cysts.
- "Cystic lesions" thought to represent cystic metastases can be considered as measurable lesions, if they meet the definition of measurability described above. However, if noncystic lesions are present in the same subject, these are preferred for selection as target lesions.

1.1.6. Lesions with Prior Local Treatment

- Tumor lesions situated in an area subjected to previous loco-regional therapy (e.g., TACE), or radiotherapy are usually not considered measurable unless there has been demonstrated progression in the lesion.

1.2. Specifications by Methods of Measurements

1.2.

1.2.1. Measurement of Lesions

All measurements should be recorded in metric notation using calipers if clinically assessed. All baseline evaluations should be performed as close as possible to the treatment start and NEVER more than 4 weeks before the beginning of the treatment.

1.2.2. Method of Assessment

The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Imaging-based evaluation should always be done rather than clinical examination unless the lesion(s) being followed cannot be imaged but are assessable by clinical examination.

CT, MRI: CT is the best currently available and reproducible method to measure lesions selected for response assessment. This guideline has defined measurability of lesions on CT scan based on the assumption that CT slice thickness is 5 mm or less. When CT scans have slice thickness greater than 5 mm, the minimum size for a measurable lesion should be twice the slice thickness. MRI is also acceptable in certain situations (e.g., for body scans).

2. Tumor Response Evaluation

2.

2.1. Assessment of Overall Tumor Burden and Measurable Disease

To assess objective response or future progression, it is necessary to estimate the overall tumor burden at baseline and use this as a comparator for subsequent measurements. In this study, only subjects with measurable disease at baseline should be included. (See definition "1.1.2")

2.2. Baseline Documentation of "Target" and "Non-target" Lesions

When > 1 measurable lesion is present at baseline, all lesions up to a maximum of 5 lesions total representative of all involved organs should be identified as target lesions and will be recorded and measured at baseline (this means in instances where subjects have only 1 or 2 organ sites involved a maximum of 2 and 4 lesions respectively will be recorded).

Target lesions should not be selected on the basis of their sizes (lesions with the longest diameter), be representative of all involved organs, but in addition should be those that lend themselves to reproducible repeated measurements. It may be the case that, on occasion, the largest lesion does not lend itself to reproducible measurement in which circumstance the next largest lesion that can be measured reproducibly should be selected.

Lymph nodes merit special mention since they are normal anatomical structures, which may be visible by imaging even if not involved by tumor. As noted above, pathological nodes that are defined as measurable and may be identified as target lesions must meet the criterion of a short axis of ≥ 15 mm by CT scan. Only the short axis of these nodes will contribute to the baseline sum. The short axis of the node is the diameter normally used by radiologists to judge if a node is involved by solid tumor. Nodal size is normally reported as 2 dimensions in the plane in which the image is obtained (for CT scan this is almost always the axial plane; for MRI the plane of acquisition may be axial, sagittal, or coronal). The smaller of these measures is the short axis. For example, an abdominal node that is reported as being 20 mm × 30 mm has a short axis of 20 mm and qualifies as a malignant, measurable node. In this example, 20 mm should be recorded as the node measurement. All other pathological nodes (those with short axis ≥ 10 mm but < 15 mm) should be considered non-target lesions. Nodes that have a short axis < 10 mm are considered non-pathological and should not be recorded or followed.

A sum of the diameters (longest for nonnodal lesions, short axis for node lesions) for all target lesions will be calculated and reported as the baseline sum diameters. If lymph nodes are to be included in the sum, then as noted above, only the short axis is added into the sum. The baseline sum diameters will be used as reference to further characterize any objective tumor regression in the measurable dimension of the disease.

All other lesions (or sites of disease) including pathological lymph nodes should be identified as

nontarget lesions and should also be recorded at baseline. Measurements are not required and these lesions should be followed as "present", "absent", or in rare cases "unequivocal progression". In addition, it is possible to record multiple non-target lesions involving the same organ as a single item on the case record form (CRF; e.g., "multiple enlarged pelvic lymph nodes" or "multiple liver metastases").

2.3. Response Evaluation Criteria (RECIST v1.1)

This section provides the definitions of the criteria based on RECIST v1.1 used to determine objective tumor response for target lesions, non-target lesions and new lesions.

2.

2.1.

2.2.

2.3.

2.3.1. Evaluation of Target Lesions

Complete Response (CR): Disappearance of all target lesions. Any pathological lymph nodes (whether target or nontarget) must have reduction in short axis to < 10 mm.

Partial Response (PR): At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters.

Progressive Disease (PD): At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: The appearance of 1 or more new lesions is also considered progression.)

Stable Disease (SD): Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum diameters while on study.

2.3.2. Special Notes on the Assessment of Target Lesions

Lymph nodes: Lymph nodes identified as target lesions should always have the actual short axis measurement recorded (measured in the same anatomical plane as the baseline examination), even if the nodes regress to below 10 mm on study. This means that when lymph nodes are included as target lesions, the "sum" of lesions may not be zero even if CR criteria are met, since a normal lymph node is defined as having a short axis of < 10 mm. For PR, SD and PD, the actual short axis measurement of the nodes is to be included in the sum of target lesions.

Target lesions that become “too small to measure”: While on study, all lesions (nodal and non-nodal) recorded at baseline should have their actual measurements recorded at each subsequent evaluation, even when very small (e.g., 2 mm). However, sometimes lesions or lymph nodes that are recorded

as target lesions at baseline become so faint on CT scan that the radiologist may not consider it appropriate to assign an exact measure and may report them as being "too small to measure". When this occurs, it is important that a value be recorded in the CRF. If it is the opinion of the radiologist that the lesion has likely disappeared, the measurement should be recorded as 0 mm. If the lesion is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned. (Note: It is less likely that this rule will be used for lymph nodes since they usually have a definable size when normal and are frequently surrounded by fat such as in the retroperitoneum; however, if a lymph node is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned in this circumstance as well). This default value is derived from the 5 mm CT slice thickness (but should not be changed with varying CT slice thickness). The measurement of these lesions is potentially nonreproducible, therefore providing this default value will prevent false responses or progressions based upon measurement error. To reiterate, however, if the radiologist is able to provide an actual measure, that should be recorded, even if it is below 5 mm.

Lesions that split or coalesce on treatment: When non-nodal lesions "fragment", the longest diameters of the fragmented portions should be added together to calculate the target lesion sum. Similarly, as lesions coalesce, a plane between them may be maintained that would aid in obtaining maximal diameter measurements of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the vector of the longest diameter in this instance should be the maximal longest diameter for the "coalesced lesion".

2.3.3. Evaluation of Nontarget Lesions

This section provides the definitions of the criteria used to determine the tumor response for the group of nontarget lesions. While some nontarget lesions may actually be measurable, they need not be measured and instead should be assessed only qualitatively at the time points specified in the protocol.

CR: Disappearance of all non-target lesions and normalization of tumor marker level. All lymph nodes must be non-pathological in size (< 10 mm short axis).

Non-CR/Non-PD: Persistence of 1 or more non-target lesion(s) and/or maintenance of tumor marker level above the normal limits.

PD: Unequivocal progression (see comments below) of existing non-target lesions (Note: The appearance of 1 or more new lesions is also considered progression).

2.3.4. Special Notes on Assessment of Progression of Nontarget Disease

The concept of progression of nontarget disease requires additional explanation as follows, when the subject also has measurable disease. In this setting, to achieve "unequivocal progression" on the

basis of the non-target disease, there must be an overall level of substantial worsening in non-target disease such that, even in the presence of SD or PR in target disease, the overall tumor burden has increased sufficiently to merit discontinuation of therapy. A modest "increase" in the size of 1 or more non-target lesions is usually not sufficient to qualify for unequivocal progression status. The designation of overall progression solely on the basis of change in non-target disease in the presence of SD or PR of target disease will therefore be extremely rare.

2.3.5. New Lesions

The appearance of new malignant lesions denotes disease progression; therefore, some comments on detection of new lesions are important. There are no specific criteria for the identification of new radiographic lesions; however, the finding of a new lesion should be unequivocal: i.e. not attributable to differences in scanning technique, change in imaging modality or findings thought to represent something other than tumor (e.g., some "new" bone lesions may be simply healing or flare of pre-existing lesions). This is particularly important when the patient's baseline lesions show PR or CR. For example, necrosis of a liver lesion may be reported on a CT scan report as a "new" cystic lesion, which it is not.

A lesion identified on a follow-up study in an anatomical location that was not scanned at baseline is considered a new lesion and will indicate disease progression. An example of this is the subject who has visceral disease at baseline and while on study has a CT or MRI brain ordered, which reveals metastases. The subject's brain metastases are considered to be evidence of PD even if they did not have brain imaging at baseline.

If a new lesion is equivocal, for example because of its small size, continued therapy and follow-up evaluation will clarify if it represents truly new disease. If repeat scans confirm there is definitely a new lesion, then progression should be declared using the date of the initial scan.

2.4. Evaluation of Overall Response

The patient's overall response assignment will depend on the findings of both target and nontarget disease and will also take into consideration the appearance of new lesions. It is assumed that at each protocol-specified time point, a response assessment occurs. The table below provides a summary of the overall response status calculation at each time point for subjects who have measurable disease at baseline.

Table 18 Overall response

Time Point Response: Subjects with Target (± Non-target) Disease			
Target Lesions	Nontarget Lesions	New Lesions	Overall Response
CR	CR	No	CR
CR	Non-CR/non-PD	No	PR
CR	Not evaluated	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD

Not all evaluated	Non-PD	No	NE
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

2.4.

2.4.1. Missing Assessment and Inevaluable Designation

When no imaging/measurement is done at all at a particular time point, the subject is not evaluable at that time point. If only a subset of lesion measurements is made at an assessment, usually the case is also considered not evaluable at that time point, unless a convincing argument can be made that the contribution of the individual missing lesion(s) would not change the assigned time point response. This would be most likely to happen in the case of PD. For example, if a subject had a baseline sum of 50 mm with 3 measured lesions and at follow-up only 2 lesions were assessed, but those gave a sum of 80 mm, the patient will have achieved PD status, regardless of the contribution of the missing lesion.

2.4.2. Special Notes on Response Assessment

When nodal disease is included in the sum of target lesions and the nodes decrease to "normal" size (< 10 mm), they may still have a measurement reported on scans. This measurement should be recorded even though the nodes are normal in order not to overstate progression should it be based on increase in size of the nodes. As noted earlier, this means that subjects with CR may not have a total sum of "zero" on the CRF.

Subjects with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be reported as "symptomatic deterioration". Every effort should be made to document objective progression even after discontinuation of treatment. Symptomatic deterioration is not a descriptor of an objective response: it is a reason for stopping study therapy. The objective response status of such subjects is to be determined by evaluation of target and nontarget disease.

For equivocal findings of progression (e.g., very small and uncertain new lesions; cystic changes or necrosis in existing lesions), treatment may continue until the next scheduled assessment. If at the next scheduled assessment, progression is confirmed, the date of progression should be the earlier date when progression was suspected.

APPENDIX 2 IMMUNE RESPONSE EVALUATION CRITERIA IN SOLID TUMORS (iRECIST)

The following text is obtained from the following reference: Lesley Seymour, Jan Bogaerts, Andrea Perrone, et al. iRECIST: guidelines for response criteria for use in trials testing immunotherapeutics. Lancet Oncol 2017; 18: e143-52.

Assessment and decision-making at progression as per RECIST 1.1

For subjects who have radiological PD assessed by the investigator per RECIST 1.1, the investigator will decide whether the subjects continue the study treatment or not until the similar imaging findings per iRECIST for subject management are obtained. The investigator should decide it based on the subject's overall clinical condition.

Clinical stability is defined as the following:

- No symptoms or signs indicating clinically significant PD
- No decrease in ECOG performance status
- No need for intensive treatment, including intensive pain relief, radiation therapy or other palliative care

Any subjects who are determined to be clinically unstable should be discontinued from study treatment at first site-assessed radiological evidence of PD. Repeat imaging examination is not required for confirmation of PD per iRECIST.

If the investigator decides to continue treatment, the subject may continue to receive the study treatment and then have the tumor assessment for re-examination 4 to 8 weeks later for confirmation of PD as assessed by the investigator per iRECIST.

iRECIST defines new response categories, including iUPD (unconfirmed PD) and iCPD (confirmed PD). For assessment as per iRECIST, the first visit showing PD based on RECIST 1.1 will be categorized as a visit (overall) response iUPD regardless of the factor causing PD. At the time of the visit, the target and non-target lesions found as per RECIST 1.1 at the baseline will be assessed in accordance with the usual practice.

New lesions will be classified as measurable or non-measurable, using the same size thresholds and rules as for baseline lesion assessment per RECIST 1.1. Of new measurable lesions, up to 5 lesions (up to 2 per organ) may be selected as New Lesions – Target (NLT). The sum of diameters of these lesions will be calculated and differentiated from the sum of diameters of target lesions at baseline. All other new lesions will be followed qualitatively as New Lesions – Non-target (NLNT).

Assessment at confirmatory imaging

On the confirmatory imaging, the subjects will be classified as progression confirmed (with an overall response of iCPD), or as showing persistent unconfirmed progression (with an overall

response of iUPD), or as showing stable disease or response (iSD/iPR/iCR).

Confirmation of disease progression

Disease progression is considered confirmed, and the overall response will be iCPD, if any of the following occurs:

- Any of the factors that are the basis for the initial iUPD show worsening.
  - For target lesions, worsening is a further increase in the sum of diameters of ≥5 mm, compared to any prior iUPD time point.
  - For non-target lesions, worsening is any significant growth in lesions overall, compared to a prior iUPD time point; this does not have to meet the "unequivocal" standard of RECIST 1.1.
  - For new lesions worsening is any of these:
    - ✓ An increase in the new lesion sum of diameters by ≥5 mm from a prior iUPD time point
    - ✓ Definite enlargement of new non-target lesions
    - ✓ Appearance of additional new lesions
- Any new factor appears that would have triggered PD by RECIST 1.1

Ongoing *iUPD*

Disease progression is considered not confirmed, and the overall response remains iUPD, if any of the following occurs:

- Absence of any of the above factors confirming PD and
- The target lesion sum of diameters (initial target lesions) remains above the initial PD threshold (by RECIST 1.1).

Additional imaging should be scheduled 4 to 8 weeks after iUPD is found during confirmatory imaging. This may be equivalent to the next visit in the original visit schedule. The assessment of the subsequent confirmation imaging proceeds in an identical manner, with possible outcomes of iCPD, iUPD, and iSD/iPR/iCR.

iUPD recovery

Disease progression is considered not confirmed, and the overall response becomes iSD/iPR/iCR, if any of the following occurs:

- Absence of any of the above factors confirming PD and
- The sum of diameters of target lesion (initial target lesions) is not greater than the initial PD threshold.

The response is classified as iSD or iPR (depending on the sum of diameters of the target lesions), or iCR if all lesions resolve.

In this case, the initial iUPD is considered to be pseudo-progression, and the level of suspicion for progression is "reset". This means that regardless of the timing of the next visit that shows radiological progression, it will be reclassified as iUPD per iRECIST, i.e., the confirmation process will be repeated until iCPD is determined.

Management after confirmatory imaging

If unconfirmed PD is assessed by investigator per iRECIST in the imaging re-examination and the subject's clinical condition is still stable, the subjects can continue the study treatment and periodically receive the imaging examination as scheduled. If PD is confirmed, subjects will discontinue the study treatment.

Detection of PD at visits after pseudo-progression resolves

After resolution of pseudo-progression (i.e., achievement of iSD/iPR/iCR), iUPD is indicated by any of the following events:

- Target Lesions

Sum of diameters reaches the PD threshold (≥20% and ≥5 mm increase from nadir) either for the first time, or after resolution of previous pseudo-progression. The nadir is always the smallest sum of diameters observed before or after the pseudo-progression during the entire trial.

- Nontarget Lesions
  - If the non-target lesion never shows unequivocal progression, the first unequivocal progression will be assessed as iUPD.
  - If the non-target lesion has previously shown unequivocal progression and has not resolved, any further significant growth of the non-target lesion (as a whole) will be assessed as iUPD.
- New Lesions
  - First appearance of new lesions
  - Appearance of additional new lesions
  - Previously identified new target lesions show an increase of ≥ 5 mm in the new lesion sum of diameters, from the nadir value of that sum
  - Previously identified non-target lesions show any significant growth

If any of the events above occur, the overall response for that visit will be assessed as iUPD, and the iUPD evaluation process (see assessment at the confirmatory imaging above) will be repeated. Disease progression must be confirmed before iCPD.

The decision process is identical to the iUPD confirmation process for the initial PD, with one exception: if new lesions occurred at a prior instance of iUPD, and at the confirmatory imaging the burden of new lesions has increased from its nadir (for new target lesions, the sum of diameters is

≥5 mm increased from its nadir), then iUPD cannot resolve to iSD or iPR. It will remain iUPD until either a decrease in the new lesion burden allows resolution to iSD or iPR, or until a confirmatory factor causes iCPD.

Appendix 3 CALCULATION METHOD OF SERUM CREATINE CLEARANCE

Men:

$$\text{CrCl (mL/min)} = \frac{[140\text{-age (years)}] \times \text{weight (kg)}}{0.818 \times \text{serum creatinine } (\mu\text{mol/L})}$$

or

$$\text{CrCl (mL/min)} = \frac{[140\text{-age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dL)}}$$

Women:

$$\text{CrCl (mL/min)} = 0.85 \times \frac{[140\text{-age (years)}] \times \text{weight (kg)}}{0.818 \times \text{serum creatinine } (\mu\text{mol/L})}$$

or

$$\text{CrCl (mL/min)} = 0.85 \times \frac{[140\text{-age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dL)}}$$

Appendix 4 NEW YORK HEART ASSOCIATION CLASSIFICATION

Category	Symptoms
I	No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, or dyspnea (shortness of breath).
II	Slight limitation of physical activity. Comfortable at rest, but ordinary physical activity results in fatigue, palpitation, dyspnea (shortness of breath).
III	Marked limitation of physical activity. Comfortable at rest, but less than ordinary activity causes fatigue, palpitation, or dyspnea.
IV	Any physical activity causes discomfort. Symptoms of heart failure at rest. If any physical activity is undertaken, discomfort increases.

Adapted from

Dolgin M, Association NYH, Fox AC, Gorlin R, Levin RI, New York Heart Association. CriteriaCommittee.Nomenclature and criteria for diagnosis of diseases of the heart and great vessels. 9th ed. Boston, MA:LippincottWilliams and Wilkins; March 1,1994.

Original source: Criteria Committee, New York Heart Association, Inc. Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis,6th edition Boston, Little, Brown and Co.1964, p114.

Appendix 5 GUIDELINES FOR ADMINISTRATION OF CAPECITABINE

The strength of capecitabine in this study is 500 mg/tablet, and the principle of administration of capecitabine is based on the following literatures: (Marsé H, et al. Management of adverse events and other practical considerations in patients receiving capecitabine (Xeloda). Eur J Oncol Nurs. 2004;8 Suppl 1:S16-30)

1. Capecitabine 1000 mg/m^2 twice daily (2000 mg/m^2 daily)

Body Surface Area (m 2)	Dose in this study (once in the morning and once in the evening, strength: 500 mg/tablet)		
	Total dose/day (mg)	Morning (tablets)	Evening (tablets)
≤1.12	2500	2	3
1.13-1.62	3000	3	3
1.63-1.87	3500	3	4
1.88-2.12	4000	4	4
≥2.13	4500	4	5

2. Capecitabine 750 mg/m^2 twice daily (1500 mg/m^2 daily)

Body Surface Area (m 2)	Dose in this study (once in the morning and once in the evening, strength: 500 mg/tablet)		
	Total dose/day (mg)	Morning (tablets)	Evening (tablets)
≤1.12	2000	2	2
1.13-1.62	2500	2	3
≥1.63	3000	3	3

3. Capecitabine 500 mg/m^2 twice daily (1000 mg/m^2 per day)

Body Surface Area (m 2)	Dose in this study (once in the morning and once in the evening, strength: 500 mg/tablet)		
	Total dose/day (mg)	Morning (tablets)	Evening (tablets)
≤1.12	1000	1	1
1.13-1.62	1500	1	2
≥1.63	2000	2	2

SPONSOR SIGNATURE PAGE

A Randomized, Double-blind, Multicenter, Phase III Clinical Study of Comparing the Efficacy and Safety of AK104 Plus Oxaliplatin and Capecitabine (XELOX) Versus Placebo Plus XELOX as First-line Treatment for Locally Advanced Unresectable or Metastatic Gastric Adenocarcinoma or Gastroesophageal Junction Adenocarcinoma

This clinical study protocol has been reviewed and approved by the following representatives of Akeso Pharmaceuticals, Inc.

Print Name	**Signature**
Title	(MM) (DD), 2021 **Date (MM/DD/YYYY)**

INVESTIGATOR SIGNATURE PAGE

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I have reviewed this protocol, and I agree to conduct this protocol in accordance with ethical principles that have their origins in the Declaration of Helsinki, International Council for Harmonisation guidelines on GCP, and any conditions required by a regulatory authority and/or IRB/IEC.

I understand that the protocol maynot be modifiedwithout written approval ofthe Sponsor. All changes to the protocol must be submitted to the applicable regulatory authority and IEC or IRB, and must be approved by the IEC or IRB prior to implementation except when necessary to eliminate immediate hazards to the subjects or when the change(s), as deemed by the Sponsor, involves only logistical or administrative changes.

I agree to make available to Sponsor personnel, their representatives, and relevant regulatory authorities, my subjects' study records in order to verify the data I have entered into the case report form. I understand my responsibilities as the Sponsor-designated Principal InvestigatorI understand that the Sponsor may decide to suspend or prematurely terminate the study at any time for whatever reason; such a decision will be communicated to me in writing. Conversely, should I decide to withdraw from the execution of the study, I will communicate my intention immediately in writing to the Sponsor.

Print Name	**Signature**
	(MM) (DD), (YYYY)
Title	**Date (MM/DD/YYYY)**